A STUDY OF RANDOM MATRICES

Sample Covariance Matrices and Spectral Clustering

AN INDEPENDENT STUDY REPORT

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Abstract

[TODO] *Write the abstract once the body is finalized. Suggested skeleton: (1) Open with a one-sentence framing of random matrix theory and why sample covariance matrices matter in high-dimensional statistics. (2) State the scope: we focus on Gaussian sample covariance matrices in Johnstone's spiked model. (3) Outline the journey: foundational probability and linear algebra, sample covariance matrices and the Wishart distribution, low-dimensional fluctuation behavior, the high-dimensional Marchenko-Pastur and Tracy-Widom laws, the spiked covariance model and its spike phase transition, eigenvector alignment beyond the threshold, and a sketch of the connection to spectral clustering. (4) Note the role of Monte Carlo simulations in illustrating each phase of the argument. (5) Close with the takeaway: the contrast between strong and weak spikes produces a clean phase transition in both eigenvalue location and eigenvector informativeness, with implications for spectral clustering. Avoid em dashes; use clean academic prose.*

Acknowledgements

[**TODO**] *Acknowledgements paragraph. Suggested mentions: Professor Gustavo Didier (advisor) for the weekly meetings, the reading recommendations (Yao, Zheng, and Bai 2015; Féral and Péché 2009), and the patient guidance through the spiked covariance literature. The second reader (TBD) for serving on the committee. Optional: family, friends, and the Tulane Department of Mathematics.*

Contents

1 Introduction

Why random matrices?

Modern data sets are large and high-dimensional. A single-cell genomics experiment may measure tens of thousands of gene expression levels in a few hundred cells. A finance application may track hundreds of asset returns over a few years of daily data. A computer-vision system may compare millions of image patches in a feature space of several hundred dimensions. In each case the number of variables p is comparable to, or larger than, the number of observations n. The most basic question one can ask of such data is how its variance is distributed across orthogonal directions, and which directions explain the most variation. Answering this question is the job of *principal components analysis*, and the algorithm reduces to computing the eigenvalues and eigenvectors of the *sample covariance matrix*

$$\widehat{\Sigma}_n \;=\; \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{X}_i^{\top}, \tag{1}$$

where the $\boldsymbol{X}_i \in \mathbb{R}^p$ are the n observed data points, assumed centered.

Classical multivariate statistics tells us what to expect when p is small and n is large. In this low-dimensional regime, $\widehat{\Sigma}_n$ is an accurate estimate of the population covariance Σ, its eigenvalues converge to the population eigenvalues, and its eigenvectors converge to the population eigenvectors. Methods that depend on this convergence, from PCA to factor analysis to the *spectral clustering* algorithms that we will reach at the end of this report, then stand on firm ground.

The classical picture breaks down when p and n are comparable. Each entry of $\widehat{\Sigma}_n$ is still an unbiased estimator of the corresponding entry of Σ (every entry is a sample average; see Section 3), but the *eigenvalues* of $\widehat{\Sigma}_n$ become inconsistent estimators of the eigenvalues of Σ: they spread into a non-trivial band of random values rather than concentrating near the population eigenvalues, and the corresponding eigenvectors may carry little or no information about the population directions. Figure 1 makes the contrast concrete. The same Gaussian sampling model produces eigenvalues that concentrate at the population value 1 when $p = 10$ and $n = 1000$, and eigenvalues that spread across a wide interval, with a well-defined limiting density, when $p = 500$ and $n = 1000$.

The mathematical framework that explains and predicts this phenomenon is *random matrix theory*, originally developed for problems in nuclear physics in the 1950s and now a central tool in high-dimensional statistics, signal processing, and machine learning (Yao et al., 2015; Vershynin, 2018; Couillet and Liao, 2022).

What this report does, and how

This report develops, from first principles, the spectral analysis of sample covariance matrices in both regimes, and uses what we learn to understand when methods like spectral clustering can recover hidden structure in high-dimensional data. The development is built bottom-up: every claim is either proved in the text, sketched with a pointer to a full proof, or cited with the cleanest reference we could find. Every theoretical claim is then illustrated with

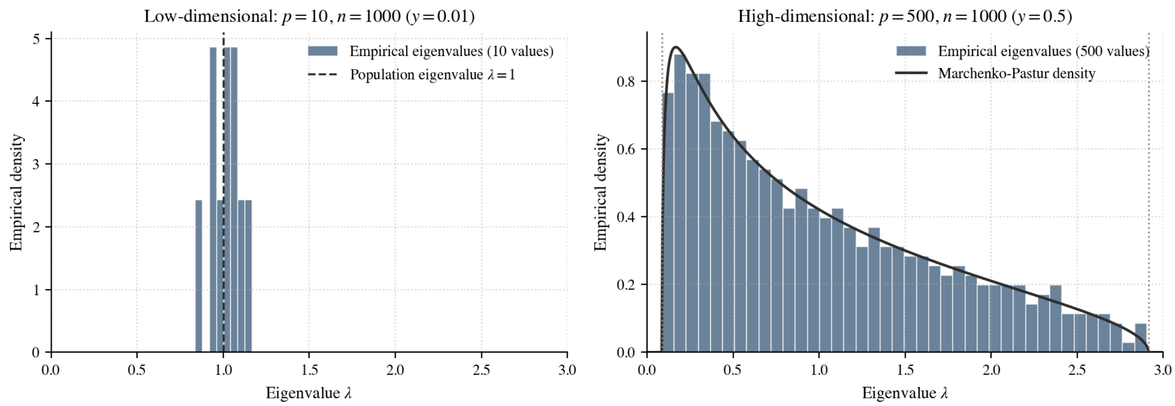

Figure 1: Eigenvalues of the sample covariance matrix $\widehat{\Sigma}_n$ when $\Sigma = I_p$ and the data are Gaussian, for two aspect ratios. *Left:* when $p = 10$ and $n = 1000$ (so $p/n = 0.01$), the ten sample eigenvalues concentrate tightly around the population value $\lambda = 1$, confirming the classical low-dimensional intuition. *Right:* when $p = 500$ and $n = 1000$ (so $p/n = 0.5$), the five hundred sample eigenvalues spread across the interval $[(1-\sqrt{y})^2, (1+\sqrt{y})^2] \approx [0.086, 2.914]$. The empirical density agrees with the Marchenko-Pastur prediction (solid curve). The phenomenon that motivates this report is already visible: in high dimensions, the sample eigenvalues are spectrally inconsistent (their limiting positions are not the population eigenvalues), and the spread is a real feature of the high-dimensional geometry, not measurement noise.

a Monte Carlo simulation in Python; the notebooks live in the `notebooks/` directory of the accompanying repository, and the figures they produce are reproduced alongside the corresponding statements.

The starting point is the random variable, the basic building block of probability theory. From there we build to moments and moment generating functions, characteristic functions, modes of convergence, the central limit theorem, random vectors with multivariate normal distributions, and the spectral theorem for symmetric matrices. Each of these objects feeds directly into the analysis of $\widehat{\Sigma}_n$ that follows.

The story arc

The mathematical journey unfolds in three movements.

1. *The low-dimensional regime* (p fixed, $n \to \infty$). The law of large numbers gives entrywise convergence of $\widehat{\Sigma}_n$ to Σ, and continuity of the characteristic polynomial then forces the sample eigenvalues to converge to the population eigenvalues. The size and shape of the fluctuations around these limits depend on whether the population eigenvalues are simple or repeated. For $\Sigma = I_p$ the matrix $\sqrt{n}(\widehat{\Sigma}_n - I_p)$ converges in distribution to a *Gaussian Orthogonal Ensemble* matrix, whose top eigenvalue has a non-Gaussian limit shaped by *eigenvalue repulsion*. For Σ with distinct eigenvalues, the delta method instead produces ordinary Gaussian fluctuations of each sample eigenvalue around its population counterpart.

2. *The high-dimensional regime* ($p, n \to \infty$ with $p/n \to y \in (0, \infty)$). Even for the null

covariance $\Sigma = I_p$, the sample eigenvalues no longer concentrate at 1; they fill out the *Marchenko-Pastur law* on the interval $[(1-\sqrt{y})^2, (1+\sqrt{y})^2]$ (Marchenko and Pastur, 1967), exactly as in the right panel of Figure 1. The largest eigenvalue converges almost surely to the upper edge $(1+\sqrt{y})^2$, but its fluctuations around that edge are not Gaussian. They live on the scale $p^{-2/3}$ and follow the *Tracy-Widom law* (Tracy and Widom, 1996; Johnstone, 2001), a distinct universality class first discovered in random matrix theory and now found in problems ranging from longest increasing subsequences to growth models in mathematical physics.

3. *Structure and a phase transition.* When the population covariance carries low-rank structure, a small number of distinguished eigenvalues riding above a flat baseline, a striking new phenomenon appears. If a signal eigenvalue exceeds a threshold determined by the aspect ratio y, an isolated outlier eigenvalue appears in the sample spectrum and the corresponding sample eigenvector aligns nontrivially with the population spike direction. If the signal is below the threshold, the spike is absorbed into the Marchenko-Pastur bulk and its eigenvector becomes asymptotically orthogonal to the population direction. This phase transition is the bridge to spectral clustering: the same picture explains how clustering algorithms recover hidden class structure from data, but only when the underlying signal-to-noise ratio crosses an analogous threshold.

Where this is applicable

The techniques developed here are not abstract curiosities. The Marchenko-Pastur law sets a baseline for detecting structure in any high-dimensional covariance estimate, with applications in array-signal processing, wireless communications, and quantitative finance (Couillet and Liao, 2022). The Tracy-Widom edge governs the null distribution of the largest principal component, which is the basis for principled testing of how many genuine signal directions are present in a data set (Johnstone, 2001). The spike phase transition determines when a low-rank signal in noise is recoverable by PCA, with direct implications for denoising and dimensionality reduction. Spectral clustering, finally, is one of the most widely used unsupervised methods for problems where the natural geometry is not Euclidean, including image segmentation, community detection in networks, and clustering of single-cell data (von Luxburg, 2007; Ng et al., 2002). The phase transition we develop in the spiked model is the cleanest mathematical lens we have for understanding when these methods succeed and when they fail.

Roadmap

Section 2 reviews probability foundations: random variables, moments and moment generating functions, characteristic functions, modes of convergence, the law of large numbers and central limit theorem, and random vectors. Section 3 develops the sample covariance matrix in both regimes, from the spectral theorem and the Wishart distribution in low dimensions through the Marchenko-Pastur and Tracy-Widom laws in high dimensions. Section 4 introduces the spiked covariance model, proves the eigenvalue phase transition, derives the eigenvector overlap formula, and sketches the connection to spectral clustering. Section 5 concludes with a summary and open questions, and three appendices collect auxiliary probabilistic results, simulation methodology, and selected Python code listings.

2 Probability and Linear Algebra Foundations

The main objects of this report are eigenvalues and eigenvectors of random covariance matrices. To study them we need two complementary languages. *Probability* explains how averages of random quantities stabilize and how they fluctuate around their limits. *Linear algebra* explains how a symmetric matrix encodes variance through eigenvalues and eigenvectors. This section develops both, starting from the random variable and building toward the spectral theorem for symmetric covariance matrices.

2.1 Random Variables and Distributions

Probability begins with a *probability space* $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the set of possible outcomes, $\mathcal{F}$ is a collection of events, and $\mathbb{P}$ assigns each event a number in $[0, 1]$. The full development imposes some technical regularity on $\mathcal{F}$ and $\mathbb{P}$ (the language of σ-algebras and measures, developed in measure-theoretic probability); we will not need to engage with it directly. We work with this triple in the background; the foreground objects are real-valued functions of the outcome.

Definition 2.1 (Random variable)**.** A *random variable* is a function $X : \Omega \to \mathbb{R}$. Strictly, X is required to satisfy the technical condition $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$, so that $\mathbb{P}(X \leq x)$ is well defined; this is called *measurability*, holds for every random variable we will encounter, and we will not verify it explicitly. The distribution of X is captured by the *cumulative distribution function* (CDF)

$$F_X(x) \;=\; \mathbb{P}(X \leq x), \qquad x \in \mathbb{R}.$$

A CDF is determined by three properties.

Proposition 2.1 (Characterization of CDFs)**.** The CDF of any random variable satisfies:

(F1) $\lim\limits_{x\to-\infty} F_X(x) = 0$ and $\lim\limits_{x\to+\infty} F_X(x) = 1$;

(F2) F_X is right-continuous: $\lim_{x\to a^+} F_X(x) = F_X(a)$;

(F3) F_X is non-decreasing: $a \leq b \Rightarrow F_X(a) \leq F_X(b)$.

Conversely, any $F : \mathbb{R} \to [0, 1]$ satisfying (F1)–(F3) is the CDF of some random variable (Casella and Berger, 2002, §1.5).

Random variables come in two main flavors. *Discrete* random variables take values in a countable set $\{x_1, x_2, \ldots\}$, and their distribution is described by the *probability mass function* (PMF) $p_X(x_i) = \mathbb{P}(X = x_i)$. *Absolutely continuous* random variables admit a nonnegative *probability density function* (PDF) f_X with

$$F_X(x) \;=\; \int_{-\infty}^{x} f_X(t)\, dt.$$

Two random variables X, Y are *independent* if $\mathbb{P}(X \in A,\, Y \in B) = \mathbb{P}(X \in A)\, \mathbb{P}(Y \in B)$ for all intervals $A, B \subseteq \mathbb{R}$. (More generally one asks the identity to hold on all events; the version on intervals suffices for real-valued random variables.)

Example 2.2 (Distributions used throughout)**.** The following appear repeatedly in later sections:

- *Bernoulli*(p): $X \in \{0, 1\}$ with $\mathbb{P}(X = 1) = p$.
- *Poisson*(λ): $\mathbb{P}(X = k) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$.
- *Normal* $\mathcal{N}(\mu, \sigma^2)$: $f_X(x) = (\sqrt{2\pi}\,\sigma)^{-1} \exp\bigl(-(x-\mu)^2/(2\sigma^2)\bigr)$.
- *Exponential*(λ): $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.
- *Cauchy*: $f_X(x) = 1/\bigl(\pi(1 + x^2)\bigr)$.

Their shapes are shown in Figure 2.

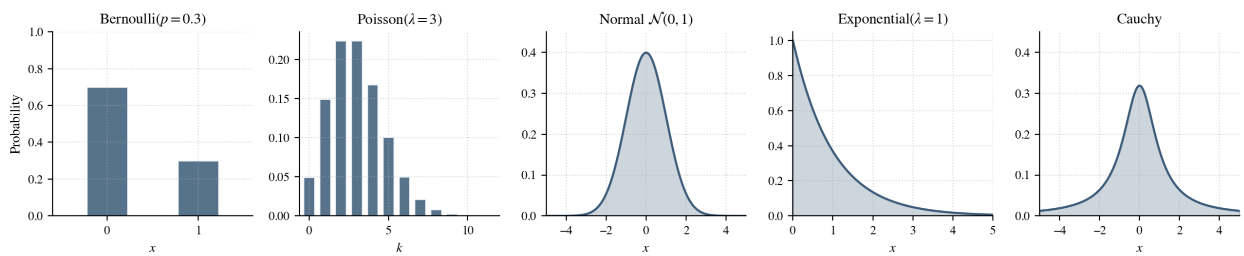

Figure 2: Shapes of the five distributions used throughout the report. Bernoulli and Poisson are discrete, displayed as probability mass functions; Normal, Exponential, and Cauchy are absolutely continuous, displayed as densities. The Normal and the Cauchy share a similar central profile but very different tails: the Cauchy density decays only like $1/x^2$, so values far from zero remain plausible. This tail behavior is what breaks the moment generating function and motivates the characteristic function below.

The Cauchy distribution will be a recurring foil: its heavy tails make some standard tools break, motivating sharper tools that do not.

2.2 Moments and Transform Methods

Moments quantify the shape of a distribution, and two ways to package the entire sequence of moments into a single function turn convergence-of-distributions questions into calculus. The first, the moment generating function, is the natural starting point. The second, the characteristic function, is the version that always exists and that powers proofs of limit theorems.

Moments and the moment generating function.

Definition 2.3 (Moments and central moments)**.** For an integer $j \geq 1$ such that $\mathbb{E}|X|^j < \infty$, the j^{th} *moment* of X is $\mathbb{E}[X^j]$ and the j^{th} *central moment* is $\mathbb{E}[(X - \mu)^j]$, where $\mu = \mathbb{E}[X]$. The second central moment is the *variance*, $\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2]$. The absolute-moment hypothesis ensures that X^j is integrable, so the bare expectation $\mathbb{E}[X^j]$ is well defined.

A generating function for the moments compresses all of this information into a single function.

Definition 2.4 (Moment generating function)**.** The *moment generating function* (MGF) of X is

$$M_X(t) \;=\; \mathbb{E}[e^{tX}], \qquad t \in \mathbb{R},$$

provided the expectation is finite on an open neighborhood $(-h, h)$ of zero.

The name is justified by the following.

Theorem 2.2 (MGFs generate moments)**.** If $M_X(t)$ is finite for $t \in (-h, h)$, then M_X is infinitely differentiable at 0 and

$$M_X^{(n)}(0) \;=\; \mathbb{E}[X^n], \qquad n = 1, 2, \ldots$$

Proof. We treat the continuous case in detail; the discrete case is identical with sums in place of integrals. The hypothesis that M_X is finite on an open interval around zero is the standard regularity condition that justifies differentiating under the integral sign in the calculation that follows; see Casella and Berger (2002, Thm. 2.3.7). Therefore,

$$\frac{d}{dt} M_X(t) \;=\; \frac{d}{dt} \int_{-\infty}^{\infty} e^{tx} f_X(x)\, dx \;=\; \int_{-\infty}^{\infty} \frac{d}{dt}\big(e^{tx}\big) f_X(x)\, dx.$$

The inner derivative is $\frac{d}{dt} e^{tx} = x\, e^{tx}$, so

$$\frac{d}{dt} M_X(t) \;=\; \int_{-\infty}^{\infty} x\, e^{tx} f_X(x)\, dx \;=\; \mathbb{E}\big[X\, e^{tX}\big].$$

Setting $t = 0$ inside the expectation gives $e^{0 \cdot X} = 1$, and hence

$$M_X'(0) \;=\; \mathbb{E}\big[X\, e^{0 \cdot X}\big] \;=\; \mathbb{E}[X].$$

Iterating the same step n times produces

$$M_X^{(n)}(t) \;=\; \mathbb{E}\big[X^n\, e^{tX}\big],$$

and evaluating at $t = 0$ yields $M_X^{(n)}(0) = \mathbb{E}[X^n]$, as claimed.

In the discrete case, the same calculation applies to the series $M_X(t) = \sum_x e^{tx} p_X(x)$; convergence of M_X on $(-h, h)$ justifies term-by-term differentiation, and the rest of the argument is unchanged. □

A second property is the source of most computational power: the MGF of a sum of independent random variables is the product of their MGFs. Indeed, if X and Y are independent,

$$M_{X+Y}(t) \;=\; \mathbb{E}[e^{t(X+Y)}] \;=\; \mathbb{E}[e^{tX}]\, \mathbb{E}[e^{tY}] \;=\; M_X(t)\, M_Y(t).$$

We compute MGFs of the canonical distributions to make this concrete.

Example 2.5 (MGFs of standard distributions)**.** We compute MGFs from the definition for three of the distributions in Example 2.2.

Bernoulli(p). The variable X takes only the values 0 and 1, so the expectation is a two-term sum:

$$M_X(t) \;=\; \mathbb{E}[e^{tX}] \;=\; e^{t\cdot 0}\,\mathbb{P}(X=0) \;+\; e^{t\cdot 1}\,\mathbb{P}(X=1) \;=\; (1-p) \;+\; p\,e^t.$$

Poisson(λ). Substitute the PMF $\mathbb{P}(X=k) = e^{-\lambda}\lambda^k/k!$ into the expectation:

$$M_X(t) \;=\; \sum_{k=0}^{\infty} e^{tk}\,e^{-\lambda}\,\frac{\lambda^k}{k!} \;=\; e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda\,e^t)^k}{k!}.$$

The inner sum is the Taylor expansion of exp evaluated at λe^t, namely $\exp(\lambda e^t)$. Combining,

$$M_X(t) \;=\; e^{-\lambda}\,e^{\lambda e^t} \;=\; \exp\bigl(\lambda(e^t-1)\bigr).$$

Normal $\mathcal{N}(\mu,\sigma^2)$. Starting from the density,

$$M_X(t) \;=\; \frac{1}{\sqrt{2\pi}\,\sigma}\int_{-\infty}^{\infty}\exp\Bigl(tx \;-\; \frac{(x-\mu)^2}{2\sigma^2}\Bigr)\,dx.$$

The trick is to complete the square in the exponent so the integrand becomes a Gaussian kernel. Expanding the quadratic and collecting terms,

$$tx \;-\; \frac{(x-\mu)^2}{2\sigma^2} \;=\; -\frac{1}{2\sigma^2}\Bigl(x-(\mu+\sigma^2 t)\Bigr)^2 \;+\; \Bigl(\mu t \;+\; \tfrac{1}{2}\sigma^2 t^2\Bigr).$$

The first piece is the kernel of a normal density with mean $\mu+\sigma^2 t$ and variance σ^2; divided by $\sqrt{2\pi}\,\sigma$ it integrates to 1. The second piece does not depend on x and comes outside the integral. Therefore,

$$M_X(t) \;=\; \exp\Bigl(\mu t \;+\; \tfrac{1}{2}\sigma^2 t^2\Bigr), \qquad t\in\mathbb{R}.$$

These three examples illustrate the technique. The multiplicative property turns difficult convolutions into easy multiplications: if $X_1,\dots,X_n$ are iid Bernoulli(p), their sum has MGF $(1-p+pe^t)^n$, which is the MGF of Binomial(n,p); independent Poissons of rates λ_1,λ_2 sum to a Poisson of rate $\lambda_1+\lambda_2$; independent normals sum to a normal with summed mean and summed variance.

Beyond computing moments, the MGF determines the distribution uniquely.

Theorem 2.3 (MGF uniqueness)**.** Let X and Y be real-valued random variables.

(1) If X and Y have the same distribution and $M_X(t)$ is finite at t, then $M_X(t) = M_Y(t)$.

(2) Conversely, if $M_X(t) = M_Y(t)$ for all t in some open neighborhood $(-h,h)$ of zero, then X and Y have the same distribution.

Proof of (1). If X and Y have the same distribution, then for any function g for which both expectations are well defined,

$$\mathbb{E}[g(X)] \;=\; \mathbb{E}[g(Y)].$$

(Both sides are computed by the same integral or sum against the common distribution function, so they agree.) Choosing $g(x) = e^{tx}$, which by hypothesis has a finite expectation under X at t, yields

$$M_X(t) \;=\; \mathbb{E}[e^{tX}] \;=\; \mathbb{E}[e^{tY}] \;=\; M_Y(t),$$

which is the claim. □

The converse direction (2) is the harder one: it says that the entire distribution of X is encoded in the MGF on any open interval around zero. The proof reduces to the uniqueness of the two-sided Laplace transform, since $M_X(t)$ is exactly that transform of the distribution evaluated at $s = -t$; see Casella and Berger (2002, Thm. 2.3.11) for the details.

There is a catch. The MGF requires e^{tX} to have a finite expectation for t near zero, which can fail if X has heavy tails.

Example 2.6 (Cauchy: no MGF)**.** For the standard Cauchy density $f(x) = 1/(\pi(1 + x^2))$, the MGF integral

$$\int_{-\infty}^{\infty} \frac{e^{tx}}{\pi(1 + x^2)}\, dx$$

diverges for every $t \neq 0$, since the integrand grows like e^{tx}/x^2 at the right tail (for $t > 0$) or at the left tail (for $t < 0$). The MGF therefore does not exist on any open neighborhood of zero. This is not a pathology; the Cauchy is a perfectly well-defined distribution, central in probability and physics.

The remedy is the characteristic function.

Characteristic functions. Replacing the real argument t of the MGF by the imaginary argument it yields an object that always exists.

Definition 2.7 (Characteristic function)**.** The *characteristic function* (CF) of a random variable X is

$$\varphi_X(t) \;=\; \mathbb{E}[e^{itX}] \;=\; \mathbb{E}[\cos(tX)] + i\,\mathbb{E}[\sin(tX)], \qquad t \in \mathbb{R}.$$

Because $|e^{itx}| = 1$ for every real x, the integrand is bounded, and so

$$|\varphi_X(t)| \;\le\; \mathbb{E}[|e^{itX}|] \;=\; 1, \qquad \varphi_X(0) = 1.$$

The CF is therefore well defined for every distribution. When X has a density, φ_X is exactly the Fourier transform of the density:

$$\varphi_X(t) \;=\; \int_{-\infty}^{\infty} e^{itx}\, f_X(x)\, dx.$$

When the MGF exists, formal substitution gives $\varphi_X(t) = M_X(it)$.

Example 2.8 (Standard normal CF)**.** For $X \sim \mathcal{N}(0,1)$, the MGF computed in Example 2.5 is

$$M_X(s) \;=\; e^{s^2/2},$$

and this expression is well defined for every complex s. Substituting $s = it$ using the identity $\varphi_X(t) = M_X(it)$ yields

$$\varphi_X(t) \;=\; M_X(it) \;=\; e^{(it)^2/2} \;=\; e^{-t^2/2}, \qquad t \in \mathbb{R}. \tag{2}$$

A direct computation from the definition of φ_X, completing the square in the exponent of the integrand, gives the same answer.

Example 2.9 (Standard Cauchy CF)**.** Let X be standard Cauchy with density $f(x) = 1/(\pi(1+x^2))$. By definition,

$$\varphi_X(t) \;=\; \int_{-\infty}^{\infty} e^{itx}\,\frac{1}{\pi(1+x^2)}\,dx \;=\; \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{e^{itx}}{1+x^2}\,dx.$$

The remaining integral is a standard Fourier-transform identity (Feller (1971, §XV.4); or any Fourier-transform table):

$$\int_{-\infty}^{\infty} \frac{e^{itx}}{1+x^2}\,dx \;=\; \pi\, e^{-|t|}, \qquad t \in \mathbb{R}.$$

Dividing by π,

$$\varphi_X(t) \;=\; e^{-|t|}, \qquad t \in \mathbb{R}. \tag{3}$$

The Cauchy CF is well defined and continuous, even though the Cauchy MGF is identically infinite for every $t \neq 0$.

Figure 3 contrasts the two cases. Although the Cauchy and Normal densities look superficially similar near the origin, their tails are very different, and that difference is exactly what the characteristic function transports into a well-behaved object.

Two algebraic properties make CFs computationally convenient. By independence, $\varphi_{X+Y}(t) = \varphi_X(t)\,\varphi_Y(t)$; by direct substitution, $\varphi_{aX+b}(t) = e^{itb}\,\varphi_X(at)$. Like the MGF, the CF determines the distribution uniquely, and an inversion formula recovers the CDF from φ_X (Billingsley, 1995, §26).

The deepest property of characteristic functions is the bridge they provide between pointwise convergence of φ_{X_n} and convergence in distribution of X_n.

Theorem 2.4 (Lévy continuity theorem)**.** Let X_n and X be random variables with characteristic functions φ_{X_n} and φ_X.

(1) If $X_n \xrightarrow{d} X$, then $\varphi_{X_n}(t) \to \varphi_X(t)$ for every $t \in \mathbb{R}$.

(2) Conversely, if $\varphi_{X_n}(t) \to \varphi(t)$ for every $t \in \mathbb{R}$ and φ is continuous at 0, then φ is the CF of some random variable X, and $X_n \xrightarrow{d} X$.

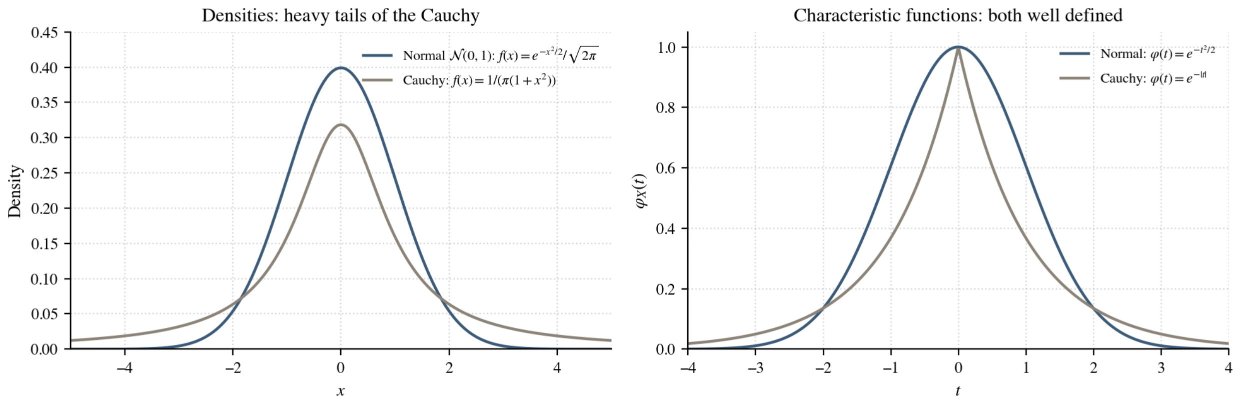

Figure 3: Left: the standard Normal and standard Cauchy densities on the same axes. The Cauchy has a slightly lower peak and noticeably heavier tails; the area in the tails beyond $|x| = 3$ is several times larger for the Cauchy than for the Normal. Right: their characteristic functions, $e^{-t^2/2}$ for the Normal and $e^{-|t|}$ for the Cauchy. Both are well-defined real functions on all of $\mathbb{R}$ and decay to zero, even though the Cauchy's MGF is identically infinite for every $t \neq 0$. This is the practical sense in which the CF rescues the heavy-tailed case.

A full proof is in Billingsley (1995, Thm. 26.3). This theorem reduces convergence in distribution to pointwise convergence of characteristic functions, a calculus question on a single complex-valued function. It is the engine behind the central limit theorem proved below.

2.3 Convergence and Limit Theorems

Many statistical quantities are averages of independent observations. Two universality results describe their behavior: the law of large numbers says averages converge to their mean, and the central limit theorem says that after rescaling their random fluctuations are universally Gaussian. Before stating either we need a precise notion of convergence for random variables, because the two laws use different ones.

Modes of convergence. A sequence of random variables can converge to a limit in several inequivalent ways. The three modes used here, from strongest to weakest, are the following.

Definition 2.10 (Three modes of convergence)**.** Let X_n, X be random variables on a common probability space.

- $X_n \xrightarrow{\text{a.s.}} X$ (*almost surely*): $\mathbb{P}\big(\lim_n X_n = X\big) = 1$.
- $X_n \xrightarrow{\mathbb{P}} X$ (*in probability*): for every $\varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$.
- $X_n \xrightarrow{d} X$ (*in distribution*): $F_{X_n}(x) \to F_X(x)$ at every continuity point of F_X.

Proposition 2.5 (Implication chain)**.** $X_n \xrightarrow{\text{a.s.}} X \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X$. The converses are false in general.

Sketch. For a.s. $\Rightarrow$ in probability: pointwise convergence holds with probability one, so for every $\varepsilon > 0$ the probability that $|X_n - X| > \varepsilon$ tends to zero. For in probability $\Rightarrow$ in distribution: the conclusion is on CDFs at continuity points and follows by a standard ε-argument; see Casella and Berger (2002, Thm. 5.5.12). □

A canonical counterexample showing that probability does not imply almost-sure convergence:

Example 2.11 ($\xrightarrow{\mathbb{P}}$ but not $\xrightarrow{\text{a.s.}}$)**.** Define a sequence $X_n \in \{0, 1\}$ by

$$\mathbb{P}(X_n = 1) \;=\; \frac{1}{n}, \qquad \mathbb{P}(X_n = 0) \;=\; 1 - \frac{1}{n},$$

and assume the X_n are independent.

Convergence in probability to 0. Fix any $\varepsilon \in (0, 1)$. Since X_n only takes the values 0 and 1, the event $\{|X_n - 0| > \varepsilon\}$ is exactly $\{X_n = 1\}$. Therefore,

$$\mathbb{P}(|X_n - 0| > \varepsilon) \;=\; \mathbb{P}(X_n = 1) \;=\; \frac{1}{n} \;\longrightarrow\; 0,$$

which is the definition of $X_n \xrightarrow{\mathbb{P}} 0$.

Failure of almost-sure convergence. Let $A_n = \{X_n = 1\}$. The events A_n are independent and their probabilities satisfy

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) \;=\; \sum_{n=1}^{\infty} \frac{1}{n} \;=\; \infty.$$

The second Borel-Cantelli lemma (Appendix A) says that for independent events whose probabilities sum to infinity, infinitely many occur with probability one. Concretely,

$$\mathbb{P}\big(X_n = 1 \text{ for infinitely many } n\big) \;=\; 1.$$

So almost every realization of the sequence keeps producing the value 1 at arbitrarily late indices, which prevents it from settling at 0. Hence $X_n \not\xrightarrow{\text{a.s.}} 0$.

A second standard fact will be useful in later sections: continuous functions preserve convergence.

Proposition 2.6 (Continuous mapping theorem)**.** If $X_n \xrightarrow{\text{a.s.}} X$ (respectively $\xrightarrow{\mathbb{P}}$, $\xrightarrow{d}$) and $g : \mathbb{R} \to \mathbb{R}$ is continuous, then $g(X_n) \xrightarrow{\text{a.s.}} g(X)$ (respectively $\xrightarrow{\mathbb{P}}$, $\xrightarrow{d}$).

Slutsky's theorem (also useful later) is recorded in Appendix A.

Markov, Chebyshev, and the law of large numbers. To prove a law of large numbers we need a way to turn information about the size of a random variable into a bound on the probability of a large deviation. Markov's inequality is the most basic such tool; Chebyshev's inequality is the version specialized to deviations from the mean.

Theorem 2.7 (Markov's inequality)**.** If $Y \geq 0$ and $a > 0$, then $\mathbb{P}(Y \geq a) \leq \mathbb{E}[Y]/a$.

Proof. On the event $\{Y \geq a\}$, the variable Y itself satisfies $Y \geq a$. Multiplying both sides by the indicator $\mathbf{1}_{\{Y \geq a\}}$,

$$Y\,\mathbf{1}_{\{Y \geq a\}} \;\geq\; a\,\mathbf{1}_{\{Y \geq a\}}.$$

Outside the event both sides equal zero, so the inequality holds everywhere on Ω. Take expectations and use that $\mathbf{1}_{\{Y \geq a\}}$ integrates to $\mathbb{P}(Y \geq a)$:

$$\mathbb{E}\big[Y\,\mathbf{1}_{\{Y \geq a\}}\big] \;\geq\; a\,\mathbb{P}(Y \geq a).$$

Finally, $Y\,\mathbf{1}_{\{Y \geq a\}} \leq Y$ (because $Y \geq 0$ and the indicator is at most 1), so

$$\mathbb{E}[Y] \;\geq\; \mathbb{E}\big[Y\,\mathbf{1}_{\{Y \geq a\}}\big] \;\geq\; a\,\mathbb{P}(Y \geq a).$$

Dividing by $a > 0$ gives the claim. □

Theorem 2.8 (Chebyshev's inequality)**.** If $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2 < \infty$, then for any $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \;\leq\; \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Squaring is strictly increasing on $[0, \infty)$, so the event $\{|X - \mu| \geq \varepsilon\}$ coincides with $\{(X - \mu)^2 \geq \varepsilon^2\}$:

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \;=\; \mathbb{P}\big((X - \mu)^2 \geq \varepsilon^2\big).$$

The variable $Y = (X - \mu)^2$ is nonnegative with $\mathbb{E}[Y] = \mathrm{Var}(X) = \sigma^2$. Applying Markov's inequality (Theorem 2.7) to Y with threshold $a = \varepsilon^2$,

$$\mathbb{P}\big((X - \mu)^2 \geq \varepsilon^2\big) \;\leq\; \frac{\mathbb{E}[(X - \mu)^2]}{\varepsilon^2} \;=\; \frac{\sigma^2}{\varepsilon^2}.$$

Combining the two displays gives the claim. □

These suffice to prove the weak law.

Theorem 2.9 (Chebyshev's weak law of large numbers)**.** Let $X_1, X_2, \ldots$ be iid with $\mathbb{E}[X_1] = \mu$ and $\mathrm{Var}(X_1) = \sigma^2 < \infty$. Let $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$. Then $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof. Compute the mean and variance of $\bar{X}_n$. By linearity of expectation,

$$\mathbb{E}[\bar{X}_n] \;=\; \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[X_i] \;=\; \mu.$$

By independence of the X_i,

$$\mathrm{Var}(\bar{X}_n) \;=\; \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}(X_i) \;=\; \frac{\sigma^2}{n}.$$

Apply Chebyshev's inequality (Theorem 2.8) to $\bar{X}_n$ with threshold ε:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \;\leq\; \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} \;=\; \frac{\sigma^2}{n\varepsilon^2} \;\longrightarrow\; 0 \qquad (n \to \infty).$$

This is the definition of $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$. □

The proof shows *why* averaging reduces variance. Each individual observation has variance σ^2, but the average has variance σ^2/n; as n grows the average becomes more tightly concentrated around μ. Averaging shrinks variance, period.

The weak law and the strong law agree on the punchline ($\bar{X}_n \to \mu$) but differ on the mode of convergence. The weak law guarantees that the probability of being far from the mean tends to zero; the strong law says something more pathwise, namely that with probability one the realized sequence $(\bar{X}_1(\omega), \bar{X}_2(\omega), \ldots)$ itself settles down to μ. The strong law also requires less than finite variance.

Theorem 2.10 (Kolmogorov's strong law of large numbers)**.** If $X_1, X_2, \ldots$ are iid with $\mathbb{E}[|X_1|] < \infty$ and $\mathbb{E}[X_1] = \mu$, then $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$.

A proof is given in Billingsley (1995, Thm. 22.1). Figure 4 visualizes this stabilization for two distributions with very different shapes but the same mean and variance.

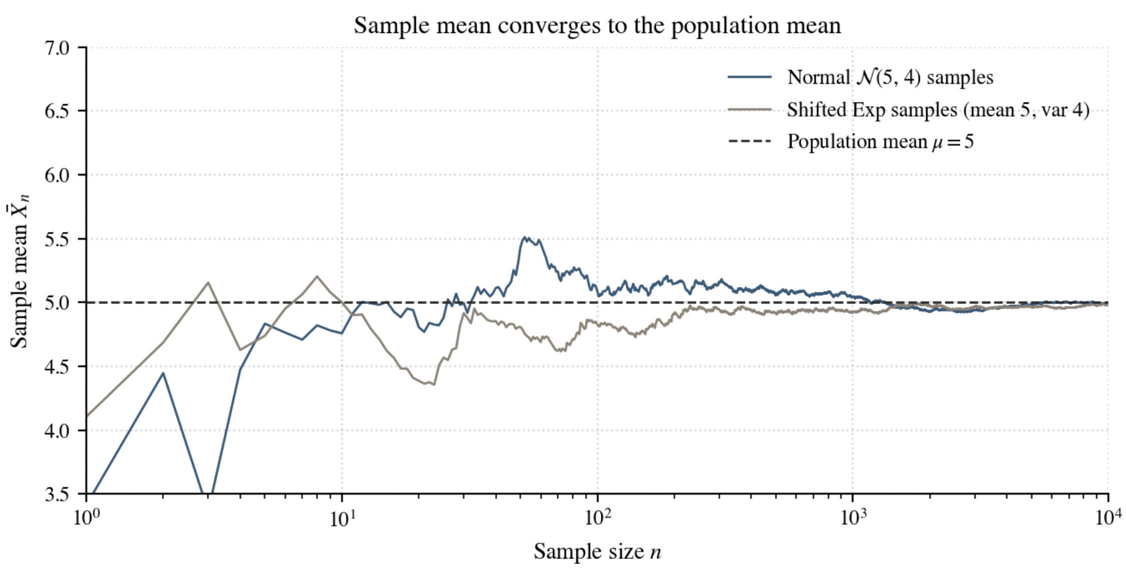

Figure 4: The sample mean $\bar{X}_n$ converges to the population mean $\mu = 5$ for two distributions with very different shapes but the same mean and variance: a normal $\mathcal{N}(5, 4)$ and an exponential rescaled and shifted to also have mean 5 and variance 4. The exponential samples fluctuate more in the early stages because their distribution is heavily skewed, but both averages settle near 5 by $n = 10000$. The choice of normal vs. shifted-exponential comparison with matched mean and variance follows Figure 1 of Genzer (2025); the figure here is generated by `notebooks/01_lln_clt_demos.py`.

The central limit theorem refines the law of large numbers by saying *at what rate* the convergence happens and *what the fluctuations look like*. The scale $\sqrt{n}$ is not arbitrary: since

$\mathrm{Var}(\bar{X}_n) = \sigma^2/n$ by the WLLN computation, the typical size of the deviation $\bar{X}_n - \mu$ is $n^{-1/2}$. Multiplying by $\sqrt{n}$ cancels this decay and keeps the random fluctuation visible at scale order one.

Theorem 2.11 (Central limit theorem)**.** Let $X_1, X_2, \dots$ be iid with $\mathbb{E}[X_1] = \mu$ and $0 < \mathrm{Var}(X_1) = \sigma^2 < \infty$. Then

$$Z_n \;:=\; \frac{\sqrt{n}\,(\bar{X}_n - \mu)}{\sigma} \;\xrightarrow{d}\; \mathcal{N}(0,1).$$

Proof. The proof has four steps: standardize, expand the characteristic function near zero, factor across the iid sum, and pass to the limit through Lévy's continuity theorem.

Step 1 (standardize). Set $Y_i = (X_i - \mu)/\sigma$. Then $\mathbb{E}[Y_i] = 0$, $\mathrm{Var}(Y_i) = 1$, and

$$Z_n \;=\; \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i.$$

Step 2 (expand the CF near zero). Let $\varphi(t) = \mathbb{E}[e^{itY_1}]$. The Taylor expansion $e^{iu} = 1 + iu - u^2/2 + o(u^2)$ as $u \to 0$, applied with $u = tY_1$, gives

$$e^{itY_1} \;=\; 1 \;+\; itY_1 \;-\; \tfrac{1}{2}t^2Y_1^2 \;+\; o(t^2Y_1^2).$$

The finite-variance assumption $\mathrm{Var}(X_1) = \sigma^2 < \infty$ is what makes this second-order expansion legitimate after taking expectations: it ensures that the remainder integrates correctly under $\mathbb{E}$. Using $\mathbb{E}[Y_1] = 0$ and $\mathbb{E}[Y_1^2] = 1$,

$$\varphi(t) \;=\; 1 \;-\; \tfrac{1}{2}t^2 \;+\; o(t^2) \qquad (t \to 0).$$

This expansion is the analytic engine of the CLT: near zero, the characteristic function of *every* standardized finite-variance random variable agrees with the standard normal CF up to second order. The CLT then says that the second-order match propagates through the iid sum to give a matching limit, which is the deep reason the Gaussian distribution appears universally.

Step 3 (factor by independence). The Y_i are independent, so the characteristic function of Z_n factors into a product of n copies of $\varphi(t/\sqrt{n})$:

$$\varphi_{Z_n}(t) \;=\; \mathbb{E}\!\left[\exp\!\Big(\frac{it}{\sqrt{n}}\sum_{i=1}^{n} Y_i\Big)\right] \;=\; \prod_{i=1}^{n} \mathbb{E}\big[e^{i(t/\sqrt{n})Y_i}\big] \;=\; \Big(\varphi\big(t/\sqrt{n}\big)\Big)^{n}.$$

Substituting the expansion from Step 2 with $t/\sqrt{n}$ in place of t,

$$\varphi_{Z_n}(t) \;=\; \Big(1 \;-\; \tfrac{t^2}{2n} \;+\; o(1/n)\Big)^{n}.$$

Step 4 (take the limit and conclude). The standard calculus identity $(1 + a_n/n)^n \to e^a$ when $a_n \to a$, applied with $a_n \to -t^2/2$, gives

$$\varphi_{Z_n}(t) \;\longrightarrow\; e^{-t^2/2} \qquad (n \to \infty).$$

By Example 2.8, $e^{-t^2/2}$ is the characteristic function of $\mathcal{N}(0,1)$. The Lévy continuity theorem (Theorem 2.4) then yields $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. □

Figure 5 illustrates the CLT for exponential samples: even though the underlying distribution is sharply skewed, the histogram of the standardized sample mean Z_n approaches the standard normal density as n grows.

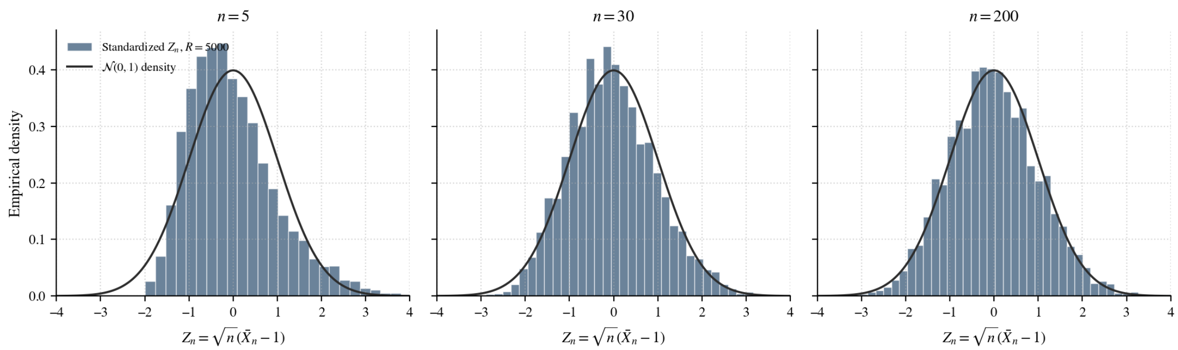

Figure 5: Central limit theorem for iid Exponential(1) samples. Each panel shows a histogram of $R = 5000$ standardized sample means $Z_n = \sqrt{n}(\bar{X}_n - 1)$ at increasing n. The histogram is sharply skewed at $n = 5$, visibly less so at $n = 30$, and indistinguishable from the standard normal density (solid line) by $n = 200$. Generated by `notebooks/01_lln_clt_demos.py`.

2.4 Random Vectors and the Multivariate Normal Distribution

Statistical observations rarely come as single numbers; they typically come as *vectors*,

$$\boldsymbol{X}_i \;=\; (X_{i1}, X_{i2}, \ldots, X_{ip})^\top,$$

where X_{ij} is, for instance, the j^{th} feature of the i^{th} observation. The vector analogues of Sections 2.1–2.3 are the joint distribution of $\boldsymbol{X}$, the mean vector and covariance matrix that summarize its first two moments, the multivariate normal as the central model, and the multivariate central limit theorem.

Random vectors and joint distributions.

Definition 2.12 (Random vector and joint CDF)**.** A *random vector* is a function $\boldsymbol{X} = (X_1, \ldots, X_p)^\top : \Omega \to \mathbb{R}^p$ whose components are random variables on a common probability space. The *joint CDF* of $\boldsymbol{X}$ is

$$F_{\boldsymbol{X}}(\boldsymbol{x}) \;=\; \mathbb{P}(X_1 \le x_1,\, \ldots,\, X_p \le x_p), \qquad \boldsymbol{x} \in \mathbb{R}^p.$$

The joint CDF is a multivariate threshold probability: it asks for the probability that *every* coordinate of $\boldsymbol{X}$ falls below the corresponding coordinate of $\boldsymbol{x}$. Unlike the individual marginal CDFs F_{X_i}, the joint CDF carries dependence information: it records not only how each coordinate behaves on its own, but how the coordinates behave together. The joint CDF obeys the multivariate analogues of the properties (F1)–(F3) of Proposition 2.1.

Proposition 2.12 (Properties of the joint CDF)**.** The joint CDF $F_{\boldsymbol{X}} : \mathbb{R}^p \to [0, 1]$ satisfies:

(F1) As $x_i \to -\infty$ for any single coordinate i, $F_{\boldsymbol{X}}(\boldsymbol{x}) \to 0$; as all coordinates tend to $+\infty$, $F_{\boldsymbol{X}}(\boldsymbol{x}) \to 1$.

(F2) $F_{\boldsymbol{X}}$ is right-continuous in each coordinate.

(F3) $F_{\boldsymbol{X}}$ is non-decreasing in each coordinate.

(F4) Marginal CDFs are recovered by limits: $F_{X_1}(x_1) = \lim_{x_2,\dots,x_p \to +\infty} F_{\boldsymbol{X}}(x_1, \dots, x_p)$, and similarly for other coordinates.

Moreover, $X_1, \dots, X_p$ are independent if and only if $F_{\boldsymbol{X}}(\boldsymbol{x}) = \prod_{i=1}^{p} F_{X_i}(x_i)$ for every $\boldsymbol{x} \in \mathbb{R}^p$.

The discrete and absolutely continuous cases admit joint PMFs and joint densities, defined exactly as in the scalar case.

Three examples of joint distributions. The next three examples illustrate the spectrum from full independence to maximal dependence to a smooth Gaussian dependence. They also show the difference between the *support* of a random vector (where it lives) and its *distribution* (how its mass is spread on its support).

Example 2.13 (Independent exponentials)**.** Let X_1, X_2 be iid with $X_i \sim \text{Exponential}(\lambda)$. Each scalar CDF is $F_X(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ (and 0 for $x < 0$). By independence,

$$F_{X_1,X_2}(x_1, x_2) \;=\; F_X(x_1)\, F_X(x_2) \;=\; (1 - e^{-\lambda x_1})(1 - e^{-\lambda x_2}), \qquad x_1, x_2 \geq 0,$$

and $F_{X_1,X_2}(x_1, x_2) = 0$ if either coordinate is negative. The mass lives on the positive quadrant $\{x_1 \geq 0,\ x_2 \geq 0\}$, and the joint density factors as $f_{X_1,X_2} = f_{X_1} f_{X_2}$.

Example 2.14 (Maximally dependent: $X_1 = X_2$ a.s.)**.** Let $X_1 \sim \text{Exponential}(\lambda)$ and set $X_2 = X_1$ almost surely. Then $\mathbb{P}(X_1 = X_2) = 1$, so the random point (X_1, X_2) lies on the diagonal $\{x_1 = x_2 \geq 0\}$ with probability one. The joint CDF is

$$\begin{aligned} F_{X_1,X_2}(x_1, x_2) \;&=\; \mathbb{P}(X_1 \leq x_1,\ X_2 \leq x_2) \\ &=\; \mathbb{P}(X_1 \leq \min\{x_1, x_2\}) \;=\; F_X(\min\{x_1, x_2\}). \end{aligned}$$

The two components have the same one-dimensional marginals as in Example 2.13, but the support of the pair collapses to a one-dimensional diagonal ray. This vector does *not* admit a joint density on $\mathbb{R}^2$, because all its mass is concentrated on a set of Lebesgue area zero. We will see shortly that it is the prototypical case of a covariance matrix that is positive semidefinite but *not* positive definite.

Example 2.15 (Bivariate normal density)**.** Let $\boldsymbol{X} = (X_1, X_2)^\top \sim \mathcal{N}(\mathbf{0}, \Sigma)$ with

$$\Sigma \;=\; \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \qquad \sigma_1, \sigma_2 > 0, \quad -1 < \rho < 1.$$

A short computation gives $\det \Sigma = \sigma_1^2 \sigma_2^2 (1 - \rho^2)$ and

$$\Sigma^{-1} \;=\; \frac{1}{\sigma_1^2 \sigma_2^2 (1 - \rho^2)} \begin{pmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{pmatrix}.$$

Substituting into the multivariate normal density (Definition 2.18 below) and simplifying yields the familiar scalar formula

$$f_{X_1,X_2}(x_1,x_2) \;=\; \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\,\exp\!\left(-\frac{1}{2(1-\rho^2)}\left[\frac{x_1^2}{\sigma_1^2} \;-\; \frac{2\rho x_1 x_2}{\sigma_1\sigma_2} \;+\; \frac{x_2^2}{\sigma_2^2}\right]\right).$$

When $\rho = 0$ the cross term vanishes and the density factors as $f_{X_1} f_{X_2}$, so $X_1 \perp X_2$. When $\rho \neq 0$ the cross term cannot be removed and the components are dependent.

Figure 6 contrasts the three cases at a glance.

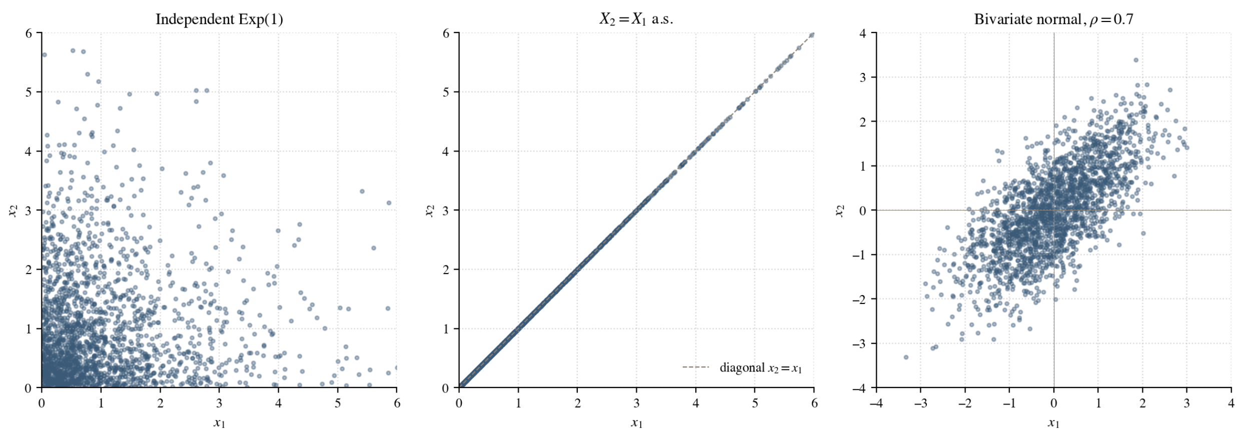

Figure 6: Three joint distributions on $\mathbb{R}^2$ with the same one-dimensional marginals. *Left:* two independent exponentials; the cloud fills the positive quadrant. *Middle:* $X_1 = X_2$ almost surely; the cloud collapses onto the diagonal line. *Right:* a bivariate normal with correlation $\rho = 0.7$; the cloud is an oblique elliptical scatter. Each panel shows 2,000 Monte Carlo samples. The middle case has no joint density on $\mathbb{R}^2$; it will be the prototype for a covariance matrix that is PSD but not PD.

Mean vector and covariance matrix.

Definition 2.16 (Mean vector and covariance matrix)**.** The *mean vector* of $\boldsymbol{X}$ is $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{X}] = (\mathbb{E}[X_1], \ldots, \mathbb{E}[X_p])^\top$. The *covariance matrix* is

$$\Sigma \;=\; \mathrm{Cov}(\boldsymbol{X}) \;=\; \mathbb{E}\big[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^\top\big],$$

a $p \times p$ matrix with entries

$$\Sigma_{ij} \;=\; \mathrm{Cov}(X_i, X_j) \;=\; \mathbb{E}[X_i X_j] \;-\; \mathbb{E}[X_i]\,\mathbb{E}[X_j].$$

Written out entrywise, the covariance matrix is

$$\Sigma \;=\; \begin{pmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1,X_2) & \cdots & \mathrm{Cov}(X_1,X_p) \\ \mathrm{Cov}(X_2,X_1) & \mathrm{Var}(X_2) & \cdots & \mathrm{Cov}(X_2,X_p) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_p,X_1) & \mathrm{Cov}(X_p,X_2) & \cdots & \mathrm{Var}(X_p) \end{pmatrix}.$$

The diagonal entries record the variability of the individual coordinates, and the off-diagonal entries record the pairwise linear association between coordinates. The matrix is automatically symmetric because $\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$.

Example 2.17 (A 2×2 covariance matrix)**.** Suppose $\boldsymbol{X} = (X_1, X_2)^\top$ has covariance matrix

$$\Sigma \;=\; \begin{pmatrix} 4 & 1.5 \\ 1.5 & 1 \end{pmatrix}.$$

The diagonal reads off $\mathrm{Var}(X_1) = 4$ and $\mathrm{Var}(X_2) = 1$; the off-diagonal reads off $\mathrm{Cov}(X_1, X_2) = 1.5$, so large values of X_1 tend to occur together with large values of X_2. For the direction $\boldsymbol{v} = (1, -1)^\top$, a direct computation gives $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = 4 - 2(1.5) + 1 = 2$, which equals $\mathrm{Var}(X_1 - X_2)$. The quadratic form on Σ at $\boldsymbol{v}$ is the variance of the corresponding linear combination of coordinates, an identity we now state and prove in general.

Covariance matrices are not arbitrary symmetric matrices. They have a built-in positivity property because every linear combination of the coordinates has a nonnegative variance, and the quadratic form on Σ literally computes that variance. This is the central property of Σ that will be exploited again and again throughout the report.

Proposition 2.13 (Σ is symmetric and PSD)**.** For any random vector with finite second moments:

(1) Σ is symmetric, $\Sigma_{ij} = \Sigma_{ji}$.

(2) Σ is positive semidefinite: $\boldsymbol{v}^\top \Sigma \boldsymbol{v} \geq 0$ for every $\boldsymbol{v} \in \mathbb{R}^p$. The identity $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = \mathrm{Var}(\boldsymbol{v}^\top \boldsymbol{X})$ shows that every quadratic form on Σ is a variance.

(3) Σ is positive definite if and only if no nontrivial linear combination $\boldsymbol{v}^\top \boldsymbol{X}$ ($\boldsymbol{v} \neq \mathbf{0}$) is almost surely constant.

Proof. Symmetry: $\Sigma_{ij} = \mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i) = \Sigma_{ji}$.

For (2), fix any $\boldsymbol{v} \in \mathbb{R}^p$ and define the scalar $Y = \boldsymbol{v}^\top \boldsymbol{X}$. Then

$$Y \;-\; \mathbb{E}[Y] \;=\; \boldsymbol{v}^\top \boldsymbol{X} \;-\; \boldsymbol{v}^\top \boldsymbol{\mu} \;=\; \boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu}),$$

and computing the variance gives

$$\mathrm{Var}(Y) \;=\; \mathbb{E}\big[(Y - \mathbb{E}[Y])^2\big] \;=\; \mathbb{E}\big[(\boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu}))^2\big].$$

Since $\boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu})$ is a scalar, its square equals $(\boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu}))(\boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu}))^\top = \boldsymbol{v}^\top (\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top \boldsymbol{v}$. Pulling the constants $\boldsymbol{v}^\top$ and $\boldsymbol{v}$ outside the expectation,

$$\mathrm{Var}(Y) \;=\; \boldsymbol{v}^\top \, \mathbb{E}\big[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top\big] \, \boldsymbol{v} \;=\; \boldsymbol{v}^\top \Sigma \boldsymbol{v}.$$

Since $\mathrm{Var}(Y) \geq 0$ for every random variable Y, we conclude $\boldsymbol{v}^\top \Sigma \boldsymbol{v} \geq 0$ for every $\boldsymbol{v}$, which is PSD.

For (3), Σ fails to be positive definite precisely when $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = 0$ for some $\boldsymbol{v} \neq \mathbf{0}$. By the identity above, this is $\mathrm{Var}(\boldsymbol{v}^\top \boldsymbol{X}) = 0$, which is the statement that $\boldsymbol{v}^\top \boldsymbol{X}$ is almost surely a constant. $\square$

Example 2.14 is the prototypical failure of positive definiteness: with $X_2 = X_1$ a.s., the vector $\boldsymbol{v} = (1, -1)^\top$ gives $\boldsymbol{v}^\top \boldsymbol{X} = X_1 - X_2 = 0$ a.s., so $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = 0$. The covariance matrix is PSD but not PD.

The identity

$$\boldsymbol{v}^\top \Sigma \boldsymbol{v} \;=\; \mathrm{Var}(\boldsymbol{v}^\top \boldsymbol{X})$$

is the key identity of this section. The left side is linear algebra; the right side is probability. Read either way, it converts a question about directions in $\mathbb{R}^p$ into a question about variances of one-dimensional projections, and vice versa.

The multivariate normal distribution. The multivariate normal distribution is the vector analogue of the ordinary normal. It is the natural model for vector observations because it is closed under linear operations and is characterized completely by its mean vector and covariance matrix. The cleanest definition uses the characteristic function.

Definition 2.18 (Multivariate normal)**.** A random vector $\boldsymbol{X} \in \mathbb{R}^p$ is *multivariate normal* with mean $\boldsymbol{\mu} \in \mathbb{R}^p$ and covariance matrix $\Sigma \in \mathbb{R}^{p \times p}$, written $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, if its characteristic function is

$$\varphi_{\boldsymbol{X}}(\boldsymbol{t}) \;=\; \exp\!\Big(i\, \boldsymbol{t}^\top \boldsymbol{\mu} \;-\; \tfrac{1}{2}\, \boldsymbol{t}^\top \Sigma \boldsymbol{t}\Big), \qquad \boldsymbol{t} \in \mathbb{R}^p. \tag{4}$$

When Σ is positive definite, $\boldsymbol{X}$ admits the density

$$f_{\boldsymbol{X}}(\boldsymbol{x}) \;=\; \frac{1}{(2\pi)^{p/2}\, |\Sigma|^{1/2}}\, \exp\!\Big(-\tfrac{1}{2}\, (\boldsymbol{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu})\Big).$$

A note on notation: in the scalar case (Example 2.2) the second argument of $\mathcal{N}(\mu, \sigma^2)$ is a variance, while in the multivariate case the second argument of $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ is a covariance *matrix*. The two-argument macro $\mathcal{N}(\,\cdot\,,\,\cdot\,)$ is the same; only the type of the second slot changes.

Equation (4) is taken as the definition because it remains valid when Σ is singular (a case that arises exactly as in Example 2.14), whereas the density form requires Σ to be invertible. The CF definition also has a clean intuitive content: the exponent $i\boldsymbol{t}^\top \boldsymbol{\mu} - \frac{1}{2} \boldsymbol{t}^\top \Sigma \boldsymbol{t}$ is the CF exponent of a univariate $\mathcal{N}(\boldsymbol{t}^\top \boldsymbol{\mu},\, \boldsymbol{t}^\top \Sigma \boldsymbol{t})$, since the scalar projection $\boldsymbol{t}^\top \boldsymbol{X}$ has mean $\boldsymbol{t}^\top \boldsymbol{\mu}$ and variance $\boldsymbol{t}^\top \Sigma \boldsymbol{t}$ (by the bridge identity above). Equation (4) therefore says

$$\boldsymbol{t}^\top \boldsymbol{X} \;\sim\; \mathcal{N}(\boldsymbol{t}^\top \boldsymbol{\mu},\, \boldsymbol{t}^\top \Sigma \boldsymbol{t}) \qquad \text{for every } \boldsymbol{t} \in \mathbb{R}^p. \tag{5}$$

That is, *every* one-dimensional projection of a multivariate normal is univariate normal. This is the most useful working characterization of $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$.

The multivariate normal is closed under affine transformations.

Proposition 2.14 (Affine closure of $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$)**.** If $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, $\boldsymbol{A} \in \mathbb{R}^{q \times p}$, and $\boldsymbol{b} \in \mathbb{R}^q$, then

$$\boldsymbol{A}\boldsymbol{X} + \boldsymbol{b} \;\sim\; \mathcal{N}(\boldsymbol{A}\boldsymbol{\mu} + \boldsymbol{b},\; \boldsymbol{A}\Sigma\boldsymbol{A}^\top).$$

Proof. We compute the characteristic function of $\boldsymbol{AX} + \boldsymbol{b}$. For $\boldsymbol{s} \in \mathbb{R}^q$,

$$\begin{aligned}\varphi_{\boldsymbol{AX}+\boldsymbol{b}}(\boldsymbol{s}) &= \mathbb{E}\big[e^{i\boldsymbol{s}^\top(\boldsymbol{AX}+\boldsymbol{b})}\big] = e^{i\boldsymbol{s}^\top\boldsymbol{b}}\,\mathbb{E}\big[e^{i(\boldsymbol{A}^\top\boldsymbol{s})^\top\boldsymbol{X}}\big] \\ &= e^{i\boldsymbol{s}^\top\boldsymbol{b}}\,\varphi_{\boldsymbol{X}}(\boldsymbol{A}^\top\boldsymbol{s}) \\ &= \exp\Big(i\,\boldsymbol{s}^\top(\boldsymbol{A\mu}+\boldsymbol{b}) - \tfrac{1}{2}\,\boldsymbol{s}^\top\boldsymbol{A}\Sigma\boldsymbol{A}^\top\boldsymbol{s}\Big).\end{aligned}$$

This is the CF of $\mathcal{N}(\boldsymbol{A\mu}+\boldsymbol{b},\, \boldsymbol{A}\Sigma\boldsymbol{A}^\top)$, so the distribution of $\boldsymbol{AX}+\boldsymbol{b}$ is identified by the uniqueness of characteristic functions. □

A second property is special to the Gaussian family. For a general distribution, zero covariance only means no linear association: $\mathrm{Cov}(X_1, X_2) = 0$ does not imply that X_1 and X_2 are independent. For a multivariate normal vector with diagonal covariance, however, the coordinates *are* independent, because the density factors into a product of marginal densities (and the characteristic function factors for the same reason in the singular case). Figure 7 shows the corresponding bivariate density contours at three correlation values.

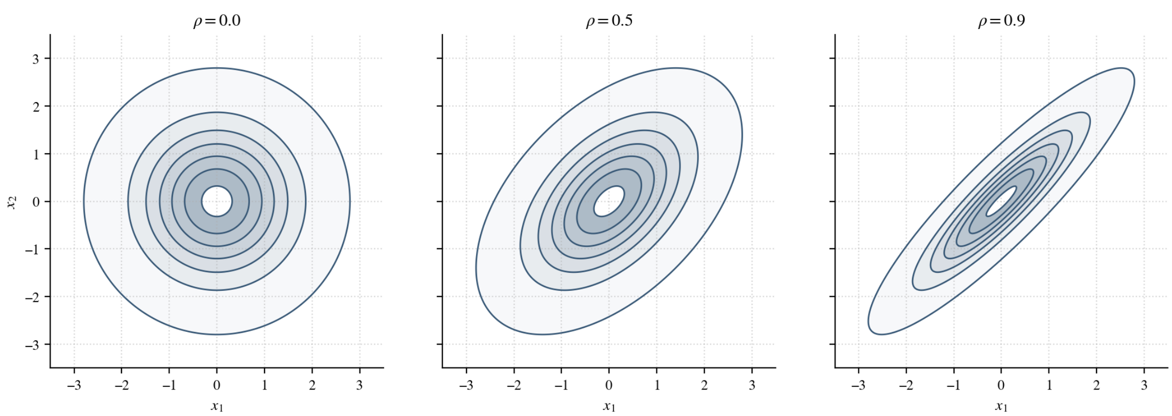

Figure 7: Density contours of the centered bivariate normal with unit variances and correlation $\rho \in \{0,\, 0.5,\, 0.9\}$. At $\rho = 0$ the contours are circles and the two coordinates are independent; as ρ increases the contours elongate along the line $x_2 = x_1$, reflecting positive linear association. The direction of elongation is the direction in which the random vector has largest variance, which the next subsection will identify as the leading eigenvector of Σ. Generated by `notebooks/01_lln_clt_demos.py`.

The multivariate central limit theorem. The univariate CLT (Theorem 2.11) extends from scalar averages to vector averages.

Theorem 2.15 (Multivariate central limit theorem)**.** *Let $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots$ be iid random vectors in $\mathbb{R}^p$ with mean $\boldsymbol{\mu}$ and covariance matrix Σ (with finite second moments). Let $\bar{\boldsymbol{X}}_n = n^{-1}\sum_{i=1}^n \boldsymbol{X}_i$. Then*

$$\sqrt{n}\,(\bar{\boldsymbol{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}(\boldsymbol{0},\, \Sigma).$$

Proof sketch. The argument is the *Cramér-Wold device*: a sequence of random vectors converges in distribution if and only if every fixed linear projection of the sequence converges in distribution to the corresponding projection of the limit. Fix $\boldsymbol{t} \in \mathbb{R}^p$ and project:

$$\boldsymbol{t}^\top \sqrt{n}\,(\bar{\boldsymbol{X}}_n - \boldsymbol{\mu}) \;=\; \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{t}^\top \boldsymbol{X}_i \;-\; \boldsymbol{t}^\top \boldsymbol{\mu}\right).$$

The scalar random variables $\boldsymbol{t}^\top \boldsymbol{X}_i$ are iid with mean $\boldsymbol{t}^\top \boldsymbol{\mu}$ and variance $\boldsymbol{t}^\top \Sigma \boldsymbol{t}$ (by the bridge identity $\boldsymbol{t}^\top \Sigma \boldsymbol{t} = \mathrm{Var}(\boldsymbol{t}^\top \boldsymbol{X}_1)$). The univariate CLT (Theorem 2.11) gives

$$\boldsymbol{t}^\top \sqrt{n}\,(\bar{\boldsymbol{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}(0,\; \boldsymbol{t}^\top \Sigma \boldsymbol{t}).$$

This is exactly the projection of $\mathcal{N}(\boldsymbol{0}, \Sigma)$ onto $\boldsymbol{t}$, by (5). Since this holds for every $\boldsymbol{t}$, the Cramér-Wold device delivers the vector convergence. □

Vector averages therefore fluctuate on the same $\sqrt{n}$ scale as scalar averages, but their limiting covariance records the dependence among coordinates.

2.5 Linear Algebra of Symmetric Matrices

The identity $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = \mathrm{Var}(\boldsymbol{v}^\top \boldsymbol{X})$ of the previous subsection ties covariance matrices to directional variance. To analyze that variance *geometrically*, we use the structure of symmetric matrices: the spectral theorem, the eigenvalue characterization of positive semidefiniteness, and the Rayleigh-quotient identification of maximal-variance directions. This subsection also records two technical results, the equivalence of norms in finite dimensions and the continuity of eigenvalues as functions of matrix entries, that will let us push matrix-level convergence to eigenvalue-level convergence.

The spectral theorem. A real $n \times n$ matrix A is *symmetric* if $A = A^\top$. Throughout this paragraph, A denotes a real symmetric matrix. Symmetry has two crucial consequences: all eigenvalues of A are real, and there exists an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A.

Theorem 2.16 (Spectral theorem for real symmetric matrices)**.** If $A \in \mathbb{R}^{n\times n}$ is symmetric, there exist an orthogonal matrix Q (meaning $Q^\top Q = I$) and a diagonal matrix $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ such that

$$A \;=\; Q\,\Lambda\,Q^\top. \tag{6}$$

The diagonal entries of Λ are the eigenvalues of A, and the columns $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ of Q form an orthonormal basis of eigenvectors: $A\boldsymbol{q}_i = \lambda_i \boldsymbol{q}_i$ and $\boldsymbol{q}_i^\top \boldsymbol{q}_j = \delta_{ij}$.

A complete proof requires careful linear algebra; see Strang (2016, §6.4). We will use the conclusion freely. The decomposition $A = Q\Lambda Q^\top$ has a clean geometric meaning: the action of A on a vector $\boldsymbol{x} \in \mathbb{R}^n$ is (i) rotate $\boldsymbol{x}$ into the eigenvector basis via $Q^\top$, (ii) scale the i^{th} coordinate by λ_i via Λ, (iii) rotate back via Q. Symmetric matrices stretch space along perpendicular directions, with stretch factors given by the eigenvalues.

Quadratic forms diagonalize.

Corollary 2.17 (Diagonalization of quadratic forms)**.** For any $\boldsymbol{x} \in \mathbb{R}^n$, setting $\boldsymbol{y} = Q^\top \boldsymbol{x}$ gives

$$\boldsymbol{x}^\top A\, \boldsymbol{x} \;=\; \sum_{i=1}^{n} \lambda_i\, y_i^2.$$

Proof. Using $A = Q\Lambda Q^\top$,

$$\boldsymbol{x}^\top A\, \boldsymbol{x} \;=\; \boldsymbol{x}^\top Q\Lambda Q^\top \boldsymbol{x} \;=\; (Q^\top \boldsymbol{x})^\top\, \Lambda\, (Q^\top \boldsymbol{x}) \;=\; \boldsymbol{y}^\top \Lambda\, \boldsymbol{y}.$$

Since $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, the last expression is $\sum_{i=1}^n \lambda_i y_i^2$. ☐

Because Q is orthogonal, the map $\boldsymbol{x} \mapsto \boldsymbol{y} = Q^\top \boldsymbol{x}$ is a bijection of $\mathbb{R}^n$ that sends $\mathbf{0}$ to $\mathbf{0}$, so $\boldsymbol{x} \neq \mathbf{0}$ if and only if $\boldsymbol{y} \neq \mathbf{0}$. The sign of $\boldsymbol{x}^\top A \boldsymbol{x}$ is therefore controlled entirely by the signs of the eigenvalues.

Eigenvalues characterize positive definiteness.

Theorem 2.18 (Eigenvalue characterization of PD/PSD)**.** Let A be a real symmetric matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$.

(1) A is positive semidefinite if and only if $\lambda_i \geq 0$ for all i.

(2) A is positive definite if and only if $\lambda_i > 0$ for all i.

Proof. We prove (1); (2) is identical with strict inequalities throughout.

PSD $\Rightarrow \lambda_i \geq 0$. Suppose A is PSD, and let $\boldsymbol{q}_k$ be a unit eigenvector with eigenvalue λ_k. Computing the quadratic form at $\boldsymbol{q}_k$,

$$\boldsymbol{q}_k^\top A\, \boldsymbol{q}_k \;=\; \boldsymbol{q}_k^\top (\lambda_k \boldsymbol{q}_k) \;=\; \lambda_k\, \boldsymbol{q}_k^\top \boldsymbol{q}_k \;=\; \lambda_k.$$

By PSD the left side is ≥ 0, so $\lambda_k \geq 0$.

$\lambda_i \geq 0 \Rightarrow$ *PSD.* Suppose every eigenvalue of A is nonnegative. For any $\boldsymbol{x} \in \mathbb{R}^n$, Corollary 2.17 gives

$$\boldsymbol{x}^\top A\, \boldsymbol{x} \;=\; \sum_{i=1}^{n} \lambda_i\, y_i^2 \;\geq\; 0,$$

where $\boldsymbol{y} = Q^\top \boldsymbol{x}$ and each $\lambda_i y_i^2 \geq 0$ by hypothesis. Hence A is PSD. ☐

Combining Theorem 2.18 with Proposition 2.13, every eigenvalue of a covariance matrix is nonnegative, and the eigenvalues are strictly positive if and only if no nontrivial linear combination of the coordinates is almost surely constant.

The diagonalization in Corollary 2.17 also identifies which directions maximize and minimize the quadratic form. The result, the Rayleigh-quotient characterization, is what justifies saying that eigenvectors are the directions of greatest variance.

Proposition 2.19 (Rayleigh quotient)**.** Let A be a real symmetric $p \times p$ matrix with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$ and corresponding orthonormal eigenvectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_p$. Then

$$\lambda_1 \;=\; \max_{\substack{\boldsymbol{v} \in \mathbb{R}^p \\ \|\boldsymbol{v}\|_2 = 1}} \boldsymbol{v}^\top A \boldsymbol{v}, \qquad \lambda_p \;=\; \min_{\substack{\boldsymbol{v} \in \mathbb{R}^p \\ \|\boldsymbol{v}\|_2 = 1}} \boldsymbol{v}^\top A \boldsymbol{v},$$

with the maxima attained at $\boldsymbol{v} = \boldsymbol{q}_1$ and $\boldsymbol{v} = \boldsymbol{q}_p$ respectively.

Proof. For any unit vector $\boldsymbol{v}$, set $\boldsymbol{y} = Q^\top \boldsymbol{v}$. The orthogonal transformation preserves length, so $\|\boldsymbol{y}\|_2 = 1$. Corollary 2.17 gives

$$\boldsymbol{v}^\top A \boldsymbol{v} \;=\; \sum_{i=1}^{p} \lambda_i \, y_i^2,$$

a convex combination of the eigenvalues (λ_i) with weights (y_i^2) summing to one. The maximum value of this combination is λ_1, attained when all the weight is on the largest eigenvalue, i.e., $\boldsymbol{y} = \boldsymbol{e}_1$ and hence $\boldsymbol{v} = Q\boldsymbol{e}_1 = \boldsymbol{q}_1$. The minimum is λ_p, attained at $\boldsymbol{v} = \boldsymbol{q}_p$. □

Applied to a covariance matrix Σ, Proposition 2.19 says that the largest eigenvalue $\lambda_1(\Sigma)$ is the maximum possible variance of a unit-norm linear combination of the coordinates, attained when that combination points along the top eigenvector. The eigenvectors of Σ are therefore the directions of greatest, second-greatest, and so on, variance of $\boldsymbol{X}$, and the eigenvalues record the variance along each direction. This is the geometric meaning of the spectral decomposition of a covariance matrix.

Worked examples in the 2×2 case. For a symmetric matrix $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, the characteristic polynomial $\det(A - \lambda I) = (a - \lambda)(d - \lambda) - b^2$ factors with eigenvalues

$$\lambda_\pm \;=\; \frac{a+d}{2} \;\pm\; \sqrt{\left(\frac{a-d}{2}\right)^2 + b^2}. \tag{7}$$

A convenient 2×2 test (a small case of Sylvester's criterion):

$$A \text{ is PD} \quad\Longleftrightarrow\quad a > 0 \;\text{ and }\; \det A = ad - b^2 > 0.$$

Example 2.19 (A positive-definite and an indefinite 2×2)**.** *Positive-definite case.* Let $A_1 = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$. Equation (7) gives $\lambda_\pm = 1 \pm \frac{1}{2}$, so

$$\lambda_+ \;=\; \tfrac{3}{2}, \qquad \lambda_- \;=\; \tfrac{1}{2},$$

both strictly positive. Sylvester's criterion agrees: $a = 1 > 0$ and $\det A_1 = 1 - \frac{1}{4} = \frac{3}{4} > 0$. One can also exhibit the positivity directly by completing the square in the quadratic form:

$$\boldsymbol{x}^\top A_1 \boldsymbol{x} \;=\; x_1^2 - x_1 x_2 + x_2^2 \;=\; \left(x_1 - \tfrac{x_2}{2}\right)^2 + \tfrac{3}{4}\, x_2^2,$$

which is a sum of squares, manifestly > 0 for $\boldsymbol{x} \neq \mathbf{0}$.

Indefinite case. Let $A_2 = \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}$. Equation (7) gives $\lambda_\pm = 1 \pm 2$, so $\lambda_+ = 3$ and $\lambda_- = -1$. The opposite signs make A_2 indefinite. Two directional certificates confirm this: $\boldsymbol{v} = (1,1)^\top$ gives $\boldsymbol{v}^\top A_2 \boldsymbol{v} = -2 < 0$, and $\boldsymbol{v} = (1,-1)^\top$ gives $\boldsymbol{v}^\top A_2 \boldsymbol{v} = 6 > 0$.

Figure 8 visualizes the geometry of the three regimes.

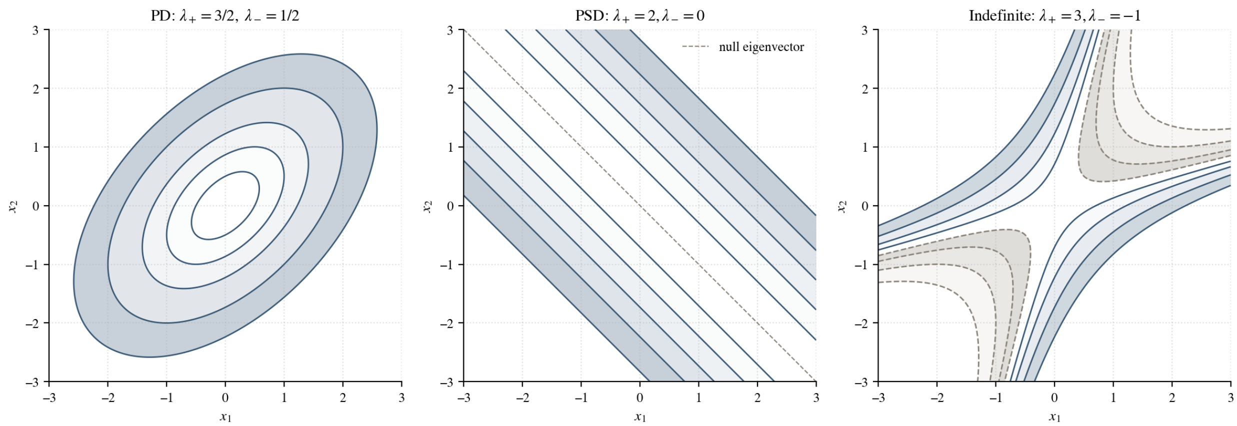

Figure 8: Level sets of $\boldsymbol{x}^\top A\, \boldsymbol{x}$ for three real symmetric 2×2 matrices, illustrating Theorem 2.18. *Left:* a positive-definite A with both eigenvalues positive; level sets are nested ellipses with axes aligned with the eigenvectors. *Middle:* a positive-semidefinite A with one zero eigenvalue; level sets degenerate into parallel strips along the null eigenvector. *Right:* an indefinite A with eigenvalues of opposite sign; level sets are hyperbolas and the quadratic form is positive in some directions and negative in others.

Vector and matrix norms. A *norm* on a real vector space V is a function $\|\cdot\| : V \to [0, \infty)$ satisfying, for all $\boldsymbol{v}, \boldsymbol{w} \in V$ and scalars α:

(N1) $\|\boldsymbol{v}\| = 0$ if and only if $\boldsymbol{v} = \mathbf{0}$ (definiteness);

(N2) $\|\alpha \boldsymbol{v}\| = |\alpha|\, \|\boldsymbol{v}\|$ (homogeneity);

(N3) $\|\boldsymbol{v} + \boldsymbol{w}\| \leq \|\boldsymbol{v}\| + \|\boldsymbol{w}\|$ (triangle inequality).

On $\mathbb{R}^p$ the standard examples are

$$\|\boldsymbol{v}\|_2 = \Big(\sum_{i=1}^p v_i^2\Big)^{1/2}, \qquad \|\boldsymbol{v}\|_1 = \sum_{i=1}^p |v_i|, \qquad \|\boldsymbol{v}\|_\infty = \max_i |v_i|;$$

each satisfies (N1)–(N3) by direct verification (the triangle inequality for $\|\cdot\|_2$ is the Cauchy-Schwarz inequality applied to $\|\boldsymbol{v} + \boldsymbol{w}\|_2^2$).

On $\mathbb{R}^{p \times p}$ we use two matrix norms.

Definition 2.20 (Frobenius and operator norms)**.** For $A \in \mathbb{R}^{p\times p}$,

$$\|A\|_F \;=\; \sqrt{\sum_{i=1}^{p}\sum_{j=1}^{p} a_{ij}^2}, \qquad \|A\|_{\mathrm{op}} \;=\; \sup_{\substack{\boldsymbol{x}\in\mathbb{R}^p\\ \|\boldsymbol{x}\|_2=1}} \|A\boldsymbol{x}\|_2.$$

The Frobenius norm is the Euclidean norm of A viewed as a vector in $\mathbb{R}^{p^2}$; it satisfies (N1)–(N3) for this reason, and

$$A_n \to A \text{ entrywise} \quad\Longleftrightarrow\quad \|A_n - A\|_F \to 0.$$

The operator norm measures the largest factor by which A stretches a unit vector; for symmetric A it equals the largest absolute eigenvalue, $\|A\|_{\mathrm{op}} = \max_i|\lambda_i|$.

Theorem 2.20 (Equivalence of norms in finite dimensions)**.** For any two norms $\|\cdot\|_a, \|\cdot\|_b$ on a finite-dimensional real vector space, there exist constants $c, C > 0$ such that

$$c\,\|\boldsymbol{v}\|_a \;\le\; \|\boldsymbol{v}\|_b \;\le\; C\,\|\boldsymbol{v}\|_a \qquad \text{for every } \boldsymbol{v}.$$

The proof is a compactness argument: the function $\boldsymbol{v} \mapsto \|\boldsymbol{v}\|_b/\|\boldsymbol{v}\|_a$ is continuous and positive on the unit sphere $\{\|\boldsymbol{v}\|_a = 1\}$, which is compact in finite dimensions; its minimum c and maximum C are strictly positive, and the inequality extends to all $\boldsymbol{v}$ by homogeneity. The practical consequence for our purposes is that for matrices in finite dimensions, every sense of convergence to zero (entrywise, Frobenius, operator) implies all of the others.

Determinant and the characteristic polynomial. The Leibniz formula expresses the determinant as a polynomial in the matrix entries:

$$\det A \;=\; \sum_{\pi\in S_p} \operatorname{sgn}(\pi) \prod_{i=1}^{p} a_{i,\pi(i)}, \tag{8}$$

where S_p is the group of permutations of $\{1,\ldots,p\}$. Since polynomials in the entries are continuous functions of the entries, so is the determinant:

$$A_n \to A \text{ entrywise} \quad\Longrightarrow\quad \det A_n \to \det A.$$

The *characteristic polynomial* of A is

$$p_A(\lambda) \;=\; \det(A - \lambda I), \qquad \lambda \in \mathbb{R},$$

whose roots are exactly the eigenvalues of A, counted with algebraic multiplicity. The coefficients of p_A are themselves polynomials in the entries of A (apply Leibniz to $A - \lambda I$), so they too are continuous functions of A.

Eigenvalue convergence. The key ingredient is that when a sequence of matrices converges, their eigenvalues converge. The deterministic statement is a standard fact about polynomial roots; we push it through the modes of convergence developed above, following Genzer (2025, Prop. 3.1).

Proposition 2.21 (Eigenvalue convergence via characteristic polynomials)**.** Let $\{\Sigma_n\}_{n\geq 1}$ and Σ_0 be $p \times p$ symmetric matrices, and write $\lambda_i(\Sigma)$ for the i^{th} eigenvalue of Σ (with some fixed ordering convention).

(1) *Deterministic.* If $\Sigma_n \to \Sigma_0$ entrywise, then $\lambda_i(\Sigma_n) \to \lambda_i(\Sigma_0)$ for every i.

(2) *Almost sure.* If $\Sigma_n \xrightarrow{\text{a.s.}} \Sigma_0$ in any matrix norm, then $\lambda_i(\Sigma_n) \xrightarrow{\text{a.s.}} \lambda_i(\Sigma_0)$.

(3) *In probability.* If $\Sigma_n \xrightarrow{\mathbb{P}} \Sigma_0$ in any matrix norm, then $\lambda_i(\Sigma_n) \xrightarrow{\mathbb{P}} \lambda_i(\Sigma_0)$.

Proof. Step 1 (deterministic). Suppose $\Sigma_n \to \Sigma_0$ entrywise. By the Leibniz formula (8), the coefficients of the characteristic polynomial $p_{\Sigma_n}(\lambda) = \det(\Sigma_n - \lambda I)$ are polynomial functions of the entries of Σ_n, so they converge to the coefficients of p_{Σ_0}. A standard result from analysis states that the roots of a monic polynomial depend continuously on its coefficients (after a consistent ordering); applied to the characteristic polynomials this gives $\lambda_i(\Sigma_n) \to \lambda_i(\Sigma_0)$ for every i.

Step 2 (almost sure). Suppose $\Sigma_n \xrightarrow{\text{a.s.}} \Sigma_0$ in some matrix norm $\|\cdot\|$. By Theorem 2.20, this is equivalent to convergence in Frobenius norm, which in turn is entrywise convergence (with probability one). Concretely, there is an event Ω_0 with $\mathbb{P}(\Omega_0) = 1$ such that for every $\omega \in \Omega_0$, $\Sigma_n(\omega) \to \Sigma_0(\omega)$ entrywise. Step 1 applies to this deterministic sequence and yields $\lambda_i(\Sigma_n(\omega)) \to \lambda_i(\Sigma_0(\omega))$ for every $\omega \in \Omega_0$, which is the statement $\lambda_i(\Sigma_n) \xrightarrow{\text{a.s.}} \lambda_i(\Sigma_0)$.

Step 3 (in probability). Suppose $\Sigma_n \xrightarrow{\mathbb{P}} \Sigma_0$ in some matrix norm. The map $\Sigma \mapsto (\lambda_1(\Sigma), \ldots, \lambda_p(\Sigma))$ is continuous (entries to coefficients to roots), so for every $\eta > 0$ there exists $\varepsilon > 0$ such that

$$\|\Sigma - \Sigma_0\| \;<\; \varepsilon \quad\Longrightarrow\quad \max_i |\lambda_i(\Sigma) - \lambda_i(\Sigma_0)| \;<\; \eta.$$

Translating to events,

$$\Big\{\max_i |\lambda_i(\Sigma_n) - \lambda_i(\Sigma_0)| \geq \eta\Big\} \;\subseteq\; \big\{\|\Sigma_n - \Sigma_0\| \geq \varepsilon\big\},$$

and taking probabilities,

$$\mathbb{P}\Big(\max_i |\lambda_i(\Sigma_n) - \lambda_i(\Sigma_0)| \geq \eta\Big) \;\leq\; \mathbb{P}(\|\Sigma_n - \Sigma_0\| \geq \varepsilon) \;\longrightarrow\; 0.$$

This is $\lambda_i(\Sigma_n) \xrightarrow{\mathbb{P}} \lambda_i(\Sigma_0)$ for every i. ☐

The deterministic chain *entries* → *characteristic-polynomial coefficients* → *roots* is the entire logical content; the random versions follow by pushing continuity through the respective modes of convergence.

The two pieces of this section now combine into a single statement about the sample covariance matrix. It is, on the one hand, a matrix of averages, controlled entrywise by the law of large numbers; and, on the other hand, a symmetric positive-semidefinite matrix, controlled geometrically by the spectral theorem and the Rayleigh quotient. The next section makes this statement precise and asks what changes when the dimension grows with the sample size.

3 Sample Covariance Matrices

The covariance matrix Σ describes the true variability of a random vector, but in data analysis it is unknown. Given observations $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$, the natural replacement is the empirical average of the outer products $\boldsymbol{X}_i \boldsymbol{X}_i^\top$. In fixed dimensions, this empirical version behaves like a classical sample average: as the number of observations grows, it converges entry by entry to the true covariance, exactly as a scalar sample mean converges to its expectation.[⋆1] When the dimension grows with the sample size (the so-called *high-dimensional regime*[⋆2]), its spectral behavior changes qualitatively, and both regimes are unpacked below.

3.1 Definition and Geometry of the Sample Covariance Matrix

Definition 3.1 (Sample covariance matrix (SCM))**.** Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ be iid copies of a random vector $\boldsymbol{X} \in \mathbb{R}^p$. In the *mean-zero* case ($\mathbb{E}[\boldsymbol{X}] = \mathbf{0}$ known), the *sample covariance matrix* is the $p \times p$ random matrix[⋆3]

$$\widehat{\Sigma}_n \;=\; \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{X}_i^\top. \tag{9}$$

In the *mean-unknown* case, the (unbiased) sample covariance matrix is

$$\widehat{\Sigma}_n \;=\; \frac{1}{n-1} \sum_{i=1}^{n} (\boldsymbol{X}_i - \bar{\boldsymbol{X}}_n)(\boldsymbol{X}_i - \bar{\boldsymbol{X}}_n)^\top, \qquad \bar{\boldsymbol{X}}_n \;=\; \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{X}_i.$$

We work in the mean-zero case throughout the rest of the report; the Gaussian models in Sections 3 and 4 all have $\boldsymbol{\mu} = \mathbf{0}$ by convention, and the matrix algebra is cleaner. The asymptotic results below carry over to the mean-unknown version with no substantive change.

Note that each entry of a SCM is a sample average.[⋆4] In fact, writing $\boldsymbol{X}_i = (X_{i1}, \ldots, X_{ip})^\top$, the (j,k) entry of $\widehat{\Sigma}_n$ is

$$(\widehat{\Sigma}_n)_{jk} \;=\; \frac{1}{n} \sum_{i=1}^{n} X_{ij}\, X_{ik}, \qquad j, k = 1, \ldots, p. \tag{10}$$

Each entry of (10)[⋆5] is therefore the sample average of the iid scalar products $W_i^{(j,k)} = X_{ij} X_{ik}$, whose common mean (since $\boldsymbol{X}_1$ has mean zero) is exactly

$$\mathbb{E}\big[X_{1j}\, X_{1k}\big] \;=\; \mathrm{Cov}(X_j, X_k) \;=\; \Sigma_{jk}.$$

By the strong law of large numbers (Theorem 2.10), every entry converges almost surely to its population counterpart, $(\widehat{\Sigma}_n)_{jk} \xrightarrow{\text{a.s.}} \Sigma_{jk}$ as $n \to \infty$. This entrywise LLN is the mechanism behind the matrix-level convergence proved in Section 3.2. The diagonal entries $(\widehat{\Sigma}_n)_{jj} = n^{-1} \sum_i X_{ij}^2$ are sample variances of individual coordinates; the off-diagonal entries are sample covariances between pairs of coordinates.

As it turns out, the entrywise formula (10) can be conveniently repackaged as a single matrix identity.[⋆6] Stack the n observations as the columns of the *data matrix*

$$\boldsymbol{X} \;=\; (\boldsymbol{X}_1 \mid \boldsymbol{X}_2 \mid \cdots \mid \boldsymbol{X}_n) \;\in\; \mathbb{R}^{p \times n}.$$

This is a $p \times n$ matrix in which the rows correspond to features (the coordinates of $\boldsymbol{X}$) and the columns correspond to observations: row j holds the values $X_{1j}, X_{2j}, \ldots, X_{nj}$ of the j^{th} feature across all n observations. The product $\boldsymbol{X}\boldsymbol{X}^\top$ is then a $p \times p$ matrix whose (j,k) entry, by the usual rules of matrix multiplication, is the dot product of the j^{th} and k^{th} feature rows of $\boldsymbol{X}$:

$$(\boldsymbol{X}\boldsymbol{X}^\top)_{jk} \;=\; \sum_{i=1}^{n} X_{ij}\,X_{ik}.$$

Dividing by n recovers equation (10), so we obtain the compact form

$$\widehat{\Sigma}_n \;=\; \frac{1}{n}\,\boldsymbol{X}\boldsymbol{X}^\top. \tag{11}$$

This is the working form of the sample covariance matrix used throughout the rest of the report. Relation (11) is very useful in establishing the following proposition.[⋆7]

Proposition 3.1 ($\widehat{\Sigma}_n$ is symmetric and PSD)**.** The sample covariance matrix $\widehat{\Sigma}_n$ is symmetric and positive semidefinite. All of its eigenvalues are therefore nonnegative (Theorem 2.18).

Proof. For the mean-zero version (11), symmetry is immediate: $(\boldsymbol{X}\boldsymbol{X}^\top)^\top = (\boldsymbol{X}^\top)^\top \boldsymbol{X}^\top = \boldsymbol{X}\boldsymbol{X}^\top$. For positive semidefiniteness, fix $\boldsymbol{v} \in \mathbb{R}^p$:

$$\boldsymbol{v}^\top \widehat{\Sigma}_n \boldsymbol{v} \;=\; \frac{1}{n}\,\boldsymbol{v}^\top \boldsymbol{X}\boldsymbol{X}^\top \boldsymbol{v} \;=\; \frac{1}{n}\,(\boldsymbol{X}^\top \boldsymbol{v})^\top (\boldsymbol{X}^\top \boldsymbol{v}) \;=\; \frac{1}{n}\,\|\boldsymbol{X}^\top \boldsymbol{v}\|_2^2 \;\geq\; 0. \tag{12}$$

The mean-unknown version uses the centered data matrix in place of $\boldsymbol{X}$; the same identity gives $\boldsymbol{v}^\top \widehat{\Sigma}_n \boldsymbol{v} = (n-1)^{-1}\|\boldsymbol{X}_c^\top \boldsymbol{v}\|_2^2 \geq 0$. $\square$

The identity (12) has a clean statistical reading. The vector $\boldsymbol{X}^\top \boldsymbol{v} \in \mathbb{R}^n$ has i^{th} entry $\boldsymbol{v}^\top \boldsymbol{X}_i$, the projection of the i^{th} observation onto the direction $\boldsymbol{v}$; dividing the squared norm of these projections by n gives the empirical second moment of the scalar sample $\boldsymbol{v}^\top \boldsymbol{X}_1, \ldots, \boldsymbol{v}^\top \boldsymbol{X}_n$. In the mean-zero case this is exactly the empirical variance of the data in the direction $\boldsymbol{v}$:

$$\boldsymbol{v}^\top \widehat{\Sigma}_n \boldsymbol{v} \;=\; \frac{1}{n}\sum_{i=1}^{n} (\boldsymbol{v}^\top \boldsymbol{X}_i)^2. \tag{13}$$

Proposition 3.1 therefore reads, in plain English, as the statement that the empirical variance in every direction is nonnegative. This is the sample analogue of the identity $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = \mathrm{Var}(\boldsymbol{v}^\top \boldsymbol{X})$ from Proposition 2.13.

Covariance matrix as geometry. Among all unit directions $\boldsymbol{v} \in \mathbb{R}^p$, some have larger empirical variance than others. The spectral theorem (Theorem 2.16) factors $\widehat{\Sigma}_n = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$ with $\boldsymbol{Q}$ orthogonal and $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$, and the Rayleigh quotient (Proposition 2.19) identifies the special directions that diagonalize the directional variance: the maximum of $\boldsymbol{v}^\top \widehat{\Sigma}_n \boldsymbol{v}$ over unit $\boldsymbol{v}$ is the largest eigenvalue λ_1, attained on the top eigenvector $\boldsymbol{q}_1$. The eigenvectors of $\widehat{\Sigma}_n$ are therefore the perpendicular *principal directions* of variation in the data, and the eigenvalues record the empirical variance in each principal direction. Figure 9 shows the population-level picture in two dimensions, with the covariance ellipse and the eigenvector directions overlaid on a data cloud.

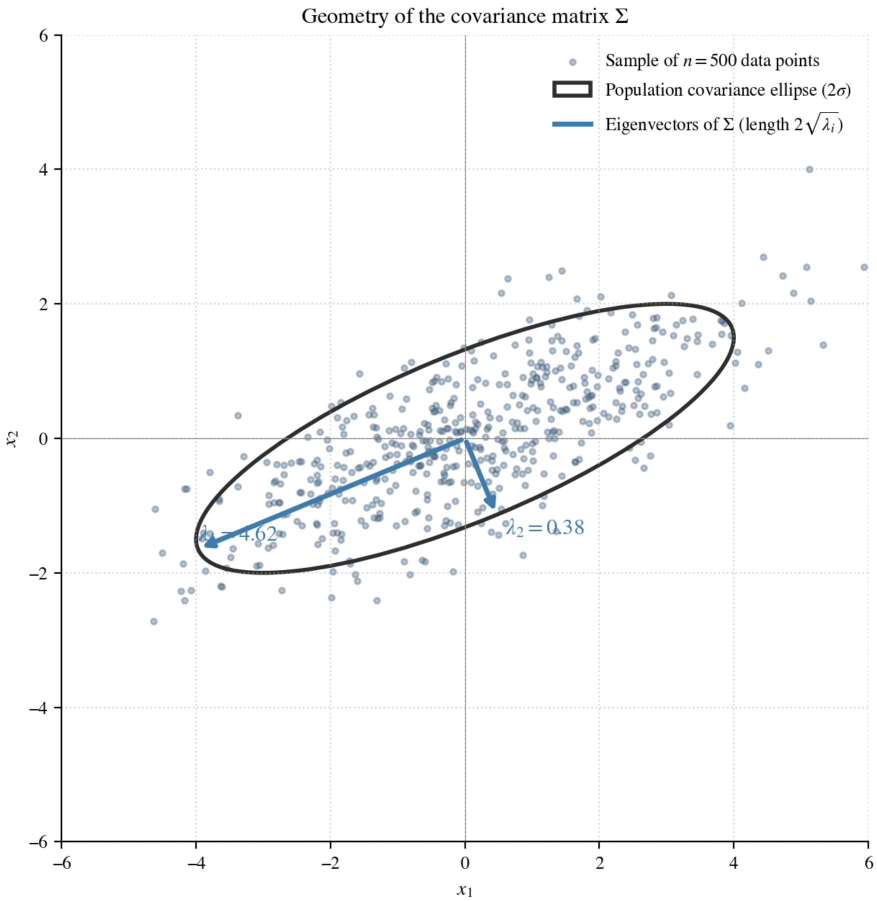

Figure 9: Geometry of the covariance matrix. The figure shows $n = 500$ iid samples from $\mathcal{N}(\mathbf{0}, \Sigma)$ with $\Sigma = \begin{pmatrix} 4 & 1.5 \\ 1.5 & 1 \end{pmatrix}$. The 2-$\sigma$ ellipse $\{\boldsymbol{x} : \boldsymbol{x}^\top \Sigma^{-1} \boldsymbol{x} = 4\}$ encloses roughly 86% of the distribution; its principal axes point in the directions of the eigenvectors of Σ, with lengths equal to $2\sqrt{\lambda_i}$. The eigenvalues λ_1, λ_2 are the variances of $\boldsymbol{X}$ projected onto the corresponding eigenvector directions. Generated by `notebooks/03_section3_demos.py`.

In fixed dimension, this matrix behaves like an ordinary average and its eigenstructure tracks that of Σ. When the dimension grows with the sample size, the eigenvalues spread away from the population eigenvalues and need separate analysis.

3.2 Low Dimensions: Consistency, Wishart, and Fluctuations

In the low-dimensional regime the dimension p is fixed and the sample size $n \to \infty$. This is the classical setting of multivariate statistics, in which the limit theorems of Section 2 apply directly to the entries of $\widehat{\Sigma}_n$. We unpack four nested questions: (i) does $\widehat{\Sigma}_n$ converge to Σ, (ii) what is the exact distribution of $\widehat{\Sigma}_n$ for Gaussian data, (iii) what is the limiting distribution of the fluctuation $\sqrt{n}(\widehat{\Sigma}_n - \Sigma)$, and (iv) what does that say about individual eigenvalues and eigenvectors of $\widehat{\Sigma}_n$?

Consistency of the matrix and its eigenvalues. The first question has the cleanest answer. Each entry of $\widehat{\Sigma}_n$ is a sample average of iid scalar products with mean Σ_{jk}, so the strong law of large numbers (Theorem 2.10) gives entrywise convergence at once. The intro figure (Figure 1, left panel) already shows the spectral consequence: with $p = 10$ and $n = 1000$, the sample eigenvalues of $\widehat{\Sigma}_n$ concentrate tightly around the population value 1.

Theorem 3.2 (Entrywise consistency of $\widehat{\Sigma}_n$)**.** Let $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots$ be iid copies of $\boldsymbol{X} \in \mathbb{R}^p$ with mean $\mathbf{0}$ and finite second moments $\mathbb{E}[X_j X_k] < \infty$. Then

$$\widehat{\Sigma}_n \xrightarrow{\text{a.s.}} \Sigma \qquad \text{entrywise, as } n \to \infty.$$

Proof. The (j,k) entry of $\widehat{\Sigma}_n$ is

$$(\widehat{\Sigma}_n)_{jk} \;=\; \frac{1}{n}\sum_{i=1}^{n} X_{ij}\, X_{ik}.$$

The random variables $\{X_{ij}X_{ik}\}_{i\geq 1}$ are iid with mean

$$\mathbb{E}[X_{1j}X_{1k}] \;=\; \Sigma_{jk}$$

(using $\boldsymbol{\mu} = \mathbf{0}$) and finite first moment by hypothesis. Kolmogorov's strong law of large numbers (Theorem 2.10) applied to this sequence yields

$$(\widehat{\Sigma}_n)_{jk} \xrightarrow{\text{a.s.}} \Sigma_{jk}.$$

Doing this for each pair (j,k) gives entrywise almost-sure convergence $\widehat{\Sigma}_n \xrightarrow{\text{a.s.}} \Sigma$. □

Combined with the eigenvalue-convergence machinery developed in Section 2.5, entrywise convergence forces convergence of the eigenvalues themselves.

Corollary 3.3 (Sample eigenvalues converge to population eigenvalues)**.** Under the hypotheses of Theorem 3.2, $\lambda_i(\widehat{\Sigma}_n) \xrightarrow{\text{a.s.}} \lambda_i(\Sigma)$ for every $i = 1, \ldots, p$.

Proof. Apply Proposition 2.21(2) (the almost-sure version of eigenvalue convergence) to the random matrix sequence $\widehat{\Sigma}_n \xrightarrow{\text{a.s.}} \Sigma$. □

Theorem 3.2 and Corollary 3.3 together give the qualitative low-dimensional picture: $\widehat{\Sigma}_n$ is a consistent estimator of Σ, and the sample eigenvalues track the population eigenvalues. The remaining questions concern the size and shape of the fluctuations.

Gaussian sample covariance and the Wishart law. For Gaussian observations, $\widehat{\Sigma}_n$ (more precisely, $n\widehat{\Sigma}_n$) has an exact distribution. This makes it a tractable starting point for finer questions than entrywise consistency.

Definition 3.2 (Wishart distribution)**.** Let $\boldsymbol{X}_1, \dots, \boldsymbol{X}_n$ be iid $\mathcal{N}(\mathbf{0}, \Sigma)$ random vectors in $\mathbb{R}^p$, and set

$$\boldsymbol{S} \;=\; \sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{X}_i^\top \;=\; n\,\widehat{\Sigma}_n.$$

The distribution of $\boldsymbol{S}$ is the *Wishart distribution* with n degrees of freedom and scale matrix Σ, written $\boldsymbol{S} \sim W_p(n, \Sigma)$. When $n \geq p$ and Σ is positive definite, $\boldsymbol{S}$ admits the density (on the cone of positive-definite matrices)

$$f_{\boldsymbol{S}}(\boldsymbol{S}) \;=\; \frac{|\boldsymbol{S}|^{(n-p-1)/2} \exp\!\big(-\frac{1}{2}\operatorname{tr}(\Sigma^{-1}\boldsymbol{S})\big)}{2^{np/2}\,|\Sigma|^{n/2}\,\Gamma_p(n/2)}, \tag{14}$$

where Γ_p is the multivariate gamma function; see Yao et al. (2015, §3.2) for a derivation.

The Wishart distribution is the multivariate analogue of the chi-square distribution, and recovers it in dimension one.

Example 3.3 (Wishart in dimension one)**.** Take $p = 1$ and $\Sigma = 1$. Then each $\boldsymbol{X}_i$ is a scalar $X_i \sim \mathcal{N}(0,1)$, so $\boldsymbol{S} = \sum_{i=1}^n X_i^2$. By the definition of the chi-square distribution, $\boldsymbol{S} \sim \chi^2_n$. Substituting $p = 1$, $\Sigma = 1$, and $\operatorname{tr}(\Sigma^{-1}\boldsymbol{S}) = \boldsymbol{S}$ into (14) confirms this:

$$f_{\boldsymbol{S}}(s) \;=\; \frac{s^{(n-2)/2}\,e^{-s/2}}{2^{n/2}\,\Gamma(n/2)}, \qquad s > 0,$$

which is exactly the χ^2_n density.

For $\Sigma = I_2$, the joint distribution of the two ordered eigenvalues $0 \leq \lambda_1 \leq \lambda_2$ of $\boldsymbol{S} \sim W_2(n, I_2)$ admits an explicit density that reveals a striking phenomenon, namely *eigenvalue repulsion*:

$$f(\lambda_1, \lambda_2) \;=\; C\,(\lambda_1\lambda_2)^{(n-3)/2}\,e^{-(\lambda_1+\lambda_2)/2}\,(\lambda_2 - \lambda_1)\,\mathbf{1}_{\{0 \leq \lambda_1 \leq \lambda_2\}}, \tag{15}$$

where $C > 0$ is a normalizing constant; a derivation via a Jacobian computation on the spectral decomposition $\boldsymbol{S} = \boldsymbol{Q}\,\operatorname{diag}(\lambda_1, \lambda_2)\boldsymbol{Q}^\top$ is given in Yao et al. (2015, Ch. 3). The key feature of (15) is the factor $(\lambda_2 - \lambda_1)$, which forces the joint density to vanish on the diagonal $\lambda_1 = \lambda_2$. This is the simplest case of *eigenvalue repulsion*: the two eigenvalues of a random symmetric matrix avoid each other. Figure 10 confirms repulsion in simulation.

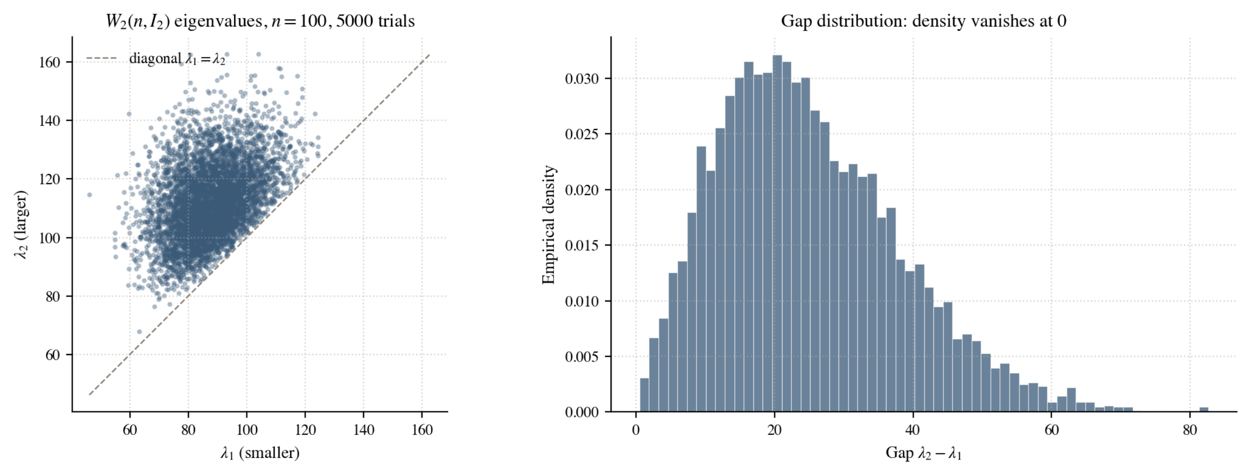

Figure 10: Eigenvalues of $W_2(100, I_2)$ computed from 5000 independent Monte Carlo trials. *Left:* scatter plot of (λ_1, λ_2) with the line $\lambda_2 = \lambda_1$ shown for reference. The cloud is visibly bounded away from the diagonal, the geometric signature of the $(\lambda_2 - \lambda_1)$ factor in (15). *Right:* histogram of the gap $\lambda_2 - \lambda_1$; the density vanishes at zero, the empirical signature of repulsion. Generated by `notebooks/03_section3_demos.py`.

The factor $(\lambda_2 - \lambda_1)$ in (15) is the simplest instance of a structure that appears in every classical random matrix ensemble: the *Vandermonde determinant*.

Definition 3.4 (Vandermonde determinant)**.** For real numbers $\lambda_1, \ldots, \lambda_p$, the *Vandermonde determinant* is

$$V(\lambda_1, \ldots, \lambda_p) \;=\; \det \begin{pmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \cdots & \lambda_p \\ \lambda_1^2 & \lambda_2^2 & \cdots & \lambda_p^2 \\ \vdots & \vdots & & \vdots \\ \lambda_1^{p-1} & \lambda_2^{p-1} & \cdots & \lambda_p^{p-1} \end{pmatrix} \;=\; \prod_{1 \le i < j \le p} (\lambda_j - \lambda_i). \tag{16}$$

The closed-form on the right is a classical identity, proved by row reduction: subtracting λ_1 times each row from the row below introduces a factor of $(\lambda_j - \lambda_1)$ in every column $j \ge 2$, the first column collapses, and the rest is induction on p. For $p = 2$ the product reduces to $V = \lambda_2 - \lambda_1$; for $p = 3$, $V = (\lambda_2 - \lambda_1)(\lambda_3 - \lambda_1)(\lambda_3 - \lambda_2)$; for $p = 4$ it is a product of six gaps; and so on. The Vandermonde is therefore the product of all signed gaps between the p eigenvalues.

The relevance for the Wishart distribution is geometric. To pass from the density (14) on the cone of positive-definite matrices to a density on the ordered eigenvalues, one diagonalizes $\boldsymbol{S} = \boldsymbol{Q}\,\mathrm{diag}(\lambda_1, \ldots, \lambda_p)\boldsymbol{Q}^\top$ and integrates out the orthogonal factor $\boldsymbol{Q}$. The Jacobian of this change of variables turns out to be exactly $|V(\lambda_1, \ldots, \lambda_p)|$ (the calculation is in Yao et al. (2015, § 3.2)). The full joint eigenvalue density of $W_p(n, I_p)$ for $n \ge p$ is therefore

$$f(\lambda_1, \ldots, \lambda_p) \;=\; C_{n,p}\, |V(\lambda_1, \ldots, \lambda_p)| \prod_{i=1}^{p} \lambda_i^{(n-p-1)/2}\, e^{-\lambda_i/2}, \qquad 0 \le \lambda_1 \le \cdots \le \lambda_p, \tag{17}$$

with a normalizing constant $C_{n,p}$ depending only on n and p. Setting $p = 2$ in (17) gives $|V| = \lambda_2 - \lambda_1$, the prefactor $\lambda_1^{(n-3)/2}\lambda_2^{(n-3)/2} = (\lambda_1\lambda_2)^{(n-3)/2}$, and the exponential $e^{-(\lambda_1+\lambda_2)/2}$ — exactly (15), with the indicator collapsing into the ordering constraint.

The Vandermonde factor vanishes on every hyperplane $\lambda_i = \lambda_j$, so configurations with near-equal eigenvalues are systematically suppressed in every dimension. This is the algebraic mechanism behind eigenvalue repulsion: the gap visible in Figure 10 for $p = 2$ generalizes to a product of $\binom{p}{2}$ pairwise gaps in higher dimensions, each contributing its own zero on the corresponding diagonal. The same Vandermonde factor appears in the joint eigenvalue density of the Gaussian Orthogonal Ensemble defined below; only the prefactor and the exponential change. Repulsion is a universal feature of the spectra of random symmetric matrices, not a quirk of the Wishart family.

Fluctuations and the role of eigenvalue multiplicity. Beyond consistency, we want to know what $\sqrt{n}(\widehat{\Sigma}_n - \Sigma)$ looks like. The limit is a Gaussian symmetric matrix whose variance structure has a standard name. The variances 2 on the diagonal and 1 off the diagonal appear so often in symmetric Gaussian matrix limits that they define a canonical ensemble.

Definition 3.5 (Gaussian Orthogonal Ensemble, 2×2 case)**.** A 2×2 random matrix

$$\boldsymbol{G} = \begin{pmatrix} G_{11} & G_{12} \\ G_{12} & G_{22} \end{pmatrix}$$

is said to belong to the *Gaussian Orthogonal Ensemble* (GOE_2) if its three independent entries are Gaussian with

$$G_{11},\, G_{22} \sim \mathcal{N}(0,\, 2), \qquad G_{12} \sim \mathcal{N}(0,\, 1),$$

and the three entries are mutually independent. More generally, a $p \times p$ GOE matrix has independent Gaussian entries on and above the diagonal with diagonal variance 2 and off-diagonal variance 1.

The variances 2 on the diagonal and 1 off the diagonal are chosen exactly so that the matrix is invariant in distribution under orthogonal conjugation $\boldsymbol{G} \mapsto \boldsymbol{O}\boldsymbol{G}\boldsymbol{O}^\top$; this is the eponymous orthogonal invariance. The top eigenvalue of GOE_2 is a non-Gaussian random variable, with a density slightly right-skewed by the repulsion that mirrors (15).

The multivariate CLT applied entry-by-entry to $\sqrt{n}(\widehat{\Sigma}_n - \Sigma)$ yields the matrix-valued statement. In the null case the limit is exactly the GOE just defined.

Proposition 3.4 (Matrix-valued CLT for $\Sigma = I_2$)**.** Let $\boldsymbol{X}_i \sim \mathcal{N}(\boldsymbol{0}, I_2)$ be iid. Then $\sqrt{n}\,(\widehat{\Sigma}_n - I_2)$ converges in distribution, as $n \to \infty$, to a GOE_2 matrix.

Proof. The three independent entries of the symmetric matrix $\sqrt{n}\,(\widehat{\Sigma}_n - I_2)$ are

$$\sqrt{n}\,((\widehat{\Sigma}_n)_{11} - 1), \quad \sqrt{n}\,((\widehat{\Sigma}_n)_{22} - 1), \quad \sqrt{n}\,(\widehat{\Sigma}_n)_{12},$$

each of which is a sum of iid centered random variables, so the multivariate central limit theorem of Section 2.4 applies. We compute the limiting variances and verify that the cross-covariances vanish.

For the diagonal entries, write $(\widehat{\Sigma}_n)_{jj} - 1 = n^{-1}\sum_i (X_{ij}^2 - 1)$. Each summand has mean zero and

$$\mathrm{Var}(X_{ij}^2) \;=\; \mathbb{E}[X_{ij}^4] \;-\; (\mathbb{E}[X_{ij}^2])^2 \;=\; 3 \;-\; 1 \;=\; 2,$$

using $\mathbb{E}[Z^4] = 3$ for $Z \sim \mathcal{N}(0,1)$. Hence the limiting variance of $\sqrt{n}\,((\widehat{\Sigma}_n)_{jj} - 1)$ is 2.

For the off-diagonal entry, write $(\widehat{\Sigma}_n)_{12} = n^{-1}\sum_i X_{i1}X_{i2}$. Each summand has mean zero (independence) and variance

$$\mathrm{Var}(X_{i1}X_{i2}) \;=\; \mathbb{E}[X_{i1}^2]\,\mathbb{E}[X_{i2}^2] \;=\; 1,$$

again by independence. Hence the limiting variance of $\sqrt{n}\,(\widehat{\Sigma}_n)_{12}$ is 1.

The cross-covariances vanish:

$$\mathrm{Cov}(X_{i1}^2,\, X_{i1}X_{i2}) \;=\; \mathbb{E}[X_{i1}^3]\,\mathbb{E}[X_{i2}] \;=\; 0,$$

because $\mathbb{E}[X_{i2}] = 0$, and analogously for the other pairs.

The three limiting variances are therefore 2, 2, 1 and the three limits are uncorrelated. By the multivariate CLT, the joint limit is $\mathcal{N}(\mathbf{0},\, \mathrm{diag}(2,2,1))$, which is exactly the entrywise distribution of a GOE_2 matrix from Definition 3.5. □

Proposition 3.4 describes the fluctuation of the *matrix* $\widehat{\Sigma}_n$, not of its individual eigenvalues. The link from matrix to eigenvalues is a first-order perturbation lemma, the multivariate mean-value theorem for simple eigenvalues.

Lemma 3.5 (First-order perturbation for simple eigenvalues)**.** Let $\boldsymbol{A}$ be a real symmetric $p \times p$ matrix with a simple eigenvalue λ and unit eigenvector $\boldsymbol{u}$. For any real symmetric $\boldsymbol{H}$ with sufficiently small $\|\boldsymbol{H}\|$, the eigenvalue $\lambda(\boldsymbol{A}+\boldsymbol{H})$ of $\boldsymbol{A}+\boldsymbol{H}$ closest to λ satisfies

$$\lambda(\boldsymbol{A}+\boldsymbol{H}) \;=\; \lambda \;+\; \boldsymbol{u}^\top \boldsymbol{H}\,\boldsymbol{u} \;+\; O(\|\boldsymbol{H}\|^2).$$

A self-contained proof requires basic implicit-function-theorem machinery; we give a direct 2×2 verification below, and refer to Abry et al. (2024, Lem. B.10) for the general statement.

Example 3.6 (Verification in the 2×2 case)**.** Take $\boldsymbol{A} = \mathrm{diag}(1,2)$, so the top eigenvalue is $\lambda = 2$ with eigenvector $\boldsymbol{u} = \boldsymbol{e}_2 = (0,1)^\top$. For a symmetric perturbation $\boldsymbol{H} = \begin{pmatrix} h_{11} & h_{12} \\ h_{12} & h_{22} \end{pmatrix}$, the 2×2 eigenvalue formula (7) applied to $\boldsymbol{A}+\boldsymbol{H}$ gives

$$\lambda_+(\boldsymbol{A}+\boldsymbol{H}) \;=\; \frac{(1+h_{11})+(2+h_{22})}{2} \;+\; \frac{1}{2}\sqrt{(1+h_{11}-2-h_{22})^2 + 4h_{12}^2}.$$

For small $\boldsymbol{H}$, Taylor-expanding the square root,

$$\sqrt{(-1+h_{11}-h_{22})^2 + 4h_{12}^2} \;=\; \sqrt{1+2(h_{22}-h_{11})+O(\|\boldsymbol{H}\|^2)} \;=\; 1+(h_{22}-h_{11})+O(\|\boldsymbol{H}\|^2).$$

Substituting back,

$$\lambda_+(\boldsymbol{A}+\boldsymbol{H}) \;=\; \frac{3+h_{11}+h_{22}}{2} \;+\; \frac{1+h_{22}-h_{11}}{2} \;+\; O(\|\boldsymbol{H}\|^2) \;=\; 2 \;+\; h_{22} \;+\; O(\|\boldsymbol{H}\|^2).$$

This is exactly Lemma 3.5 with $\boldsymbol{u}^\top\boldsymbol{H}\boldsymbol{u} = \boldsymbol{e}_2^\top\boldsymbol{H}\boldsymbol{e}_2 = h_{22}$.

With Proposition 3.4 and Lemma 3.5 in hand, the dichotomy between repeated and distinct population eigenvalues is straightforward.

Repeated case: $\Sigma = I_2$. Lemma 3.5 does *not* apply, because both eigenvalues equal 1. Instead, the top-eigenvalue map $\boldsymbol{A} \mapsto \lambda_{\max}(\boldsymbol{A})$ is a continuous function from symmetric matrices to $\mathbb{R}$ (this is the eigenvalue-continuity statement of Proposition 2.21(1)). By the continuous mapping theorem (Proposition 2.6), applied to Proposition 3.4,

$$\sqrt{n}\,(\lambda_{\max}(\widehat{\Sigma}_n) - 1) \;\xrightarrow{d}\; \lambda_{\max}(\boldsymbol{G}),$$

where $\boldsymbol{G} \sim \mathrm{GOE}_2$. The right side is the top eigenvalue of a GOE_2 matrix, which is *not* Gaussian; it has the slightly right-skewed density shaped by repulsion.

Distinct case: $\Sigma = \mathrm{diag}(1, 2)$. The population eigenvalues are simple. Lemma 3.5 applies with $\boldsymbol{u} = \boldsymbol{e}_2$, giving

$$\lambda_2(\widehat{\Sigma}_n) - 2 \;=\; (\widehat{\Sigma}_n)_{22} - 2 \;+\; O_p(\|\widehat{\Sigma}_n - \Sigma\|^2).$$

Multiplying by $\sqrt{n}$ and using $(\widehat{\Sigma}_n - \Sigma)_{22} = O_p(n^{-1/2})$ (so the quadratic remainder is $O_p(n^{-1/2})$ and vanishes in the limit),

$$\sqrt{n}\,(\lambda_2(\widehat{\Sigma}_n) - 2) \;=\; \sqrt{n}\,((\widehat{\Sigma}_n)_{22} - 2) \;+\; o_p(1). \tag{18}$$

The entry $(\widehat{\Sigma}_n)_{22} = n^{-1}\sum_i X_{i2}^2$ with $X_{i2} \sim \mathcal{N}(0, 2)$ has, by the univariate CLT, $\sqrt{n}\,((\widehat{\Sigma}_n)_{22} - 2) \xrightarrow{d} \mathcal{N}(0,\, \mathrm{Var}(X_{i2}^2))$, and

$$\mathrm{Var}(X_{i2}^2) \;=\; \mathbb{E}[X_{i2}^4] \;-\; 4 \;=\; 12 \;-\; 4 \;=\; 8,$$

using $\mathbb{E}[Z^4] = 3\sigma^4 = 3 \cdot 4 = 12$ for $Z \sim \mathcal{N}(0, 2)$. Combining with (18),

$$\sqrt{n}\,(\lambda_2(\widehat{\Sigma}_n) - 2) \;\xrightarrow{d}\; \mathcal{N}(0,\, 8). \tag{19}$$

The limit is now an ordinary Gaussian. An identical argument applied to the smaller eigenvalue gives $\sqrt{n}\,(\lambda_1(\widehat{\Sigma}_n) - 1) \xrightarrow{d} \mathcal{N}(0, 2)$, and the two limits are independent because the relevant entries of $\widehat{\Sigma}_n$ are uncorrelated.

Figure 11 confirms this dichotomy with normal Q-Q plots from a Monte Carlo simulation.

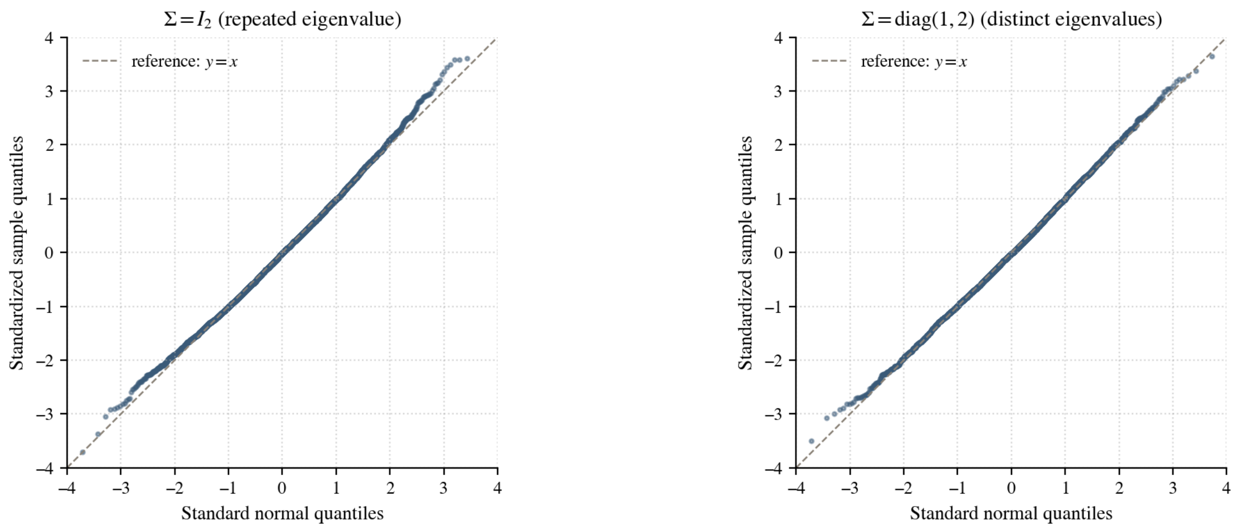

Figure 11: Normal Q-Q plots for the standardized top sample eigenvalue $\sqrt{n}\,(\lambda_{\max}(\widehat{\Sigma}_n) - \lambda_{\max}(\Sigma))$, 5000 Monte Carlo trials at $n = 500$. *Left:* $\Sigma = I_2$ (repeated population eigenvalues). The visible curvature away from the dashed reference line, especially at the right tail, is the signature of the non-Gaussian GOE_2 top-eigenvalue limit. *Right:* $\Sigma = \mathrm{diag}(1, 2)$ (distinct population eigenvalues). The points lie close to the reference line, confirming the Gaussian limit of equation (19). Generated by `notebooks/03_section3_demos.py`.

Eigenvectors and spectral separation. The same dichotomy applies to eigenvectors. In the distinct case $\Sigma = \mathrm{diag}(1, 2)$, the population eigenvectors are $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$, uniquely determined up to sign. The same first-order perturbation that produced (18) also produces, after a sign convention $\langle \widehat{\boldsymbol{u}}_\ell, \boldsymbol{e}_\ell \rangle \geq 0$,

$$\widehat{\boldsymbol{u}}_\ell \xrightarrow{\text{a.s.}} \boldsymbol{e}_\ell \qquad (\ell = 1, 2).$$

The general statement is in Didier and Pipiras (2012, Lem. B.2).

In the repeated case $\Sigma = I_2$ there is no canonical limiting eigenbasis: every direction in $\mathbb{R}^2$ is a population eigenvector. The sample eigenvectors may not converge, even when the sample matrix itself does. This is illustrated in the following deterministic counterexample.[⋆8]

Example 3.7 (Sample eigenvectors can fail to converge under repeated eigenvalues)**.** Define a sequence of 2×2 symmetric matrices

$$\boldsymbol{S}_n = \boldsymbol{O}_n \begin{pmatrix} 1 + \frac{1}{n} & 0 \\ 0 & 1 - \frac{1}{n} \end{pmatrix} \boldsymbol{O}_n^\top, \qquad \boldsymbol{O}_n = \begin{pmatrix} \cos\theta_n & -\sin\theta_n \\ \sin\theta_n & \cos\theta_n \end{pmatrix}, \quad \theta_n = \frac{n\pi}{2}.$$

Then $\boldsymbol{S}_n \to I_2$ in operator norm because $\|\boldsymbol{S}_n - I_2\|_{\mathrm{op}} = 1/n \to 0$. The eigenvectors of $\boldsymbol{S}_n$, however, are the columns of $\boldsymbol{O}_n$; they rotate by $\pi/2$ at every step and never settle. The matrix converges; its eigenvectors do not.

The lesson of Example 3.7 is that eigenvectors require *spectral separation*: without a gap between population eigenvalues, the eigenvectors are not stable under perturbation. This theme returns in Section 4, where only above a critical signal level do sample eigenvectors carry reliable information about the population spike direction.

3.3 High Dimensions: Marchenko-Pastur Law and the Tracy-Widom Edge

[⋆9]The *high-dimensional regime* is the situation in which the SCM parameters satisfy

$$\frac{p}{n} \longrightarrow y \in (0,\infty), \qquad \text{as } n \to \infty.$$

The ratio y is called the *aspect ratio* of the data.[⋆10]

In high dimensions, the picture of Section 3.2 breaks down qualitatively. The basic phenomenon was already visible in Figure 1 of the introduction: with $p = 500$ and $n = 1000$, the sample eigenvalues spread across the interval $[0.086, 2.914]$ even when $\Sigma = I_p$. In other words, by comparison to the low-dimensional regime, they no longer converge to $\lambda_i(\Sigma) = 1$.[⋆11]

In high dimensions, the relevant observable may no longer be an individual eigenvalue, but the entire range of them, also called the *bulk*.[⋆12] Individual eigenvalues nonetheless remain of interest: in the second half of this section the largest one, $\lambda_{\max}(\widehat{\Sigma}_n)$, takes center stage.[⋆13] To study the eigenvalues in the bulk, we first introduce the *empirical spectral distribution.*

Definition 3.8 (Empirical spectral distribution)**.** The *empirical spectral distribution* (ESD) of a $p \times p$ symmetric matrix A with eigenvalues $\lambda_1, \dots, \lambda_p$ is the probability measure

$$F_A = \frac{1}{p}\sum_{i=1}^{p} \delta_{\lambda_i},$$

placing equal mass $1/p$ at each eigenvalue.

The Marchenko-Pastur law. The shape of the limiting ESD depends only on the aspect ratio y of the data, and is the same across a wide range of entry distributions.[⋆14]

Theorem 3.6 (Marchenko-Pastur law (Marchenko and Pastur, 1967))**.** Let $\boldsymbol{X} \in \mathbb{R}^{p\times n}$ have iid entries with mean zero, unit variance, and finite fourth moment, and let $\widehat{\Sigma}_n = (1/n)\,\boldsymbol{X}\boldsymbol{X}^\top$. As $p, n \to \infty$ with $p/n \to y \in (0,\infty)$, the ESD $F_{\widehat{\Sigma}_n}$ converges weakly almost surely to the *Marchenko-Pastur law* MP_y. When $0 < y < 1$, this distribution is defined by the p.d.f.

$$f_{\mathrm{MP}_y}(\lambda) = \frac{1}{2\pi y\lambda}\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}, \qquad \lambda \in [\lambda_-, \lambda_+], \tag{20}$$

where $\lambda_\pm = (1 \pm \sqrt{y})^2$. If $y > 1$, the law carries an additional atom at zero of mass $1 - 1/y$.[⋆15]

A complete proof passes through the Stieltjes transform of $F_{\widehat{\Sigma}_n}$ and is beyond the scope of this report; see Yao et al. (2015, Thm. 2.5) or Vershynin (2018, Thm. 3.6). The hypothesis covers Gaussian entries (the main case we simulate) and many other distributions with finite fourth moment.[⋆16]

Figure 12 shows the law for four aspect ratios. The shape changes dramatically with y: for y small the support is a narrow band around $\lambda = 1$ (the low-dimensional regime is recovered as $y \to 0$); for y near one the density piles up at zero; for $y > 1$ the matrix $\widehat{\Sigma}_n$ becomes rank-deficient and $F_{\widehat{\Sigma}_n}$ carries an explicit Dirac mass at zero on top of the continuous bulk. A high-resolution single-realization companion plot is included in Appendix B, Figure 25.

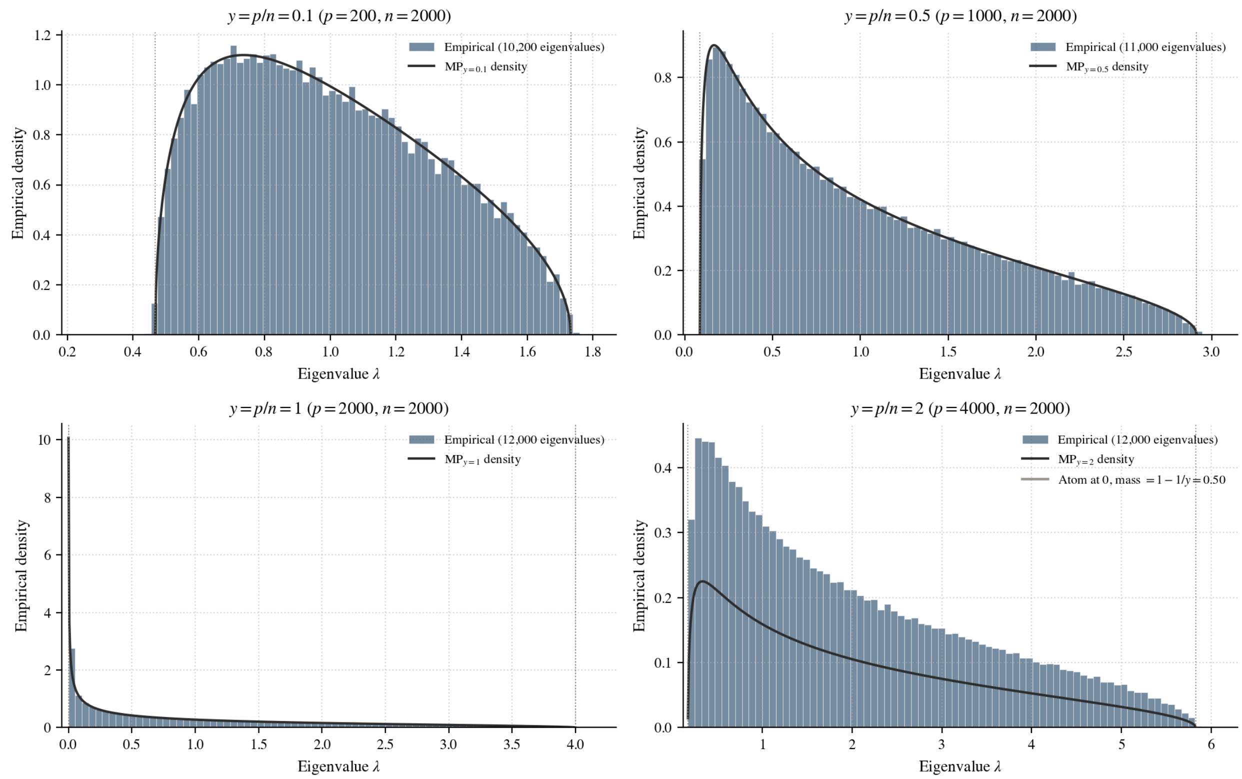

Figure 12: The Marchenko-Pastur law for four aspect ratios $y \in \{0.1, 0.5, 1, 2\}$. Each panel pools eigenvalues from many independent realizations of $\widehat{\Sigma}_n = (1/n)\,\boldsymbol{X}\boldsymbol{X}^\top$ with iid $\mathcal{N}(0,1)$ entries (at least 10,000 eigenvalues per panel), overlaid on the theoretical MP_y density (20) (solid curve). For $y > 1$ (lower right) the law has an atom at zero of mass $1 - 1/y$; the marker at $\lambda = 0$ records this. Generated by `notebooks/03_section3_demos.py`.

Three features of the MP law are worth recording.

- For $0 < y < 1$,[⋆17] the support is the interval $[(1-\sqrt{y})^2, (1+\sqrt{y})^2]$, of width $4\sqrt{y}$. As $y \to 0$ the support collapses to the point $\lambda = 1$, recovering the low-dimensional consistency of Section 3.2.
- Near the upper edge λ_+, the density vanishes like a square root: $f_{\mathrm{MP}_y}(\lambda) \sim C(\lambda_+ - \lambda)^{1/2}$. This square-root edge is the input to the Tracy-Widom heuristic below.[⋆18]
- When $y > 1$ the atom at zero has mass $1 - 1/y$; the matrix $\boldsymbol{X}\boldsymbol{X}^\top$ has rank at most $n < p$, so $p - n$ of its eigenvalues are deterministically zero.

The largest sample eigenvalue converges to the upper edge of the bulk.[⋆19] In the following theorem, we describe this convergence for the top eigenvalue of SCMs in the high-dimensional regime.[⋆20]

Theorem 3.7 (Convergence of the top eigenvalue (Baik and Silverstein, 2006))**.** Under the hypotheses of Theorem 3.6,

$$\lambda_{\max}(\widehat{\Sigma}_n) \xrightarrow{\text{a.s.}} \lambda_+ = (1+\sqrt{y})^2, \qquad n \to \infty.$$

Its importance notwithstanding, Theorem 3.7 settles *where* $\lambda_{\max}(\widehat{\Sigma}_n)$ lives in the limit but not *how* it fluctuates around λ_+. Unlike in low dimensions, the fluctuations of $\lambda_{\max}(\widehat{\Sigma}_n)$ in high dimensions cannot be Gaussian. In fact, the geometry on the two sides of $\lambda_{\max}$ is fundamentally different, which can be explained schematically as follows.[⋆21]

- *To the left,* the other $p-1$ sample eigenvalues are densely packed and repel each other via the Vandermonde factor in (16), exerting an upward "pressure" that resists $\lambda_{\max}$ drifting back into the bulk.
- *To the right,* the support of the MP density ends at λ_+; past the edge there is nothing, so excursions above λ_+ are unconstrained by neighbouring eigenvalues.

Figure 13 makes this geometry tangible: across hundreds of realizations, the second- and third-largest eigenvalues sit packed tightly against λ_+ from below, while $\lambda_{\max}$ itself scatters substantially on both sides of the edge. The increasing width of the rows as we move from $\lambda_{\max-5}$ up to $\lambda_{\max}$ is the Vandermonde repulsion in action, holding consecutive eigenvalues apart while pushing the top one outward.

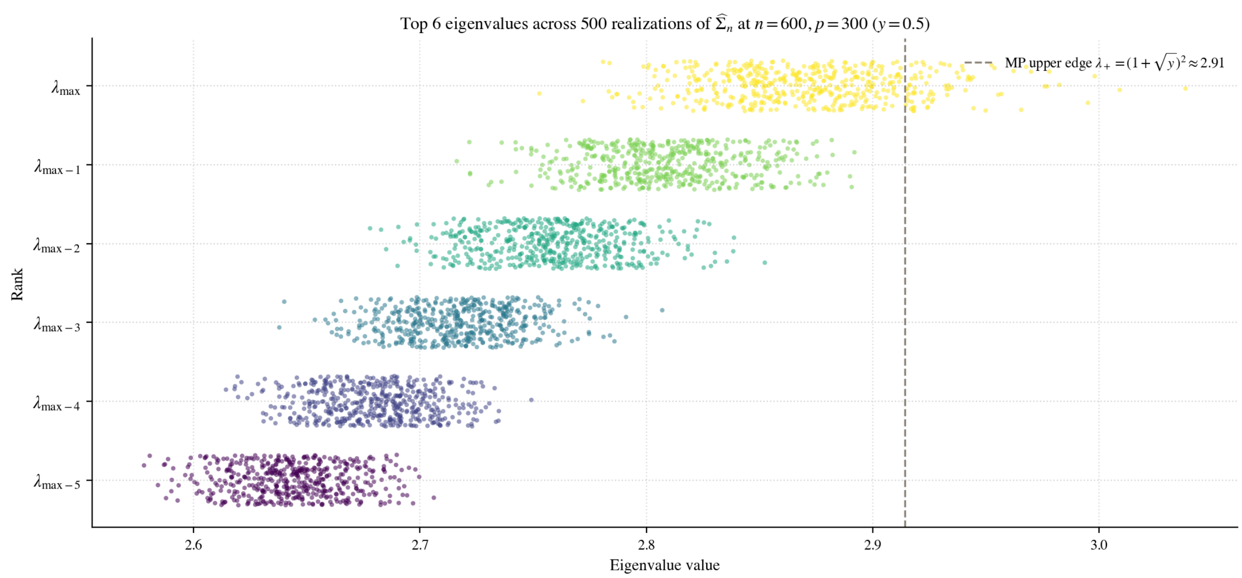

Figure 13: Top six eigenvalues of $\widehat{\Sigma}_n$ across $R = 500$ realizations at $n = 600$, $p = 300$ ($y = 1/2$). Each row shows the empirical locations of the eigenvalue of that rank; the rows are vertically jittered for visibility. The dashed line marks the MP upper edge λ_+. The rows below $\lambda_{\max}$ cluster against λ_+ from below and never overshoot it; only $\lambda_{\max}$ scatters past the edge. This asymmetric "one-sided pressure" is the geometric source of the TW_1 right skew. Generated by `notebooks/03_section3_demos.py`.

Two regimes, two different restoring mechanisms: the limit is right-skewed, with mean ≈ -1.207 and standard deviation ≈ 1.268. It is the *Tracy-Widom Type 1 distribution*, discovered by Tracy and Widom (1996) as the top-eigenvalue limit of the GOE; Wolchover (2014) is an accessible non-technical introduction.

The technical definition of the Tracy-Widom Type 1 distribution is provided next.[⋆22]

Definition 3.9 (Tracy-Widom Type 1 distribution)**.** The *Tracy-Widom Type 1* law TW_1 is the probability distribution on $\mathbb{R}$ with cumulative distribution function

$$F_1(s) \;=\; \exp\!\left(-\frac{1}{2}\int_s^{\infty}\Big[q(x) \;+\; (x-s)\,q(x)^2\Big]\,dx\right), \qquad s \in \mathbb{R}, \tag{21}$$

where q is the unique *Hastings-McLeod solution* (Hastings and McLeod, 1980) of the Painlevé II equation

$$q''(x) \;=\; x\,q(x) \;+\; 2\,q(x)^3, \qquad q(x) \;\sim\; \mathrm{Ai}(x) \;\text{ as } x \to +\infty, \tag{22}$$

with Ai the Airy function.

TW_1 has no closed-form density: the CDF (21) is defined indirectly through the Painlevé II ODE, and the Hastings-McLeod boundary condition pins down the unique solution that decays correctly at $+\infty$. The same distribution arises in problems far from random matrices, the longest increasing subsequence of a random permutation (Baik et al., 1999) being the canonical example. That one law recurs across so many unrelated models is an instance of *universality*.[⋆23] We use TW_1 here only through its name and its empirical density, plotted in Figure 15.

Heuristic for the $p^{-2/3}$ edge scale. A short calculation using only the square-root edge of the MP density makes this $p^{-2/3}$ scaling plausible.[⋆24] Near the upper edge,

$$f_{\mathrm{MP}_y}(\lambda) \;\sim\; C\,(\lambda_+ - \lambda)^{1/2} \qquad (\lambda \uparrow \lambda_+).$$

The largest of p iid draws from this density sits, heuristically, where the mass beyond it equals $1/p$:

$$\frac{1}{p} \;\sim\; \int_{\lambda_{\max}}^{\lambda_+} f_{\mathrm{MP}_y}(\lambda)\,d\lambda \;\sim\; \tfrac{2C}{3}\,(\lambda_+ - \lambda_{\max})^{3/2},$$

so $\lambda_+ - \lambda_{\max} \sim p^{-2/3}$. The exponent $3/2$ in the integrated square-root density inverts to the exponent $2/3$ in the fluctuation scale.

The Johnstone (2001) edge theorem. The heuristic above predicts the right *scale* of the fluctuations, but not their precise centering. To compare $\lambda_{\max}$ to a definite distribution we need explicit finite-sample constants μ_{np} (which pins the centering to the edge) and σ_{np} (which sets the $p^{-2/3}$ scale). Johnstone (2001) supplied both, in the work that, alongside that of Johansson (2000), gave the first proof that the largest eigenvalue of a null sample covariance matrix has a Tracy-Widom limit.[⋆25]

Theorem 3.8 (Tracy-Widom edge for the null sample covariance, Johnstone, 2001)**.** Under the hypotheses of Theorem 3.6, define

$$\mu_{np} = \frac{(\sqrt{n-1}+\sqrt{p})^2}{n}, \qquad \sigma_{np} = \frac{\sqrt{n-1}+\sqrt{p}}{n}\left(\frac{1}{\sqrt{n-1}}+\frac{1}{\sqrt{p}}\right)^{1/3}. \tag{23}$$

Then

$$\frac{\lambda_{\max}(\widehat{\Sigma}_n)-\mu_{np}}{\sigma_{np}} \xrightarrow{d} \mathrm{TW}_1,$$

where TW_1 is the distribution of Definition 3.9.

The constants line up with the heuristic: $\mu_{np} \to (1+\sqrt{y})^2 = \lambda_+$, so the centering picks out the MP upper edge in the limit, and σ_{np} is of order $n^{-2/3}$, so the scaling absorbs the $p^{-2/3}$ edge scale derived just above. The same theorem can be written with the rate factored out, as in Genzer (2025, Prop. 3.9): $n^{2/3}(\lambda_{\max}(\widehat{\Sigma}_n)-\mu_{np})/\tilde{\sigma} \xrightarrow{d} \mathrm{TW}_1$ with $\tilde{\sigma} \to (1+\sqrt{y})(1+1/\sqrt{y})^{1/3}$; algebraically the two forms are identical. We use Theorem 3.8 as a black box here; the same constants will be reused in Section 4 when we ask which population eigenvalues exceed this null TW_1 baseline.

Simulation. Write $z = (\lambda_{\max}(\widehat{\Sigma}_n)-\mu_{np})/\sigma_{np}$ for the Johnstone-standardized top eigenvalue, so that $z \xrightarrow{d} \mathrm{TW}_1$ by Theorem 3.8. Figure 14 shows the empirical distribution of z collapsing to a common shape as p, n grow at fixed aspect ratio. Figure 15 fixes a single large (n,p) and compares the histogram of z directly to the TW_1 density, with a Q-Q plot of z against $\mathcal{N}(0,1)$ on the side.

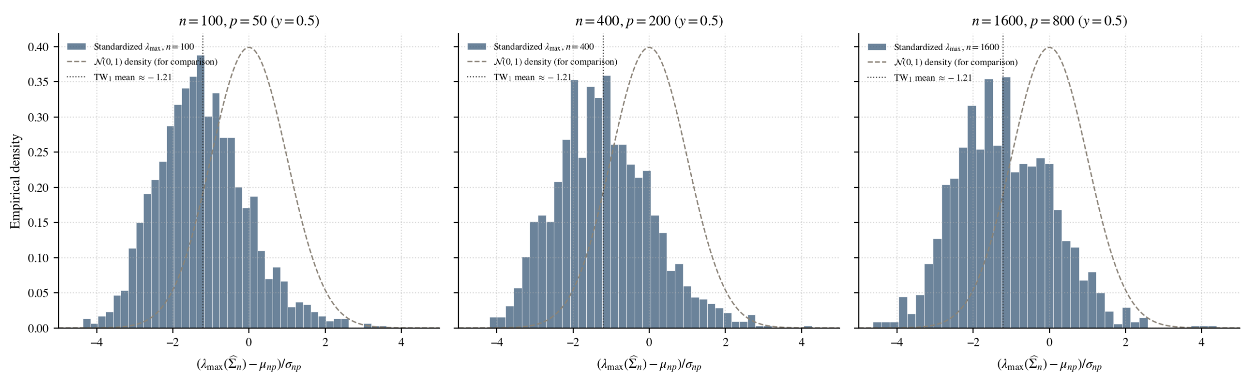

Figure 14: Histogram of $z = (\lambda_{\max}(\widehat{\Sigma}_n)-\mu_{np})/\sigma_{np}$ at three growing (n,p) pairs with fixed aspect ratio $y = 1/2$; $R = 1500$ Monte Carlo trials per panel. The standard normal density (dashed) and the TW_1 mean (≈ -1.21, dotted) are shown for reference. Generated by `notebooks/03_section3_demos.py`.

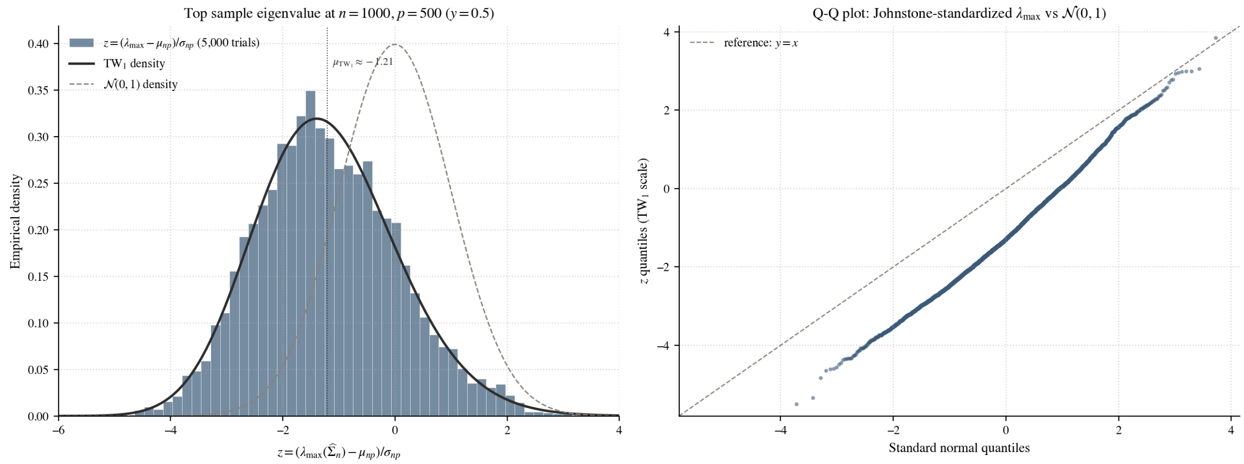

Figure 15: Direct comparison of the empirical $z = (\lambda_{\max} - \mu_{np})/\sigma_{np}$ at $n = 1000$, $p = 500$ ($y = 1/2$, $R = 5000$ trials) with the limiting Tracy-Widom distribution. *Left:* histogram of z against the theoretical TW_1 density (solid) and the standard normal density (dashed); the TW_1 bulk is centred on its mean ≈ -1.21, well to the left of the Gaussian. *Right:* Q-Q plot of z against $\mathcal{N}(0,1)$ quantiles. The curve sits below the $y = x$ reference at the bulk (TW_1's negative mean) and crosses above it at the upper extreme (TW_1's heavier right tail). Generated by `notebooks/03_section3_demos.py`.

Quantitative tests of normality. The visual non-normality in Figures 14 and 15 can be backed up by classical goodness-of-fit tests. Two natural choices are the *Kolmogorov-Smirnov* test, with statistic

$$D_R \;=\; \sup_x |F_R(x) - F(x)|$$

measuring the largest gap between the empirical CDF F_R of an R-sample and a reference CDF F, and the *Shapiro-Wilk* test, whose null hypothesis is that the sample comes from *some* normal distribution, without specifying its mean or variance. Applied to the $R = 5000$ standardized samples z from Figure 15 (further centred to mean 0 and scaled to variance 1 before the KS comparison against $\mathcal{N}(0,1)$, so that the test is sensitive to shape rather than location), both reject normality decisively:

$$D_{5000} = 0.029, \quad p = 3.5 \times 10^{-4} \qquad \text{(Kolmogorov-Smirnov)},$$

$$W = 0.995, \quad p = 5.1 \times 10^{-12} \qquad \text{(Shapiro-Wilk)}.$$

The Shapiro-Wilk statistic W is close to 1 (the value attained by a perfect normal sample), but at $R = 5000$ trials even a small departure shows up at machine-precision p-values; the KS statistic is similarly tiny in absolute terms yet far outside the null distribution. For comparison, a control sample of 5000 genuine $\mathcal{N}(0,1)$ draws yields $D = 0.017$ ($p \approx 0.12$) and $W = 0.9997$ ($p \approx 0.69$), neither of which rejects normality. The TW_1-distributed top eigenvalue is therefore non-Gaussian by every standard diagnostic, visual and quantitative.

The contrast between the two regimes is sharp. In fixed dimension, the sample covariance matrix behaves like an ordinary average: its entries converge to those of Σ, and at a simple population eigenvalue the corresponding sample eigenvalue fluctuates Gaussianly on the $\sqrt{n}$ scale via Lemma 3.5. At a repeated population eigenvalue the limiting object is instead a finite-dimensional GOE matrix, with the non-Gaussian top-eigenvalue law that reflects eigenvalue repulsion. In high dimensions, by contrast, the sample eigenvalues no longer collapse to the population eigenvalues. Even for $\Sigma = I_p$, they fill the Marchenko-Pastur interval $[\lambda_-, \lambda_+]$, and the largest eigenvalue sits at the right edge λ_+ with Tracy-Widom TW_1 fluctuations on the $p^{-2/3}$ scale.

The next section inserts signal into this null picture by allowing a small number of population eigenvalues to rise above the unit background, and asks when the corresponding sample eigenvalue and eigenvector can recover that signal.

4 Spiked Covariance Matrices and the Spike Phase Transition

The Marchenko-Pastur and Tracy-Widom results of Section 3.3 describe the sample spectrum under the null hypothesis $\Sigma = I_p$. Real datasets carry structure: a small number of distinguished directions account for most of the variance (regulatory pathways in single-cell genomics, market factors in finance, edge orientations in image processing). The simplest model that captures this picture, introduced by Johnstone (2001) and developed systematically by Yao et al. (2015, Ch. 11), allows a small number of population eigenvalues to exceed the noise baseline of 1. The questions this section asks are when such a *spike* produces a detectable outlier in the sample spectrum (Section 4.1–4.2), what the corresponding fluctuations look like (Section 4.3), and what the sample eigenvector reveals about the population spike direction (Section 4.4). The answer is a clean phase transition with an explicit threshold determined by the aspect ratio y (Baik et al., 2005; Baik and Silverstein, 2006).

4.1 Johnstone's Spiked Model and the Spike-Location Map

We fix the model first.

Definition 4.1 (Johnstone's spiked covariance model)**.** Fix an integer $m \geq 0$ (the number of *spikes*) and real numbers

$$\alpha_1 \;\geq\; \alpha_2 \;\geq\; \cdots \;\geq\; \alpha_m \;>\; 0, \qquad \alpha_j \neq 1.$$

The *Johnstone spiked covariance* of dimension $p \geq m+1$ is

$$\Sigma \;=\; \operatorname{diag}(\alpha_1, \ldots, \alpha_m, \underbrace{1, \ldots, 1}_{p-m \text{ times}}) \;\in\; \mathbb{R}^{p\times p}. \tag{24}$$

Throughout this section, $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ are iid $\mathcal{N}(\mathbf{0}, \Sigma)$ and $\widehat{\Sigma}_n = (1/n)\sum_i \boldsymbol{X}_i \boldsymbol{X}_i^\top$ is the sample covariance, with $p, n \to \infty$, $p/n \to y \in (0,1)$, and m fixed.

This is the simplest case of the *generalised spiked population model* of Yao et al. (2015, § 11.2), in which the population and sample covariances inherit the block structure

$$\Sigma \;=\; \begin{pmatrix} \Lambda & 0 \\ 0 & \boldsymbol{V}_p \end{pmatrix}, \qquad \widehat{\Sigma}_n \;=\; \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix}.$$

The fixed-size block $\Lambda = \operatorname{diag}(\alpha_1, \ldots, \alpha_m)$ carries the spikes; the $(p-m)\times(p-m)$ baseline block $\boldsymbol{V}_p$ has empirical spectral distribution converging to a limit H, and an eigenvalue of Λ is a *spike* when it lies outside the support of H (Yao et al., 2015, Def. 11.2). The sample blocks S_{ij} arise from the partition $\boldsymbol{X}_i = \Sigma^{1/2}\boldsymbol{Y}_i$, $\boldsymbol{Y}_i \sim \mathcal{N}(\mathbf{0}, I_p)$ (Yao et al., 2015, Eqs. (11.6)–(11.8)). Johnstone's model (§ 11.2.1) is the specialisation $\boldsymbol{V}_p = I_{p-m}$, so $H = \delta_1$ and "spike" reduces to $\alpha_j \neq 1$. Spikes split into *large* ($\alpha_j > 1$) and *small* ($\alpha_j < 1$); we focus on the large case throughout.

The structural question is whether a large spike produces an outlier in the sample spectrum. Figure 16 shows the answer empirically: at $y = 1/2$ and $n = 1000$, a spike $\alpha = 2.5$ lifts

$\lambda_{\max}(\widehat{\Sigma}_n)$ to ≈ 3.42, well above the MP edge $\lambda_+ \approx 2.91$; a spike $\alpha = 1.4$ leaves $\lambda_{\max}$ at ≈ 2.90, indistinguishable from the edge. Both spikes exceed 1, yet only one is detectable. The dividing threshold depends on y in a precise way.

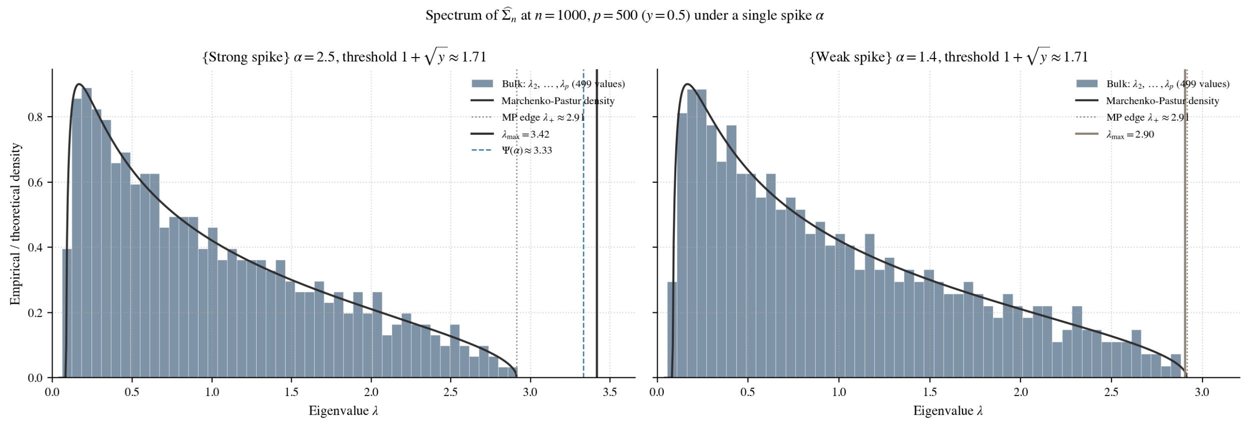

Figure 16: Empirical spectra of $\widehat{\Sigma}_n$ at $n = 1000$, $p = 500$ ($y = 1/2$) under a single spike. Histograms show $\lambda_2, \dots, \lambda_p$ with the MP density (solid) and edge $\lambda_+ \approx 2.91$ (dotted); the coloured vertical line marks $\lambda_{\max}$. *Left:* $\alpha = 2.5$ produces an outlier near $\Psi(\alpha) \approx 3.33$ (dashed; Definition 4.2). *Right:* $\alpha = 1.4$ is absorbed at λ_+. Generated by `notebooks/04_section4_demos.py`.

The threshold is encoded in a single algebraic object: the value at which a separated sample eigenvalue would have to sit so that the perturbed bulk still matches the MP shape. The derivation passes through the Stieltjes transform and is the $H = \delta_1$ specialisation of Yao et al. (2015, Eq. (11.15)). We take it as given.

Definition 4.2 (Spike-location map)**.** For $\alpha > 1$ and aspect ratio $y \in (0, 1)$,

$$\Psi(\alpha) \;=\; \alpha \;+\; \frac{y\,\alpha}{\alpha - 1}. \tag{25}$$

The whole phase transition is encoded in the shape of Ψ alone.

Proposition 4.1 (Properties of the spike-location map)**.** For each fixed $y \in (0, 1)$, the map $\Psi : (1, \infty) \to \mathbb{R}$ defined by (25) satisfies:

(1) $\Psi'(\alpha) \;=\; 1 \;-\; \dfrac{y}{(\alpha - 1)^2}$.

(2) Ψ has a unique minimum on $(1, \infty)$ at $\alpha = 1 + \sqrt{y}$. On $(1, 1 + \sqrt{y})$ the map is strictly decreasing; on $(1 + \sqrt{y}, \infty)$ it is strictly increasing.

(3) The minimum value is exactly the Marchenko-Pastur upper edge:

$$\Psi(1 + \sqrt{y}) \;=\; (1 + \sqrt{y})^2 \;=\; \lambda_+.$$

Proof. For (1), differentiate:

$$\Psi'(\alpha) \;=\; 1 \;+\; y \cdot \frac{(\alpha - 1) - \alpha}{(\alpha - 1)^2} \;=\; 1 \;-\; \frac{y}{(\alpha - 1)^2}.$$

For (2), $\Psi'(\alpha) = 0$ gives $(\alpha-1)^2 = y$, so $\alpha = 1 \pm \sqrt{y}$; on $(1, \infty)$ the relevant root is $\alpha = 1+\sqrt{y}$. For $1 < \alpha < 1 + \sqrt{y}$, $(\alpha - 1)^2 < y$ and $\Psi' < 0$; for $\alpha > 1 + \sqrt{y}$, $\Psi' > 0$. For (3), substituting $\alpha = 1 + \sqrt{y}$ into (25) and using $y/\sqrt{y} = \sqrt{y}$,

$$\Psi(1 + \sqrt{y}) \;=\; (1 + \sqrt{y}) \;+\; \sqrt{y}\,(1 + \sqrt{y}) \;=\; (1 + \sqrt{y})^2 \;=\; \lambda_+.$$

□

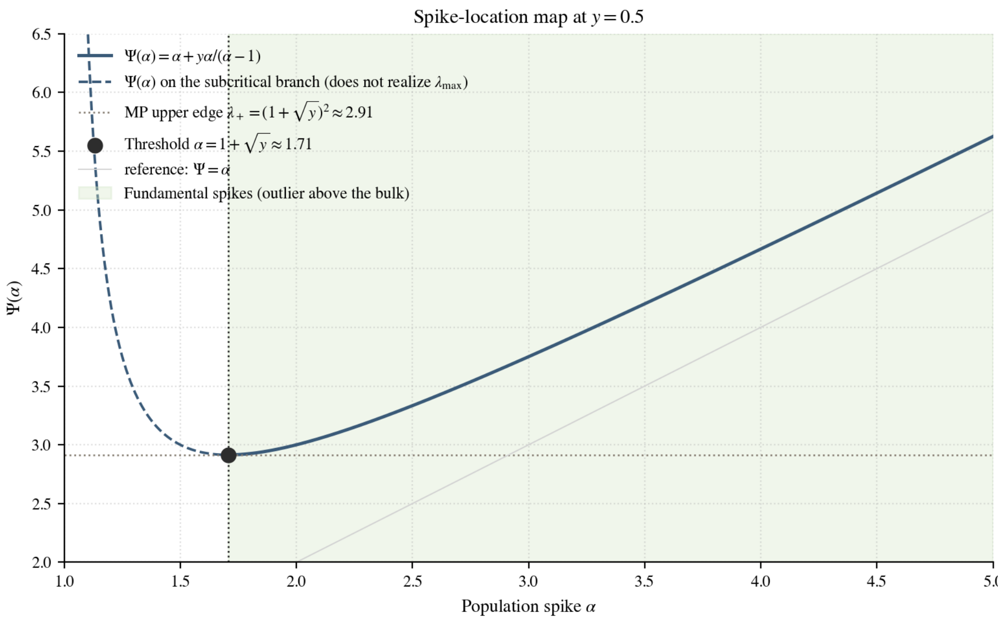

Figure 17: The spike-location map Ψ on $(1, \infty)$ at $y = 1/2$. The U-shape bottoms out at the threshold $\alpha = 1 + \sqrt{y} \approx 1.71$, where Ψ touches the MP edge $\lambda_+ \approx 2.91$. The shaded region marks the fundamental spikes (where $\Psi(\alpha)$ predicts $\lambda_{\max}$); on the non-fundamental branch (dashed) the prediction is not realised. Generated by `notebooks/04_section4_demos.py`.

Following Yao et al. (2015, Thm. 11.3), we call a large spike *fundamental* if $\Psi'(\alpha) > 0$ (equivalently $\alpha > 1+\sqrt{y}$) and *non-fundamental* otherwise. The two regimes correspond exactly to the two panels of Figure 16: the fundamental spike $\alpha = 2.5$ lifts $\lambda_{\max}$ to $\Psi(2.5) \approx 3.33$; the non-fundamental spike $\alpha = 1.4$ lands at λ_+. Section 4.2 makes the convergence precise.

4.2 The Spike Phase Transition for Eigenvalues

With Ψ in hand the phase transition has a clean statement.

Theorem 4.2 (Spike phase transition, large case)**.** Under the assumptions of Definition 4.1, fix a large population spike $\alpha_j > 1$ of multiplicity m_j, and let J_j index the m_j sample eigenvalues associated with it. As $p, n \to \infty$ with $p/n \to y \in (0, 1)$:

(1) if $\alpha_j > 1 + \sqrt{y}$ (fundamental),

$$\lambda_i(\widehat{\Sigma}_n) \xrightarrow{\text{a.s.}} \Psi(\alpha_j) \;=\; \alpha_j \;+\; \frac{y\alpha_j}{\alpha_j - 1}, \qquad i \in J_j;$$

(2) if $1 < \alpha_j \leq 1 + \sqrt{y}$ (non-fundamental),

$$\lambda_i(\widehat{\Sigma}_n) \xrightarrow{\text{a.s.}} \lambda_+ \;=\; (1+\sqrt{y})^2, \qquad i \in J_j.$$

This is the Johnstone specialisation of Yao et al. (2015, Cor. 11.4), itself a consequence of the general phase-transition theorem (Yao et al., 2015, Thm. 11.3); the original results for sample covariance matrices are due to Baik et al. (2005); Baik and Silverstein (2006). The proof passes through the Stieltjes transform of the unspiked bulk: an outlier sample eigenvalue is the unique solution λ of $s_F(\lambda) = -1/\alpha_j$ outside the support of F_y, and substituting the explicit Stieltjes transform of the Marchenko-Pastur law recovers $\lambda = \Psi(\alpha_j)$. When $\Psi'(\alpha_j) > 0$ (i.e. $\alpha_j > 1 + \sqrt{y}$, by Proposition 4.1) such a λ exists outside the bulk and case (1) applies; otherwise the equation has no outside-the-bulk solution and the associated sample eigenvalue snaps to the upper edge λ_+, giving case (2). We use Theorem 4.2 as a black box.

Example 4.3 (Numerical predictions at $y = 1/2$)**.** With $y = 1/2$, the threshold is $1+\sqrt{1/2} \approx 1.71$ and the upper edge is $\lambda_+ \approx 2.91$. For $\alpha = 2.5$ (fundamental),

$$\Psi(2.5) \;=\; 2.5 + \frac{(0.5)(2.5)}{2.5-1} \;=\; 2.5 + \tfrac{5}{6} \;\approx\; 3.33.$$

For $\alpha = 1.4 < 1.71$ (non-fundamental), Theorem 4.2(2) gives the limit $\lambda_+ \approx 2.91$ directly, regardless of the precise value of α as long as it stays below the threshold. The two panels of Figure 16 already showed the top sample eigenvalue near these limits at a single (n, p); Figure 18 now traces the convergence as n grows.

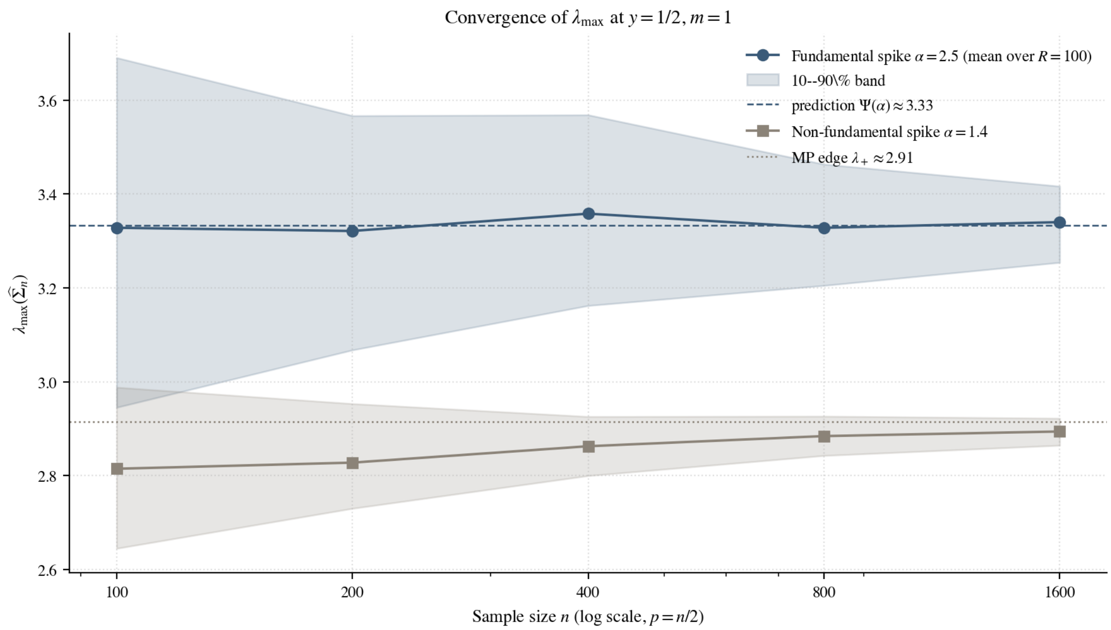

Figure 18: Smoothed convergence of $\lambda_{\max}(\widehat{\Sigma}_n)$ for a single spike ($m = 1$) at $y = 1/2$, across $n \in \{100, 200, 400, 800, 1600\}$ with $R = 100$ replications per setting. Solid curves are sample means; shaded bands are 10–90% empirical quantiles. The fundamental $\alpha = 2.5$ curve settles toward $\Psi(\alpha) \approx 3.33$ (dashed); the non-fundamental $\alpha = 1.4$ curve settles toward the bulk edge $\lambda_+ \approx 2.91$ (dotted). Generated by `notebooks/04_section4_demos.py`.

Both curves track Theorem 4.2 cleanly, and the convergence becomes visible by $n = 800$; the residual width of the bands shrinks at the slow $O(n^{-1/2})$ rate that Section 4.3 will quantify. When the population covariance carries more than one large spike the theorem applies to each α_j separately, with the associated sample eigenvalues converging independently to either $\Psi(\alpha_j)$ or λ_+ according to the fundamental/non-fundamental dichotomy; a simulation with one strong and one weak spike, for instance, produces a top sample eigenvalue near $\Psi(\alpha_1)$ and a second-largest eigenvalue near λ_+. Theorem 4.2 is, however, only an almost-sure statement; it does not say how $\lambda_{\max}$ fluctuates around its limit. Section 4.3 addresses that.

4.3 Fluctuations of Spiked Eigenvalues

Theorem 4.2 pins down the almost-sure limits but says nothing about how the corresponding sample eigenvalue fluctuates around them. The two regimes turn out to inherit qualitatively different limit laws and qualitatively different scales: on the fundamental side, the fluctuations are Gaussian on the $\sqrt{n}$ scale; on the non-fundamental side, they are Tracy-Widom on the much finer edge scale of Section 3.3.

Theorem 4.3 (Gaussian fluctuations for a simple fundamental spike)**.** Under the assumptions of Definition 4.1, let $\alpha > 1 + \sqrt{y}$ be a fundamental spike of multiplicity $m_j = 1$, and let

$\lambda_i(\widehat{\Sigma}_n)$ be the associated sample eigenvalue. Then

$$\sqrt{n}\,\big(\lambda_i(\widehat{\Sigma}_n) - \Psi(\alpha)\big) \xrightarrow{d} \mathcal{N}(0,\, \sigma_\alpha^2), \qquad \sigma_\alpha^2 \;=\; 2\,\alpha^2\,\Psi'(\alpha) \;=\; \frac{2\alpha^2\big[(\alpha-1)^2 - y\big]}{(\alpha-1)^2}. \tag{26}$$

This is the Johnstone simple-spike specialisation of Yao et al. (2015, Thm. 11.11 and Eq. (11.36)); the explicit variance σ_α^2 uses the fact that we are in the real Gaussian case, so the fourth-cumulant correction $\beta_y = \mathbb{E}|y_{11}|^4 - 3$ appearing in the general statement of Theorem 11.11 vanishes. Féral and Péché (2009, Thm. 1.6) give a clean direct proof for diagonal spiked Gaussian models. The heuristic mirrors Section 3.2: a separated simple sample eigenvalue is a smooth functional of $\widehat{\Sigma}_n$, and a delta-method argument lifts the entrywise $\sqrt{n}$-CLT for $\widehat{\Sigma}_n$ to a $\sqrt{n}$-Gaussian limit for the eigenvalue. The Gaussian limit fails once the spike has multiplicity $m_j \geq 2$, in which case the limiting joint distribution of the corresponding sample eigenvalues is a finite-dimensional GOE-type object (Yao et al., 2015, Eq. (11.35)); we do not treat that case.

Theorem 4.4 (Edge fluctuations for a non-fundamental spike)**.** Under the assumptions of Definition 4.1, let $1 < \alpha < 1 + \sqrt{y}$ be a strictly non-fundamental spike. With the Johnstone constants μ_{np} and σ_{np} of Theorem 3.8,

$$\frac{\lambda_{\max}(\widehat{\Sigma}_n) - \mu_{np}}{\sigma_{np}} \xrightarrow{d} \mathrm{TW}_1. \tag{27}$$

The result is implicit in the spectral universality at the edge proved by Baik and Silverstein (2006) and is the natural analogue of the null Johnstone edge fluctuation (Johnstone, 2001): a subcritical spike does not separate, the corresponding sample eigenvalue is pinned to the bulk's right edge, and the Tracy-Widom law that governs the null edge continues to govern this near-critical edge. The critical case $\alpha = 1 + \sqrt{y}$ is genuinely different: the limit is neither Gaussian nor TW_1 but an interpolating crossover law studied by Baik et al. (2005), which we examine empirically at the end of this subsection.

Example 4.4 (Numerical predictions at $y = 1/2$)**.** At $y = 1/2$, $\Psi'(2.5) = 1 - 0.5/(1.5)^2 \approx 0.778$, so Theorem 4.3 predicts

$$\sigma_\alpha^2 \;=\; 2 \cdot (2.5)^2 \cdot 0.778 \;\approx\; 9.72,$$

i.e. $\lambda_{\max} - \Psi(2.5)$ should look like $\mathcal{N}(0,\, 9.72/n)$ at large n. For the non-fundamental spike $\alpha = 1.4$, Theorem 4.4 predicts the TW_1 shape after Johnstone standardisation; the centring and scaling constants μ_{np}, σ_{np} are the same as in Theorem 3.8. At $n = 800$, $p = 400$, $\mu_{np} \approx 2.91$, only slightly below the asymptotic edge $\lambda_+ \approx 2.91$.

The visual contrast between the two regimes is already clear at modest n. Figure 19 overlays the empirical distributions of $\lambda_{\max}(\widehat{\Sigma}_n)$ at $n = 800$, $R = 1000$, under the two spike values of Example 4.4.

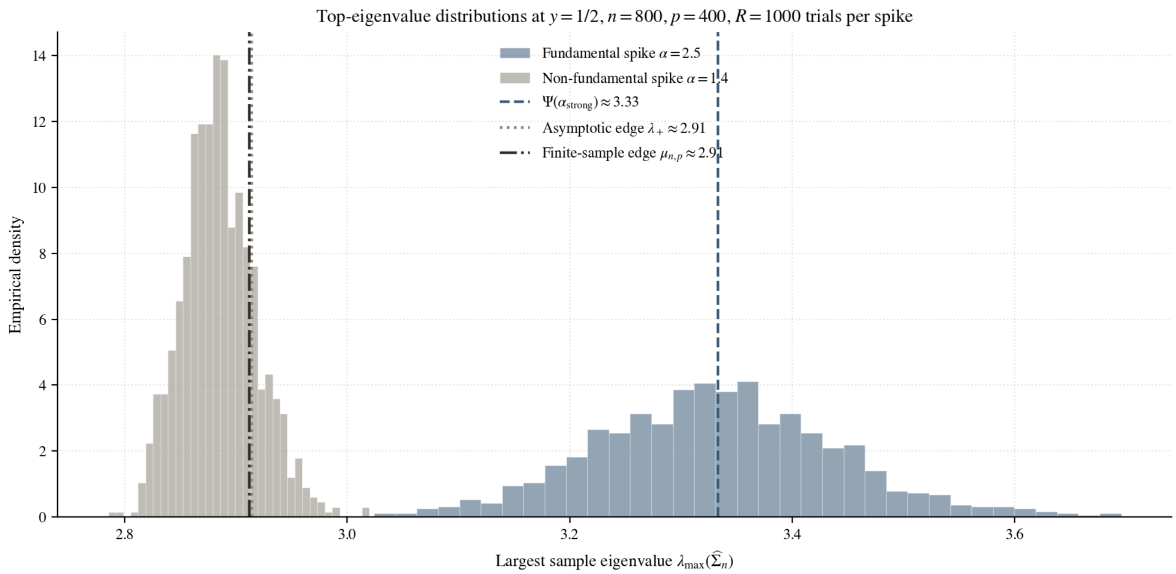

Figure 19: Empirical distributions of $\lambda_{\max}(\widehat{\Sigma}_n)$ at $y = 1/2$, $n = 800$, $p = 400$, with $R = 1000$ replications per spike. The fundamental-spike histogram ($\alpha = 2.5$) sits to the right and is centred at $\Psi(\alpha_{\text{strong}}) \approx 3.33$ (dashed); the non-fundamental histogram ($\alpha = 1.4$) sits to the left, sharply concentrated and slightly to the left of the asymptotic edge $\lambda_+ \approx 2.91$ (dotted), in line with the finite-sample Johnstone centring $\mu_{n,p} \approx 2.91$ (dash-dot). Generated by `notebooks/04_section4_demos.py`.

Three features of Figure 19 match the theory cleanly. First, the two histograms are disjoint: the fundamental and non-fundamental cases place $\lambda_{\max}(\widehat{\Sigma}_n)$ in qualitatively different regions of the spectrum. Second, the fundamental-spike histogram is roughly symmetric and visibly wider, consistent with the Gaussian variance $\sigma^2_{2.5}/n \approx 9.72/800 \approx 0.012$ of Example 4.4. Third, the non-fundamental histogram is asymmetric, much tighter, and sits slightly to the *left* of the asymptotic edge λ_+; this is not a contradiction with Theorem 4.2 (which guarantees only almost-sure convergence as $n \to \infty$), but exactly the finite-n signature of the Tracy-Widom limit, which is centred at the finite-sample edge $\mu_{n,p}$ and has a negative mean of ≈ -1.21 after Johnstone standardisation.

To verify the shapes themselves we move to the standardised statistics suggested by Theorems 4.3 and 4.4. Figure 20 shows them at $n = 1500$, where the limiting laws emerge cleanly.

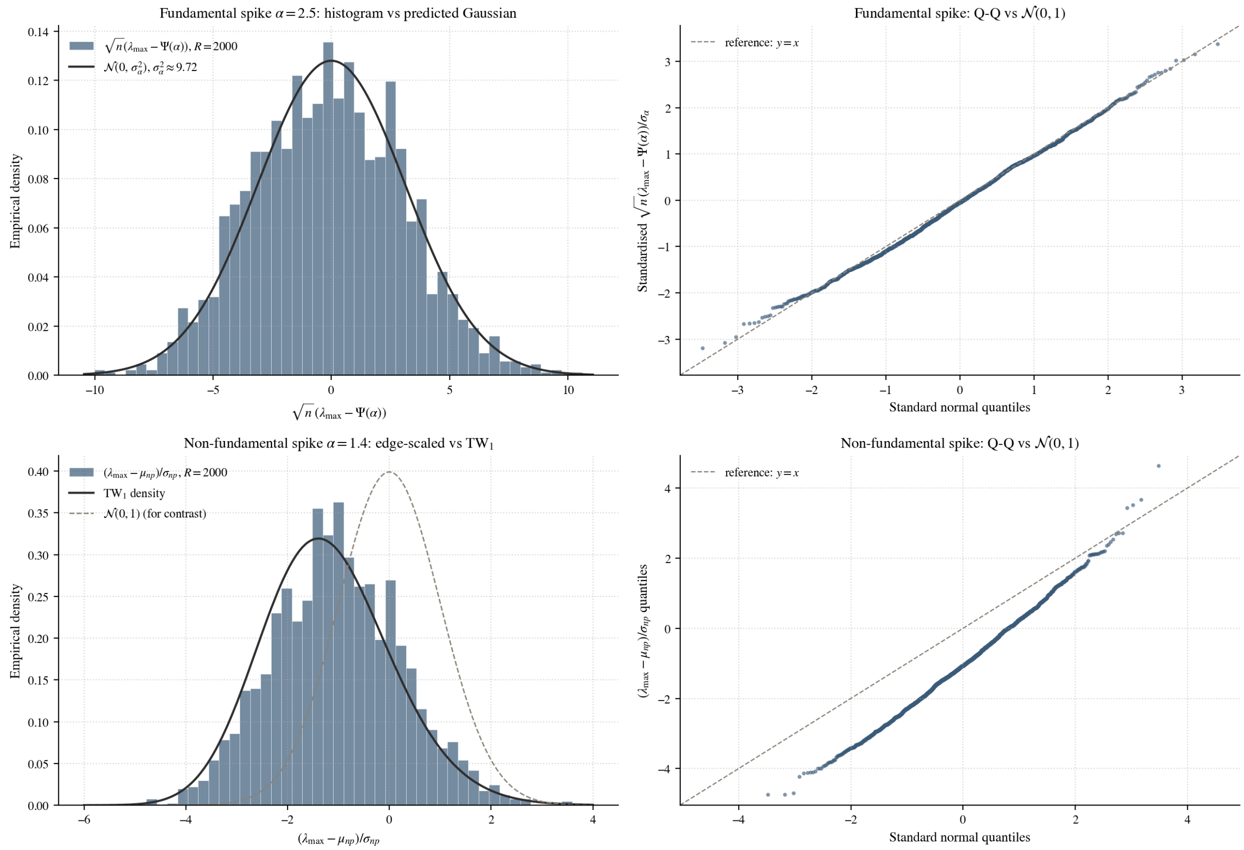

Figure 20: Standardised top-eigenvalue fluctuations at $n = 1500$, $p = 750$ ($y = 1/2$), with $R = 2000$ trials per regime. *Top row, fundamental spike* $\alpha = 2.5$: the histogram of $\sqrt{n}(\lambda_{\max} - \Psi(\alpha))$ tracks the predicted $\mathcal{N}(0,\, \sigma_\alpha^2)$ density of Theorem 4.3, and the Q-Q plot against $\mathcal{N}(0, 1)$ lies on the $y = x$ reference. *Bottom row, non-fundamental spike* $\alpha = 1.4$: the Johnstone-standardised statistic $(\lambda_{\max} - \mu_{np})/\sigma_{np}$ tracks the TW_1 density and lies clearly to the left of $\mathcal{N}(0, 1)$ (shown for contrast); the Q-Q plot against $\mathcal{N}(0, 1)$ bends in the characteristic TW_1 shape. Generated by `notebooks/04_section4_demos.py`.

The four panels of Figure 20 confirm the predictions quantitatively. On the fundamental side, the empirically standardised statistic has mean ≈ -0.04 and standard deviation ≈ 1.01; the Shapiro-Wilk statistic is $W = 0.998$ (its perfect-normal value is 1) and a small p-value of 0.026 reflects the very large sample size $R = 2000$ of the test rather than any meaningful departure from normality. On the non-fundamental side, the Johnstone-standardised statistic has mean ≈ -1.02 and standard deviation ≈ 1.27, close to the theoretical TW_1 values of -1.21 and 1.27; a Kolmogorov-Smirnov test against $\mathcal{N}(0, 1)$ returns $D = 0.377$ with $p < 10^{-250}$, confirming that the Johnstone-standardised statistic is decisively non-Gaussian and consistent instead with TW_1. The two regimes therefore inherit not only different almost-sure limits, as in Theorem 4.2, but different fluctuation laws on different scales: smooth $\sqrt{n}$-Gaussian fluctuations from a separated outlier, versus the $p^{-2/3}$-Tracy-Widom signature of an edge eigenvalue.

The critical window. Theorems 4.3 and 4.4 cover two fixed cases, one on each side of the threshold $\alpha_c = 1 + \sqrt{y}$. What happens right at the threshold is less clear-cut, and two facts about the map Ψ (Proposition 4.1) explain why. First, the prediction curve meets the edge *flatly*: $\Psi(\alpha_c) = \lambda_+$ and $\Psi'(\alpha_c) = 0$, so just above the threshold the outlier has barely pulled away from the edge. Second, the Gaussian spread $\sigma_\alpha^2 = 2\alpha^2\Psi'(\alpha)$ shrinks to zero as α drops toward α_c. So at any finite n the change cannot be sudden; it is spread over a range of α around α_c.

Figure 21 shows the full distribution of $\lambda_{\max}(\widehat{\Sigma}_n)$ at each spike value. Below the threshold the violins are small, sit on the edge line, and look nearly the same from one α to the next: a subcritical spike does not separate, so its exact value barely matters, and the slight differences that show are magnified by the zoomed-in vertical axis, which spans only a thin range of eigenvalues. Above the threshold the violins climb along $\Psi(\alpha)$, grow taller as the spread increases, and change from a lopsided edge shape into a symmetric, bell-shaped Gaussian. The change happens gradually, not as a jump.

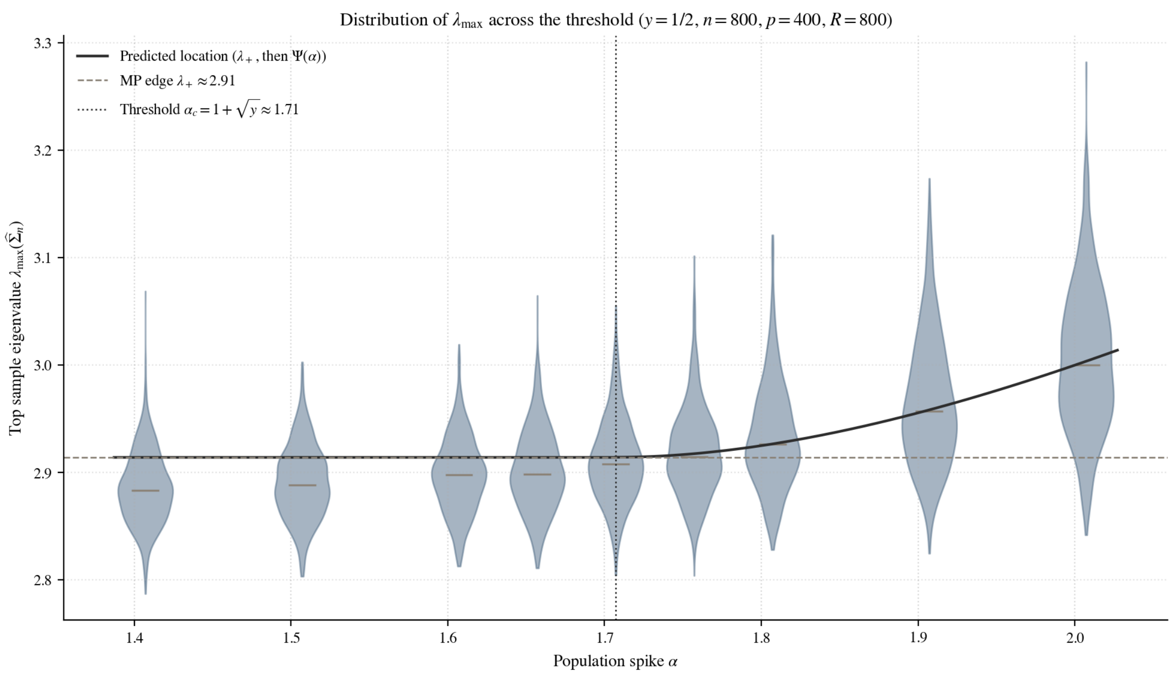

Figure 21: Distribution of the top sample eigenvalue $\lambda_{\max}(\widehat{\Sigma}_n)$ at each spike value α, at $y = 1/2$, $n = 800$, $p = 400$, with $R = 800$ replications (each violin is a smoothed histogram; the bar is the median). The solid curve is the prediction of Theorem 4.2: the edge λ_+ for $\alpha \le \alpha_c$, then $\Psi(\alpha)$. Below the threshold the violins are nearly identical and look slightly different only because the vertical axis is zoomed in; above it they climb, widen, and become symmetric. Generated by `notebooks/04_section4_demos.py`.

Figure 22 looks at this coincidence head-on, overlaying the raw distributions just below and just above α_c along with the one exactly at α_c. At the small distance $\delta = |\alpha - \alpha_c| = 0.05$ all three curves lie almost on top of one another: a spike just above the threshold looks the same as one just below. At the large distance $\delta = 0.30$ the two sides have pulled apart, the above-threshold curve shifted to the right while the below-threshold curve stays at the edge.

So near α_c the below- and above-threshold distributions are essentially the same, and they become clearly different only away from it.

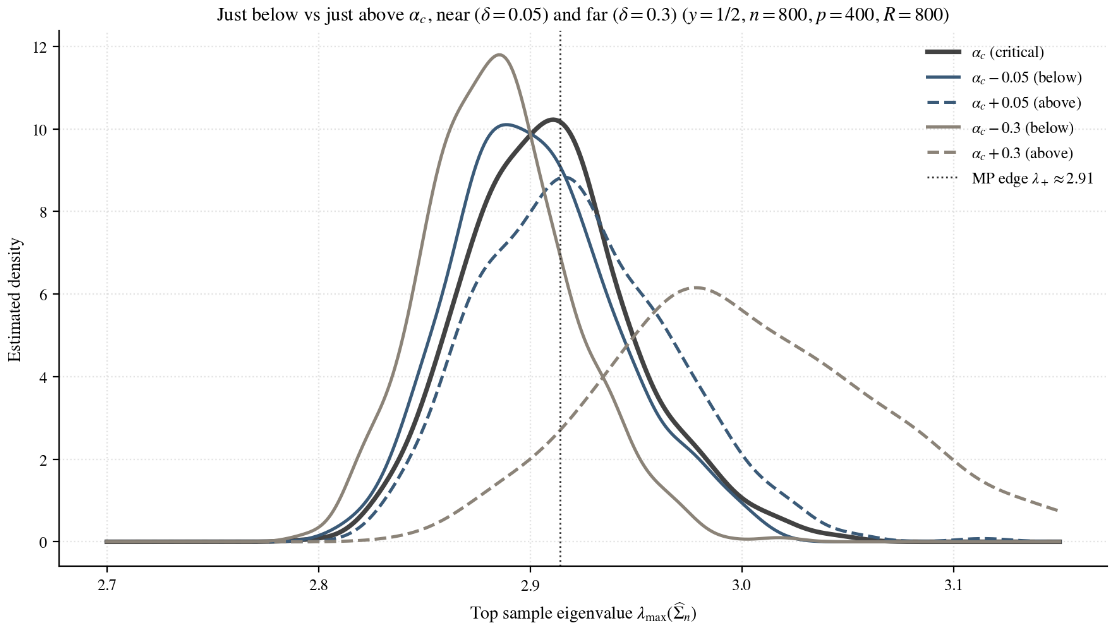

Figure 22: Raw distributions of $\lambda_{\max}(\widehat{\Sigma}_n)$ just below ($\alpha_c - \delta$, solid), just above ($\alpha_c + \delta$, dashed), and exactly at the threshold (α_c, thick), at $y = 1/2$, $n = 800$, $p = 400$, $R = 800$. At the small distance $\delta = 0.05$ all three curves nearly coincide; at the large distance $\delta = 0.30$ the above-threshold curve has shifted right while the below-threshold curve stays at the edge λ_+. The horizontal axis is the eigenvalue itself, with no rescaling. Generated by `notebooks/04_section4_demos.py`.

So the transition is not the Tracy-Widom shape below the threshold slowly bending into the Gaussian shape above it. Instead, close to α_c both sides look like a single in-between shape, and the two different behaviours appear only once the spike is well away from the threshold. That in-between shape is a single bump rather than two, and was studied by Baik et al. (2005); our simulations only illustrate it, and we do not work it out here.

The location result of Theorem 4.2 and the fluctuation result of Theorem 4.3 together tell us *where* a detectable spike lives in the sample spectrum and *how* it fluctuates there; they say nothing yet about whether the corresponding sample *eigenvector* carries information about the population spike direction. The next subsection takes up that question.

4.4 Eigenvector Alignment Beyond the Threshold

Theorem 4.2 tells us *that* a fundamental spike is visible in the sample spectrum, and Theorem 4.3 how its eigenvalue fluctuates. Neither says whether the corresponding sample *eigenvector* points anywhere near the true population spike direction. In the one-spike model that direction is $\boldsymbol{e}_1 = (1, 0, \ldots, 0)^\top$, the axis carrying the inflated variance α. Writing $\widehat{\boldsymbol{u}}_1$ for the unit eigenvector of the top sample eigenvalue, we measure how well it recovers $\boldsymbol{e}_1$ by the *squared alignment*

$$|\langle \widehat{\boldsymbol{u}}_1, \boldsymbol{e}_1 \rangle|^2 \;=\; \cos^2\theta \;\in\; [0, 1],$$

where θ is the angle between the two directions; squaring removes the harmless sign ambiguity in $\widehat{\boldsymbol{u}}_1$. A value near 1 means near-perfect recovery; a value near 0 means the sample eigenvector is essentially noise.

Yao et al. (2015) answer this by splitting the sample eigenvector into the coordinates that carry signal and those that carry only noise.

Theorem 4.5 (Signal-block length, Yao et al., 2015, Thm. 11.5)**.** Let α_j be a fundamental spike ($\Psi'(\alpha_j) > 0$), and let $\widehat{\boldsymbol{u}}$ be a unit sample eigenvector whose eigenvalue converges to $\Psi(\alpha_j)$. Split its coordinates as $\widehat{\boldsymbol{u}} = (\widehat{\boldsymbol{u}}_{(1)}, \widehat{\boldsymbol{u}}_{(2)})$, with $\widehat{\boldsymbol{u}}_{(1)}$ the m spike directions and $\widehat{\boldsymbol{u}}_{(2)}$ the $p-m$ baseline directions. Then $\widehat{\boldsymbol{u}}_{(1)}$ converges almost surely to a population spike eigenvector for α_j (the direction $\boldsymbol{e}_1$ when the spike is simple), and the two blocks have squared lengths

$$\|\widehat{\boldsymbol{u}}_{(1)}\|^2 \xrightarrow{\text{a.s.}} d^2 \;=\; \frac{\alpha_j\,\Psi'(\alpha_j)}{\Psi(\alpha_j)}, \qquad \|\widehat{\boldsymbol{u}}_{(2)}\|^2 \xrightarrow{\text{a.s.}} 1 - d^2. \tag{28}$$

In words: of the unit length of $\widehat{\boldsymbol{u}}$, a fraction d^2 lands on the true spike directions and the remaining $1-d^2$ leaks into noise. Specialising to Johnstone's model makes the fraction explicit.

Corollary 4.6 (Johnstone model, Yao et al., 2015, Cor. 11.6)**.** For $\Sigma = \mathrm{diag}(\alpha, 1, \ldots, 1)$ and a fundamental spike $\alpha > 1 + \sqrt{y}$,

$$d^2 \;=\; \frac{(\alpha-1)^2 - y}{(\alpha-1)(\alpha-1+y)}. \tag{29}$$

For a *simple* spike ($m = 1$) the signal block is the single coordinate $\boldsymbol{e}_1$, so $\|\widehat{\boldsymbol{u}}_{(1)}\|^2 = |\langle \widehat{\boldsymbol{u}}_1,\, \boldsymbol{e}_1\rangle|^2$ and (29) is a formula for the squared alignment itself. Calling it $\rho(\alpha, y)$,

$$|\langle \widehat{\boldsymbol{u}}_1,\, \boldsymbol{e}_1\rangle|^2 \xrightarrow{\text{a.s.}} \rho(\alpha,y) \;=\; \frac{1 - \dfrac{y}{(\alpha-1)^2}}{1 + \dfrac{y}{\alpha-1}} \quad (\alpha > 1+\sqrt{y}), \qquad \text{and} \xrightarrow{\text{a.s.}} 0 \quad (1 < \alpha \le 1+\sqrt{y}). \tag{30}$$

The two expressions (28) and (30) are the same quantity: with $\Psi(\alpha) = \alpha\big(1 + y/(\alpha-1)\big)$ and $\Psi'(\alpha) = 1 - y/(\alpha-1)^2$,

$$\frac{\alpha\,\Psi'(\alpha)}{\Psi(\alpha)} \;=\; \frac{1 - y/(\alpha-1)^2}{1 + y/(\alpha-1)} \;=\; \rho(\alpha, y).$$

This ties the two phase transitions together exactly: the alignment carries the same factor $\Psi'(\alpha)$ that sets the eigenvalue fluctuations in Theorem 4.3, so the eigenvalue and the eigenvector become informative at the very same threshold. At $\alpha = 1 + \sqrt{y}$ we have $\Psi'(\alpha) = 0$ and the alignment is 0; as $\alpha \to \infty$, $\rho \to 1$ and the recovery becomes perfect.

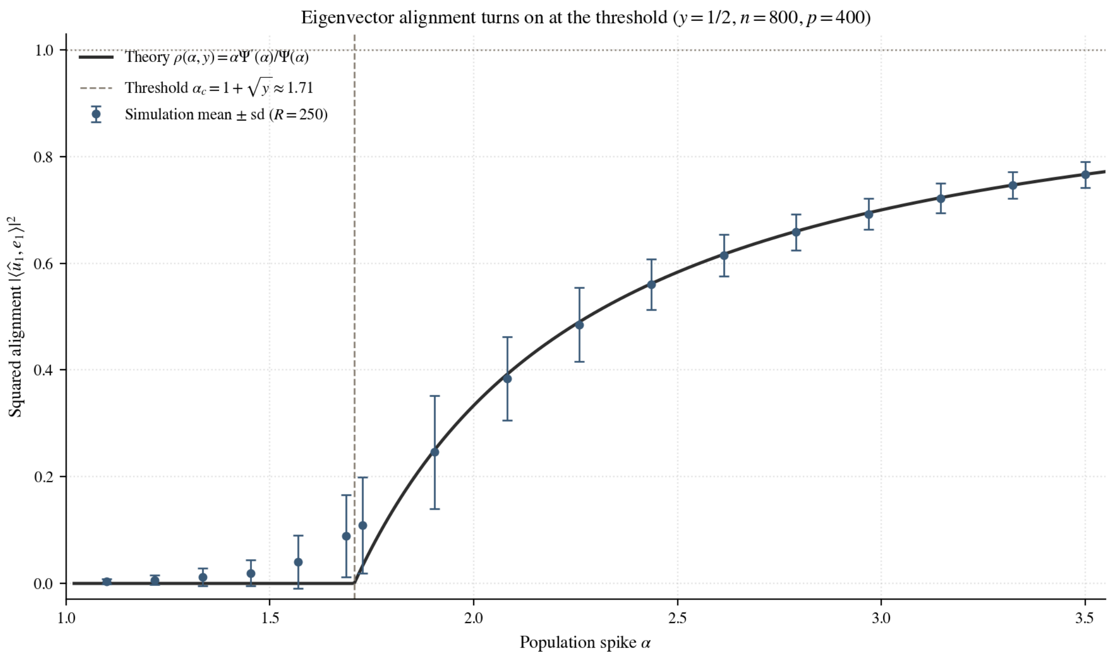

Figure 23: Squared alignment $|\langle \widehat{\boldsymbol{u}}_1, \boldsymbol{e}_1 \rangle|^2$ against the spike α at $y = 1/2$, $n = 800$, $p = 400$ ($R = 250$ replications per point). Below the threshold $\alpha_c \approx 1.71$ the top eigenvector is essentially noise (alignment ≈ 0); above it the alignment turns on and climbs toward 1, tracking the prediction $\rho(\alpha, y)$ of (30). Generated by `notebooks/04_section4_demos.py`.

Figure 23 confirms (30) at $y = 1/2$: the simulated alignment stays at zero until the threshold and then rises along $\rho(\alpha, y)$, reaching $\rho(2.5, 1/2) \approx 0.58$ at $\alpha = 2.5$ and approaching 1 for large spikes. Detecting a spike and recovering its direction thus share one threshold, but the eigenvector recovers the direction only partially unless the spike is well above it.

The aspect ratio y sets how far above is “well above”. Because the threshold $1 + \sqrt{y}$ grows with y, a higher-dimensional problem (larger $y = p/n$) needs a stronger spike to reach the same alignment. Figure 24 shows this directly.

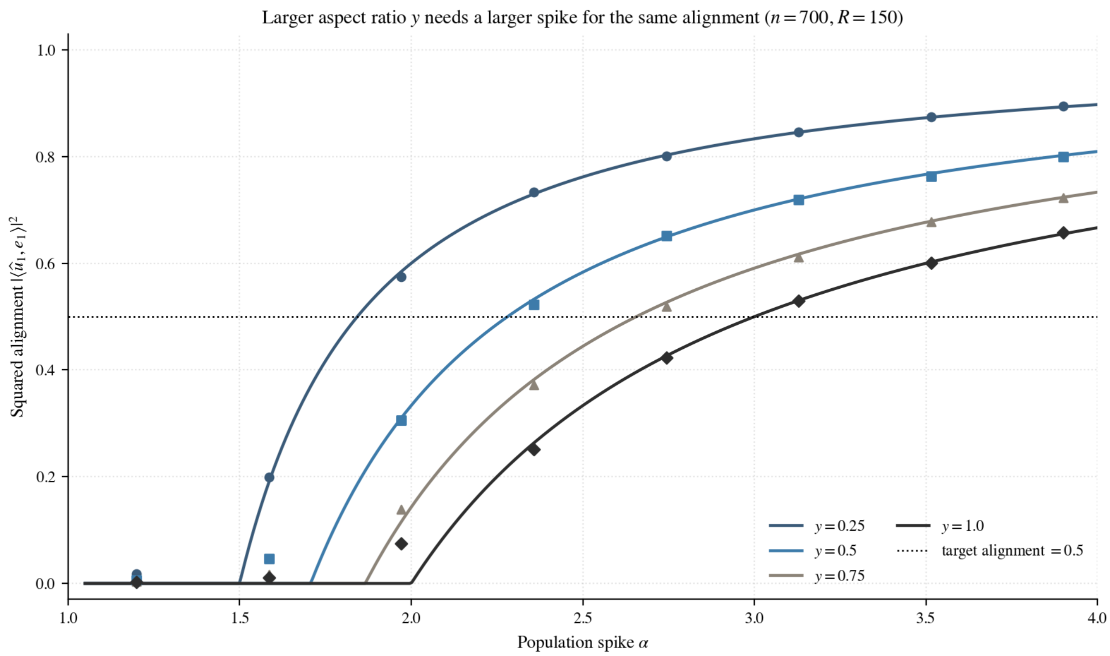

Figure 24: Squared alignment against α for four aspect ratios y (theory curves of (30), with simulation means at $n = 700$, $R = 150$ overlaid). As y grows the curve shifts to the right: to reach a fixed alignment, say 0.5 (dotted line), a larger y requires a larger α. Generated by `notebooks/04_section4_demos.py`.

Reading off the target line in Figure 24 answers the practical question: to hold the alignment at 0.5, a problem at $y = 1/4$ needs only a modest spike, whereas at $y = 1$ the required spike is substantially larger. In high dimensions a signal must be proportionally stronger to be not merely detectable but *recoverable*.

Sections 4.1–4.4 now give a complete account of a single spike: when it is detectable ($\alpha > 1 + \sqrt{y}$), where its eigenvalue settles ($\Psi(\alpha)$), how that eigenvalue fluctuates (Gaussian, on the $n^{-1/2}$ scale), and how well its eigenvector is recovered ($\rho(\alpha, y)$). The same signal-versus-noise threshold reappears, in a slightly different guise, when one moves from covariance matrices to the similarity matrices used for clustering, the subject of the final subsection.

4.5 Sketch of the Connection to Spectral Clustering

[TODO] *Content to fill in (from Gustavo's email + v5 + von Luxburg 2007): - Setup: data points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$, kernel similarity $K_{ij} = f(\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2/p)$. - Definition: degree matrix $D = \operatorname{diag}(K\mathbf{1})$ and normalized Laplacian-style kernel $L = nD^{-1/2}KD^{-1/2}$. - Heuristic decomposition: $L =$ noise bulk $+$ low-rank class signal, in direct analogy with $\widehat{\Sigma}_n =$ noise $+$ spike. - Conclusion: class structure is recoverable from L precisely when the signal-to-noise ratio crosses an analogue of the spike threshold; the informative eigenvectors of L then provide the clustering. Cite von Luxburg (2007) for the algorithmic side and Couillet and Benaych-Georges (2016) for the high-dimensional analysis. - Toy simulation: notebook 08, a two-class Gaussian mixture and the top non-trivial eigenvector of L.*

5 Conclusion and Outlook

[TODO] *Content to fill in: (a) Summary table: the strong vs. weak spike contrast, three rows: eigenvalue limit, eigenvector overlap, fluctuation scale.*

— Spike strength — Eigenvalue limit — Overlap with $\boldsymbol{e}_1$ — Fluctuation — —————— —-——————————-————————————————-—————————-— — Strong $\alpha > 1 + \sqrt{y}$ — $\Psi(\alpha)$ — $\rho(\alpha, y) > 0$ — Gaussian, $\sqrt{n}$ — — Weak $1 < \alpha \leq 1 + \sqrt{y}$ — $(1 + \sqrt{y})^2$ — 0 — Edge / Tracy-Widom, $p^{-2/3}$ —

(b) One-paragraph synthesis: the spike phase transition is the cleanest quantitative example of when high-dimensional inference recovers structure and when it does not. It explains, in a controlled toy model, why methods like PCA and spectral clustering succeed only when signal exceeds a threshold determined by the aspect ratio.

(c) Outlook: open questions to flag. - The critical case $\alpha = 1 + \sqrt{y}$. The current literature does not yet contain a clean limit theorem; cite open work. - Exact variance $\sigma^2_{\alpha,y}$ in the strong-spike CLT. - Beyond Johnstone's case: general spiked models with non-flat bulk (cite Baik and Silverstein (2006); Benaych-Georges and Nadakuditi (2011)). - Rigorous spectral clustering derivations beyond the toy model; cite Couillet and Benaych-Georges (2016).

(d) Closing sentence: the goal of this thesis was to develop the foundational picture from one end of the dimension spectrum to the other; the open questions above point to where the field is going.

A Auxiliary Results

[TODO] *Content to fill in:*

A.1 Borel-Cantelli Lemmas. - First Borel-Cantelli: if $\sum_n \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$. - Second Borel-Cantelli (independence version): if events are independent and $\sum_n \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$. - Used in Section 2.3 for the convergence-in-probability but not-almost-surely counterexample.

A.2 Slutsky's theorem. - Statement and a one-paragraph proof or reference to Casella and Berger (2002, Thm. 5.5.17).

A.3 Continuous mapping theorem. - Statement: if $X_n \xrightarrow{d} X$ and g is continuous on a set of $\mathbb{P}_X$-probability one, then $g(X_n) \xrightarrow{d} g(X)$.

A.4 Multivariate delta method. - Statement and reference; used in Sections 3.2 and 4.3.

A.5 Characteristic functions and the Lévy continuity theorem. - Definition: $\varphi_X(t) = \mathbb{E}[e^{itX}]$; always exists. - Examples: $\mathcal{N}(0,1)$ gives $e^{-t^2/2}$, Cauchy gives $e^{-|t|}$. - Lévy: $X_n \xrightarrow{d} X \iff \varphi_{X_n}(t) \to \varphi_X(t)$ pointwise, with continuity at 0 sufficient for the converse. - Proof of the classical CLT via $(1 + t/n)^n \to e^t$ expansion.

B Simulation Methodology

Single-realization Marchenko-Pastur companion plot. The varying-aspect-ratio figure in Section 3.3 (Figure 12) pools eigenvalues from many independent realizations of $\widehat{\Sigma}_n$ to make the limiting Marchenko-Pastur shape sharp. For comparison, Figure 25 below shows the histogram of a *single* realization at $p = 500$, $n = 1000$ ($y = 1/2$). Even one realization is already enough for the empirical eigenvalue distribution to match the limiting Marchenko-Pastur density visually.

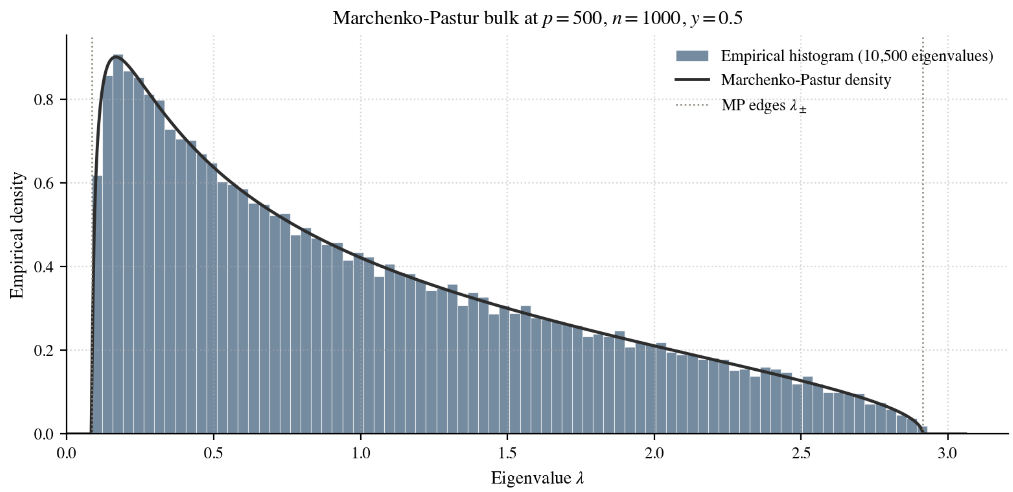

Figure 25: Single-realization Marchenko-Pastur fit at $p = 500$, $n = 1000$ ($y = 1/2$). The histogram of all 500 eigenvalues from one draw of $\widehat{\Sigma}_n$ closely matches the $\mathrm{MP}_{1/2}$ density. The dotted lines mark the support edges $\lambda_\pm = (1 \pm \sqrt{y})^2 \approx 0.086, 2.914$. Companion to Figure 12 in Section 3.3. Generated by `notebooks/03_section3_demos.py`.

[TODO] *Content to fill in:*

B.1 General reproducibility conventions. - All notebooks use Python 3, with ***`numpy`****,* ***`matplotlib`****, and* ***`scipy.stats`****. - Random number generation:* ***`rng = np.random.default_rng(seed)`****, with explicit* ***`seed`*** *at the top of every notebook. - Eigendecomposition:* ***`np.linalg.eigh`*** *on symmetric inputs; the top eigenpair is taken as* ***`evals[-1]`****,* ***`evecs[:, -1]`****. - Plot styling: shared helper module* ***`notebooks/thesis_style.py`*** *applies the six-color palette and sans-serif math labels.*

B.2 Per-notebook parameter table. - Notebook 01 (LLN/CLT demos): $n \in \{100, 1000, 10000\}$*, 500 trials. - Notebook 02 (Wishart and repulsion):* $n = 100$*,* $p = 2$*, 5000 trials. - Notebook 03 (fixed-dim fluctuations):* $n = 500$*, 5000 trials, both* $\Sigma = I_2$ *and* $\Sigma = \mathrm{diag}(1, 2)$*. - Notebook 04 (Marchenko-Pastur & TW):* $n = 1000$*,* $y = 1/2$*, 1 trial for bulk density, 1000 trials for top eigenvalue. - Notebook 05 (spiked eigenvalues, spike phase transition):* $y = 1/2$*,* $n \in \{100, 200, 400, 800, 1200\}$*, 200 trials per* n*,* $\alpha \in \{1.4, 2.5\}$*. - Notebook 06 (fluctuation Q-Q):* $n = 1200$*,* $y = 1/2$*, 500 trials,* $\alpha = 2.5$*. - Notebook 07 (eigenvector overlap):* $n = 1200$*,* $y = 1/2$*, 500 trials,* $\alpha \in \{1.4, 2.5\}$*. - Notebook 08 (spectral clustering toy): two-class Gaussian mixture,* $n = 600$ *per class,* $p \in \{50, 200\}$*.*

B.3 Numerical caveats. - For the strong-spike fluctuation Q-Q plot, we standardize empirically (subtract sample mean, divide by sample standard deviation) because the exact variance $\sigma^2_{\alpha,y}$ requires constants from Yao et al. (2015, Thm. 11.11). - For the weak spike, we do not attempt an exact Tracy-Widom fit because the finite-sample centering and scaling constants are delicate; we report the qualitative shape of the histogram and defer the rigorous fit to future work. - The realized aspect ratio p/n differs from the target y when yn is not integer; figures use the realized ratio in the bulk-edge formula $(1+\sqrt{p/n})^2$ for consistency.

C Selected Code Listings

[TODO] *Include 4 to 6 of the most informative snippets from the notebooks (not full notebooks). Each listing should be self-contained at the function level, captioned, and referenced from the main text where the corresponding figure appears.*

Suggested listings to include here:

- *C.1 One-spike Johnstone simulation (driving function for Section 4.2). Source:* `notebooks/04_sectio`
- *C.2 Eigenvector overlap simulation (Section 4.4). Source:* `notebooks/07_eigenvector_overlap.ipy`
- *C.3 Strong-spike fluctuation simulation (Section 4.3). Source:* `notebooks/06_fluctuation_qq_plots`
- *C.4 Marchenko-Pastur bulk simulation (Section 3.3). Source:* `notebooks/04_marchenko_pastur_trac`

Each listing will be wrapped in a `lstlisting` *environment using the* `thesisPython` *style defined in* `style/thesis.sty`*, with a caption and a* `label` *for cross-referencing from the main text.*

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Patrice Abry, B.C̃ooper Boniece, Gustavo Didier, and Herwig Wendt. On high-dimensional wavelet eigenanalysis. *Annals of Applied Probability*, 2024. See Lemma B.10 (mean-value theorem for eigenvalues).

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Revision Log: Response to G. Didier's Comments

This log appears in the preprint only and is suppressed in the final version (set `\preprintfalse` *in* `main.tex`*). Each green star ⋆ in the body marks a point where a review comment was incorporated and links to the matching entry below; each entry links back to its place in the text.*

⋆1 **§3, opening.** *Didier:* "Meaning?", on "behaves like a classical sample average". *Incorporated:* Spelled out that the SCM converges entry by entry to the true covariance by the law of large numbers, exactly as a scalar sample mean converges to its expectation. ↪ text

⋆2 **§3, opening.** *Didier:* Named the growing-dimension setting "the so-called high-dimensional regime". *Incorporated:* Adopted the phrase as a parenthetical on first mention. ↪ text

⋆3 **Definition 3.1.** *Didier:* Introduce the abbreviation "(SCM)" and call $\widehat{\Sigma}_n$ a *random* matrix. *Incorporated:* Both adopted; "SCM" is now used throughout. ↪ text

⋆4 **after Definition 3.1.** *Didier:* Rephrase as flowing prose: "Note that each entry of a SCM is a sample average. In fact, . . ." *Incorporated:* Replaced the run-in `paragraph` heading with his sentence. ↪ text

⋆5 **equation (10).** *Didier:* Cross-reference the entrywise formula explicitly. *Incorporated:* Added "of (10)" and removed a duplicate `label` on the equation that was breaking compilation. ↪ text

⋆6 **before equation (11).** *Didier:* "As it turns out, the entrywise formula . . . can be conveniently repackaged . . ." *Incorporated:* Replaced the "Matrix form" heading with his sentence. ↪ text

⋆7 **before Proposition 3.1.** *Didier:* "Relation (11) is very useful in establishing the following proposition." *Incorporated:* Replaced the "Symmetry and PSD" heading with his lead-in. ↪ text

⋆8 **§3.2, before Example 3.7.** *Didier:* Soften to "may not converge . . . illustrated in the following counterexample". *Incorporated:* Adopted; also fixed the "counterexampl" typo. ↪ text

⋆9 **§3.3, opening.** *Didier:* "I'm trying to streamline the text." *Incorporated:* Restructured the subsection opening along the lines of his subsequent edits. ↪ text

⋆10 **§3.3.** *Didier:* Define the high-dimensional regime up front via $p/n \to y$. *Incorporated:* Moved the formal definition to the front of the subsection. ↪ text

⋆11 **§3.3.** *Didier:* "In high dimensions, the picture . . . breaks down . . . they no longer converge to $\lambda_i(\Sigma) = 1$." *Incorporated:* Adopted the rewording. ↪ text

⋆12 **§3.3.** *Didier:* Introduce the term "the bulk" for the range of eigenvalues. *Incorporated:* Defined *bulk* on first use (comma, no dash). ↪ text

⋆13 **§3.3.** *Didier:* "You might want to consider the distribution of an individual eigenvalue, such as the top eigenvalue." *Incorporated:* Added a sentence noting that $\lambda_{\max}(\widehat{\Sigma}_n)$ is the focus of the rest of the section, so individual eigenvalues do remain of interest. ↪ text

⋆14 **before Theorem 3.6.** *Didier:* "This is a repetition." *Incorporated:* Deleted the repeated clause about a non-trivial limit when $\Sigma = I_p$. ↪ text

⋆15 **Theorem 3.6.** *Didier:* State the density only for $0 < y < 1$. *Incorporated:* Qualified the p.d.f. with "When $0 < y < 1$ . . . ". ↪ text

⋆16 **after Theorem 3.6.** *Didier:* "'the form of the limit . . . is the same' " is redundant. *Incorporated:* Removed the redundant clause. ↪ text

⋆17 **features list.** *Didier:* Prepend "For $0 < y < 1$" to the support description. *Incorporated:* Adopted. ↪ text

⋆18 **features list.** *Didier:* "Nice." (on the square-root edge). *Incorporated:* No change required; positive feedback noted. ↪ text

⋆19 **before Theorem 3.7.** *Didier:* "Exits?", questioning "the largest sample eigenvalue exits the bulk". *Incorporated:* Reworded to "converges to the upper edge of the bulk", since at the null it settles at the boundary rather than leaving it. ↪ text

⋆20 **before Theorem 3.7.** *Didier:* Add a sentence announcing the top-eigenvalue convergence theorem. *Incorporated:* Adopted. ↪ text

⋆21 **after Theorem 3.7.** *Didier:* Rewrite the "where vs. how" passage, adding "Unlike in low dimensions . . . cannot be Gaussian." *Incorporated:* Adopted; also fixed the "notwithstanding,Theorem" spacing. ↪ text

⋆22 **before Definition 3.9.** *Didier:* Add a lead-in to the TW_1 definition. *Incorporated:* Adopted. ↪ text

⋆23 **after Definition 3.9.** *Didier:* "I don't understand the previous sentence" (". . . universality is what gives the family its name"). *Incorporated:* Rewrote it: the recurrence of one law across unrelated models is an instance of *universality* (the name itself is after Tracy and Widom). ↪ text

⋆24 **$p^{-2/3}$ heuristic.** *Didier:* "At this point, we have not said anything about exponents." *Incorporated:* Replaced "the exponent" with the explicit "$p^{-2/3}$ scaling". ↪ text

⋆25 **before Theorem 3.8.** *Didier:* Johnstone (and Johansson) gave the first *proof* of TW limits for SCMs, not only the normalization sequences. *Incorporated:* Credited the first proof and added Johansson (2000). ↪ text